West Bengal State Council of Technical & Vocational Education and Skill Development
(Technical Education Division)



Syllabus
of

Diploma in Electrical Engineering [EE]

Part-II (3rd Semester)

Revised 2022

3rd Semester

Sl. No	Category of course	Code No	Course Title	Credits	Marks	Total Contact Hours per Week	
						L	P
1	Program Core Course		Introduction to Electric Generation Systems	3	100	3	0
2	Program Core Course		Introduction to Electric Generation Systems Laboratory	1	100	0	2
3	Program Core Course		Electrical Circuits	3	100	3	0
4	Program Core Course		Electrical Circuits Laboratory	1	100	0	2
5	Program Core Course		Electrical and Electronic Measurement	3	100	3	0
6	Program Core Course		Electrical and Electronic Measurement Laboratory	1	100	0	2
7	Program Core Course		DC Machines and Transformers	3	100	3	0
8	Program Core Course		DC Machines and Transformers Laboratory	1	100	0	2
9	Program Core Course		Analog and Digital electronics	3	100	3	0
10	Program Core Course		Analog and Digital electronics Laboratory	1	100	0	2
11	Internship		Internship-I	1	100	0	
	TOTAL			**21**	**1100**	**15**	**10**

- Student **contact hrs./ week =25**
- Theory and practical periods of 60 minutes each
- Abbreviation: L: Lecture class; P: Practical class
- **For Theoretical subjects**: Internal Assessment (40 Marks): Mid semester class test: 20 Marks; Quizzes, viva-voce, Assignment: 10 Marks; Attendance: 10; External Assessment: 60 Marks.
- **For Practical/ Sessional Subjects**: Internal Assessment-60 Marks [Continuous Evaluation:50; Class Attendance:10]; End Semester Assessment-40 Marks [Assignment on the day of Viva-voce and Practical Report submission:20; Viva-voce:20]
- To make the students more familiar with software, effort should be made to prepare laboratory report (like graph; data table etc.) in soft format in addition with traditional hard copy wherever possible.

Course Code	:	
Course Title	:	INTRODUCTION TO ELECTRIC GENERATION SYSTEMS
Semester		Third
Number of Credits	:	3 (L: 3, P: 0)
Prerequisites	:	NIL
Course Category	:	PC
Full Marks		100 [Internal :40 Marks+ External: 60 Marks]

Course Objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences –

- Maintain the efficient operation of various conventional electric power generating plants.

Contents (Theory):		Hrs./Unit
Unit : 1	**Coal Based Thermal Power Plant** 1.1 Selection of site 1.2 Layout and working with block diagram 1.3 Features and Function of the following equipments – (a) Boiler (b) Economiser (c) Super Heater & Reheater (d) Air Pre-Heater (e) F.D. Fan & I.D. Fan, Chimney (f) Steam Turbine (g) Condenser (h) Feed Water Heater and feed water pump (i) De-aerator (j) Spray pond & Cooling Tower (k) D.M. Water plant (l) Coal Handling Plant (m) Ash Handling (n) ESP (o) concept of zero discharge system. 1.4 Basic concept of Rankine Cycle 1.5 Fuel combustion: Stokers, Pulverisers, Fluidised Bed Combustion (concept only) 1.6 Basic concept of Super Critical Boiler 1.7 Classification of coal, Calorific Value, selection of coal for power generation 1.8 Merits & Demerits of coal power plant 1.9 Thermal, Electrical and Overall Efficiency, simple numerical problems. 1.10 List of important coal power plants in India with their capacities	**12**
Unit : 2	**Nuclear Power Plant** 2.1 Selection of site 2.2 Nuclear Fission & Fusion, Chain Reaction, Half-Life period 2.3 Layout and working with block diagram 2.4 Construction and Working of Nuclear Reactor. Features and Function of the following equipments – (a) Core (b) Fuel rod (c) Moderator (d) Control rod (e) Thermal shielding (f) Reflector 2.5 Nuclear fuels : Fissile & Fertile materials, Types of reactors 2.6 Disposal of nuclear waste 2.7 Merits & Demerits of nuclear power plant 2.8 List of important nuclear power plants in India with their capacities	**07**

Unit: 3	**Gas and Diesel Power Plant** **(A) Gas Turbine Power Plant** 3.1 Selection of site 3.2 Layout and working with block diagram 3.3 Fuels and different elements used in gas turbine power plant 3.4 Merits and Demerits of Gas Turbine Power Plant **(B) Diesel Electric Power Plant** 3.5 Layout and working with block diagram 3.6 Working of different elements used in diesel power plant 3.7 Merits & Demerits of Diesel Electric Power Plant 3.8 Field of application	**06**
Unit: 4	**Large Hydro Power Plants** 4.1 Selection of site 4.2 Layout and working with schematic diagram 4.3 Function of different components – Storage reservoir, Dam, Spillway,Penstock, Surge Tank etc. 4.4 Types of Water Turbines and their construction,basic operation of each type and their use(concept only) 4.5 Pumped Storage Plant 4.6 Merits & Demerits 4.7 Power calculation and plant related simple numerical problems 4.8 List of important Large Hydro Plants in India with their capacities	**08**
Unit: 5	**Economics of Power Generation and Interconnected Power Station** 5.1 Related Terms – Connected Load, Firm Power, Cold Reserve,Hot Reserve,Operating Reserve, Spinning Reserve, Average Demand, Maximum Demand 5.2 Base Load Plant & Peak Load Plant 5.3 Load Curve, Load Duration Curve, Integrated Duration Curve, Mass Curve, simple numerical problems 5.4 Factors affecting the cost of generation – Demand Factor, Load Factor, Diversity Factor, Plant Use Factor, Plant Capacity Factor. Simple numerical problems 5.5 Different types of Tariff system - Flat Rate, Block Rate, Two Part, Three Part, simple numerical problems 5.6 Significance of Interconnected Power System 5.7 Choice of size and number of units, combined operation of power stations 5.8 Reasons and impact of grid system faults, State Grid and National Grid 5.9 Brownout and Blackout	**12**
	Total	**45**

References:

1. P. K. Nag, Power Plant Engineering, McGraw Hill, New Delhi, ISBN: 978-9339204044
2. Tanmoy Deb, Electrical Power Generation, Khanna Publishing House, Delhi (Ed. 2018) ISBN-10- 9789386173379 ; ISBN: 13- 978-9386173379
3. B.R.Gupta, Generation of Electrical Energy, S. Chand & Co. New Delhi,
4. J.B. Gupta, A Course in Electrical Power- S. K Kataria and Sons, New Delhi. ISBN:13- 978-9350143742; ISBN:10-9350143747
5. Soni, Gupta, Bhatnagar, A Course in Electrical Power. – Dhanpat Rai and Sons
6. V K. Mehta, Principles of Power System – S. Chand & Company Ltd., New Delhi.ISBN-13- 978-8121924962; ISBN:10-9788121924962

Course Outcomes

After completion of this course, the students will be able to-

1. Explain the features, functioning, merits and demerits of Coal Based Thermal Power Plant.
2. Describe the construction, merits and demerits of Nuclear Power Plant.
3. Identify elements of gas turbine Power plant and Diesel Electric Power Plant through appropriate block diagram.
4. Select the merits and demerits of gas turbine Power Plant and Diesel Electric Power Plant for application in real field.
5. Explain the layout and functioning, merits and demerits of Large Hydro Power Plant with respect to it's various components.
6. Identify various factors associated in the economic use of Power generation and interconnected Power Station.

Internal Assessment (40 Marks)		
Mid Semester Class Test:20 Marks	Quizzes, viva-voce, Assignment: 10 Marks	Attendance: 10
External Assessment (End Semester Examination:60 Marks)		
GROUP	UNIT	
A	1	
B	2,3	
C	4,5	

Course Code	:	
Course Title	:	INTRODUCTION TO ELECTRIC GENERATION SYSTEMS LABORATORY
Semester		Three
Number of Credits	:	1 (L: 0, P: 2)
Prerequisites	:	NIL
Course Category	:	PC
Full marks		100 [Internal :60; External:40]

Course Objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences –

- Maintain the efficient operation of various conventional electric power generating plants.

Practicals

At least EIGHT are to be done

1. Identify the routine maintenance parts of the coal fired thermal power plant after watching a video programme
2. Identify the routine maintenance parts of the gas fired thermal power plant after watching a video programme
3. Identify the routine maintenance parts of the diesel generator power plant after watching a video programme
4. Identify the routine maintenance parts of the nuclear fired thermal power plant after watching a video programme
5. Identify the routine maintenance parts of the large hydro power plant after watching a video programme
6. Study on the different types of Boiler used in coal based thermal power plant
7. Study on different types of Nuclear Reactor used in nuclear power plant
8. Study on different types of Water Turbines used in large hydro power plant
9. Draw Load Curve, Load Duration Curve and Mass curve of your institute
10. Calculate the total energy cost in a (i) Residential (ii) Commercial and (iii) Industrial Bill.

Course Outcomes

The theory, practical experiences and relevant soft skills associated with the course are to be taught and implementatecd so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

a) Analyze the optimized working of the coal power plant.

b) Analyze the optimized working of the nuclear power plant.

c) Explain the optimized working of the gas power plant.

d) Explain the optimized working of the diesel power plant.

e) Analyze the optimized working of the large hydro power plant

f) Calculate the cost of electricity for different class of consumers.

E X A M I N A T I O N S C H E M E (SESSIONAL)

1. **Continuous Internal Assessment of 60 marks** is to be carried out by the teachers throughout the Third Semester. **Distribution of marks:** Continuous evaluation:50 Marks; Class Attendance: 10 Marks
2. **External Assessment (end Semester examination) of 40 marks** shall be held at the end of the Third Semester on the entire syllabus. Assignment on the day of Viva-voce and practical report submission:20; Viva-voce:20

Course Code	:	
Course Title	:	ELECTRIC CIRCUITS
Number of Credits	:	3 (L: 3, P: 0)
Semester		Third
Prerequisites	:	NIL
Course Category	:	PC
Full Marks		100 [Internal :40; External:60]

Course Objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences –

- Maintain electrical systems applying AC and DC circuit fundamentals.

Contents (Theory):		Hrs./Unit
Unit : 1	**Network Theorems (Statement, procedure, areas of applications and limitations of all theorems)** 1.1. Mesh Analysis and Node Analysis 1.2. Star/delta and delta/star transformation 1.3. Superposition theorem. 1.4. Thevenin's theorem. 1.5. Norton's theorem 1.6. Maximum power transfer theorem 1.7. Related Numerical problems.	**10**
Unit : 2	**Single Phase A.C Circuits** 2.1 Generation of alternating voltage. 2.2 Phasor representation of sinusoidal quantities 2.3 R, L, C circuit elements its voltage and current response. 2.4 R-L, R-C, R-L-C combination of A.C series and parallel circuit, impedance, reactance, impedance triangle, Power factor, active power, reactive power, apparent power, power triangle and vector diagram. 2.5 Resonance, Bandwidth, Quality factor and voltage magnification in series and parallel R- L-C circuit. 2.6 Related Numerical problems.	**09**
Unit: 3	**Three Phase A.C Circuits** 3.1 Phasor and complex representation of three phase supply 3.2 Phase sequence and polarity 3.3 Types of three-phase connections, Relationship between Phase and line quantities in three phase star and delta system with derivation. 3.4 Concept of balanced and unbalanced load, neutral shift in unbalanced load 3.5 Three phase power, active, reactive and apparent power in star and delta system. 3.6 Related Numerical problems.	**08**

Unit: 4	**Transient Analysis** 4.1 Introduction 4.2 Simple R-L Circuit supplied from a DC voltage source 4.3 Simple R-C circuit supplied from a DC voltage source. 4.4 Time Constant. 4.5 Related Numerical problems.	**06**
Unit: 5	**Laplace Transform** 5.1 Definition & Properties. 5.2 Laplace Transform of Unit Step, Impulse, Ramp, Exponential, Sine, Cosine Function. Initial value and Final Value Theorem. 5.3 Applications of Laplace Transformations for solving differential equations describing simple electrical circuits 5.4 Related Numerical problems.	**08**
Unit: 6	**Two port network** 6.1 Open circuit Impedance and Short circuit Admittance parameters, 6.2 Transmission parameters and their Inter relations. (Simple Numerical)	**04**
	Total	**45**

References:

1. B.L. Theraja, Basic Electrical Engineering Volume – S. Chand; ISBN-13 : 978-8121924405
2. Mahmood Nahvi & JosephA Edminister, Schaum's outlines Electric circuits, McGrawhill Education (India)Pvt. Ltd., ISBN-13 : 978-9389538908
3. D Roy Choudhury, Networks and Systems, Publisher: NEW AGE, ISBN-13 : 978-8122427677; ISBN-10 : 9788122427677
4. A.Chakraborty, Circuit Theory Analysis and Synthesis. – Dhanpat Rai & Co.
5. S P Ghosh & A KChakraborty, Network Analysis & Synthesis – McGrawhill Education (India)Pvt. Ltd., ISBN-13- 9780070144781;ISBN-10- 9780070144781
6. S Salivahanan; Circuit theory Analysis and Synthesis; Pearson India Education Service Pvt Ltd; ISBN: 978- 93-539-4818-4
7. Gargi Basu; Introduction to circuit and Network; Platinum Publishers; ISBN: 978-8189874-46-9

Course Outcomes

After completion of this course, the students will be able to-

1. Explain the statement, procedure, areas of application and limitations of Network Theorems.
2. Describe the generation, phasor diagram of sinusoidal quantities, R, L, C series and parallel combination of Single-Phase AC Circuits.
3. Analyze circuits and systems by their standard parameters to identify their characteristics in Three Phase AC circuits
4. Explain Transient Analysis of R-L and R-C circuits supplied from DC Voltage Source and concept of time constant.
5. Apply Laplace Transformation to solve various real-life problem in Electrical Circuit Networks.

Internal Assessment (40 Marks)		
Mid Semester Class Test:20 Marks	Quizzes, viva-voce, Assignment: 10 Marks	Attendance: 10

External Assessment (End Semester Examination:60 Marks)	
GROUP	UNIT
A	1,6
B	2,3
C	4,5

Course Code	:	
Course Title	:	ELECTRIC CIRCUITS LABORATORY
Semester		Three
Number of Credits	:	1 (L: 0, P: 2)
Prerequisites	:	NIL
Course Category	:	PC
Full Marks		100 [Internal 60 Marks; External:40 Marks]

Course Objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences –

- Maintain electrical systems applying AC and DC circuit fundamentals.

Practicals

At least EIGHT are to be done

1) To verify Kirchhoff's Current Law and Kirchhoff's Voltage Law.
2) Use dual trace oscilloscope to determine A.C voltage and current response in given R, L, C circuit.
3) Use voltmeter, ammeter, wattmeter to determine active, reactive and apparent power consumed in given R-L / R-C series circuit. Draw phasor diagram.
4) Use voltmeter, ammeter, wattmeter to determine active, reactive and apparent power consumed in given R-L-C series circuit. Draw phasor diagram.
5) Use variable frequency supply to create resonance in given series and parallel R-L-C circuit or by using variable inductor or variable capacitor
6) Use voltmeter, ammeter, wattmeter to determine current, p.f. , active, reactive and apparent power in R-L / R-C parallel A.C. circuit.
7) Use voltmeter, ammeter, wattmeter, p.f meter to determine current, p.f., active, reactive and apparent power for given R-L-C parallel circuit with series connection of resistor and inductor in parallel with capacitor
8) Use voltmeter, ammeter, wattmeter, p.f meter to determine line and phase quantities of voltage and current for balanced three phase star and delta connected load and calculate active, reactive, and apparent power. Draw phasor diagram.
9) Use voltmeter, ammeter, wattmeter, p.f meter to determine line and phase quantities of

voltage and current for unbalanced three phase star and delta connected load and calculate active, reactive, and apparent power. Draw phasor diagram

10) Use voltmeter, ammeter to determine current through the given branch of an electric network by applying mesh analysis.
11) Use voltmeter, ammeter to determine current through the given branch of an electric network by applying node analysis.
12) Use voltmeter, ammeter to determine current through the given branch and voltage across the given element of circuit by applying superposition theorem.
13) Use voltmeter, ammeter to determine equivalent circuit parameter in a given circuit by applying Thevenin's theorem
14) Use voltmeter, ammeter to determine equivalent circuit parameter in a given circuit by applying Norton's theorem
15) Use voltmeter, ammeter to determine load resistance for maximum power transfer for a given circuit by applying maximum power transfer theorem.

Course Outcomes

The theory, practical experiences and relevant soft skills associated with the course are to be taught and implementatecd so that the student demonstrates the following industry-oriented COs associated with the above-mentioned competency:

a) Troubleshoot problems related to single phase A.C series circuits.

b) Troubleshoot problems related to single phase A.C parallel circuits.

c) Troubleshoot problems related to three phase circuits.

d) Use principles of circuit analysis to troubleshoot electric circuits.

e) Apply network theorems to troubleshoot electric circuits.

E X A M I N A T I O N S C H E M E (SESSIONAL)

1. Continuous Internal Assessment of 60 marks is to be carried out by the teachers throughout the Third Semester. **Distribution of marks:** Continuous evaluation:50 Marks; Class Attendance: 10 Marks

2. External Assessment (end Semester examination) of 40 marks shall be held at the end of the Third Semester on the entire syllabus. Assignment on the day of Viva-voce and practical report submission:20; Viva-voce:20

Course Code	:	
Course Title	:	**ELECTRICAL AND ELECTRONIC MEASUREMENTS**
Semester		**Third**
Number of Credits	:	3 (L:3, P:0)
Prerequisites	:	NIL
Course Category	:	PC
Full Marks		100 Marks [Internal :40 Marks + External: 60 Marks]

Course Objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences –

1. Know relevant measuring instrument in different electrical & electronics applications.
2. Know working of relevant measuring instruments in analog & digital applications.

Contents (Theory):		Hrs./Unit
Unit : 1	**Fundamentals of Measurements:** 1.1 Purpose of measurement and significance of measurement. 1.2 Definition & brief explanations of: - Range, sensitivity, true & indicated value, Errors (including limiting errors), Resolutions, Accuracy, Precision and instrument efficiency. 1.3 Classification of instrument systems: - 1.3.1 Null and deflection type instruments 1.3.2 Absolute and secondary instruments 1.3.3 Analog and digital instruments 1.3.4 Static and dynamic characteristics, types of errors 1.4 Calibration of instruments: Necessity and procedure 1.5 Classification of measuring instruments: - Indicating, Recording and Integrating instruments. 1.5.1 Essential requirements of indicating instruments	**05**
Unit : 2	**Measurement of voltage and current:** 2.1 Construction, working principle, salient features, merits and demerits of – i) Permanent Magnet Moving Coil (PMMC) type Instrument ii) Moving Iron (MI) type Instrument iii) Dynamometer type Instrument 2.2 AC voltmeter: Rectifier type (half wave and full wave) – their construction, working principle, salient features. 2.3 Different Methods of range extension of Ammeter and Voltmeter & related problems. 2.4 Calibration of Ammeter and Voltmeter. 2.5 CT: Construction, working principle, different errors and their reduction, Accuracy class, Burden on CT, Specifications, Precautions in the use of CT, application. 2.6 PT: Working principle, Errors (concept only), Accuracy class, Burdens, Specifications, Precautions in the use of PT, application. 2.7 Clamp-on meter: Construction, working principle, application.	**09**

Unit: 3	**Measurement of Electric Power:** 3.1 Dynamometer type wattmeter: Construction and working principle, Multiplying factor and extension of range, Different types of errors and their compensation. 3.2 Measurements of active power in three phase circuit for balanced load by one wattmeter method, two wattmeter method, three wattmeter method, related problems. 3.3 Measurement of active power in three phase circuit for unbalanced load. 3.4 Effect of power factor variation on wattmeter readings in two wattmeter method – Numerical problems. 3.5 Measurement of reactive power in three phase circuit. 3.6 Maximum Demand indicator - Construction and working principle.	**08**
Unit: 4	**Measurement of Electric Energy:** 4.1 Single phase and three phase electronic energy meter: Constructional features and working principle, Different types of errors and their compensation. 4.2 Constructional feature & working principle of single phase and three-phase induction type energy meter. Different types of errors and their compensation. 4.3 Calibration of single-phase electronic energy meter using direct loading.	**07**
Unit: 5	**Measurement of Circuit Parameter, CRO and Other Meters:** 5.1 Measurement of resistance: 5.1.1 Low resistance: Kelvin's double bridge 5.1.2 Medium Resistance: Voltmeter and ammeter method 5.1.3 High resistance: Megger 5.1.4 Ohm meter: Series, Shunt type 5.2 Measurement of inductance using Anderson bridge. 5.3 Measurement of capacitance using Schering bridge. 5.4 Earth tester: Working and connection diagram, application. 5.5 Digital Multimeter: Working principle and basic block diagram, application. 5.6 L-C-R meter: Working and basic block diagram, application. 5.7 Digital Frequency meter: Working and basic block diagram, application. 5.8 Phase sequence indicator: Working and basic block diagram, application. 5.9 Power factor meter: Single phase and Three phase dynamometer type: Working principle and circuit diagram. 5.10 Synchroscope: Working principle and circuit diagram, application. 5.11 Tri-vector meter: Working principle, application. 5.12 Cathode Ray Oscilloscope (CRO): 5.12.1 Single beam/single trace, Dual trace CRO 5.12.2 Digital storage Oscilloscope**:** Basic block diagram, working. 5.12.3 Cathode ray tube, electrostatic deflection, vertical amplifier, horizontal amplifier, time base generator. 5.12.4 Measurement of voltage, amplitude, time period, frequency, phase angle using CRO. Lissajou figure. 5.12.5 Specifications of CRO. 5.13 Signal generator: Working and basic block diagram, Applications. 5.14 Function generator: Working and basic block diagram, function of symmetry, Applications.	**16**
	Total	**45**

References:

1. Sawhney A.K., Electrical and Electronics Measurements and Instrumentation.,DhanpaiRai and Sons, New Delhi, ISBN : 9780000279744
2. Mittle V. N., Basic Electrical Engineering, McGraw-Hill New Delhi, ISBN : 978-0-07-0088572-5
3. Edward Hughes, Electrical Technology, Pearson Education, New Delhi, ISBN-13: 978-0582405196
4. Rajput R.K., Electrical and Electronic Measurement and Instrumentation, S.Chand and Co. New Delhi, ISBN : 9789385676017
5. Theraja B. L., Theraja A. K., A Text Book of Electrical Technology Vol-I(Basic Electrical Engg.), S.Chand and Co. New Delhi, ISBN: 9788121924405
6. Suryanarayna N.V., Electrical Measurements and Measuring Instruments, S.Chand and Co. New Delhi , ISBN :8121920116
7. Kalsi H S; Electronic Instrumentation; McGraw-Hill New Delhi; ISBN13:978-0-07-070206-6; ISBN10:0-07-070206-3

Course Outcomes:

After completion of this course, the students will be able to-

1. Explain the features, functioning and classification of Measuring instruments.
2. Describe the functioning and classification of different types of measuring instrument for measuring voltage and current.
3. Identify the construction, features and application of instrument for measurement of electric power.
4. Describe the features and application of Energy Meter.
5. Explain the functioning and application of different types of electrical instruments including CRO for measuring various electrical parameters.

Internal Assessment (40 Marks)		
Mid Semester Class Test:20 Marks	Quizzes, viva-voce, Assignment: 10 Marks	Attendance: 10 Marks

External Assessment (End Semester Examination:60 Marks)	
GROUP	UNIT
A	1,2
B	3,4
C	5

Course Code	:	
Course Title	:	**ELECTRICAL AND ELECTRONIC MEASUREMENTS LABORATORY**
Semester		Third
Number of credits	:	1 (L:0, P:2)
Prerequisites	:	NIL
Course Category	:	PC
Full marks		100 Marks [Internal :60 Marks; External:40 Marks]

Course Objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences –

1. Use relevant measuring instrument in different electrical & electronics applications.
2. Use working of relevant measuring instruments in analog & digital applications.
3. Know the troubleshooting of different electrical & electronic instruments.

List of Practicals: (At least EIGHT are to be performed)

1. Use Clamp-on meter / digital multi-meter for measurement of AC/DC current, AC/DC voltage.
2. Extend range of ammeter and voltmeter by using (i) shunt and multiplier (ii) CT and PT.
3. Use single wattmeter for measurement of active and reactive power of three phase balanced load.
4. Use two watt-meters for measuring active power of three-phase balanced load.
5. Calibrate single phase electronic energy meter by direct loading.
6. Troubleshoot single phase electronic energy meter.
7. Use Kelvin's double bridge for measurement of low resistance.
8. Use voltmeter and ammeter method / Wheatstone bridge for measurement of medium resistance.
9. Use Megger for measurement of insulation resistance.
10. Use earth tester for measurement of earth resistance.
11. Measure unknown inductance using Anderson bridge.
12. Measure unknown capacitance using Schering bridge.
13. Use CRO for the Measurement of voltage, frequency, phase angle.
14. Use Tri-vector meter for measuring kW, kVAr and kVA of a power line.

Course Outcomes

The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry-oriented COs associated with the above-mentioned competency:

a) Check the working of the electrical measuring instrument.
b) Use different types of measuring instruments for measuring voltage and current.
c) Use different types of measuring instruments for measuring electric power
d) Use different types of measuring instruments for measuring electric energy.
e) Use different types of electrical instruments for measuring various ranges of electrical parameters.
f) Know the use of CRO, Signal generator, Function generator.

E X A M I N A T I O N S C H E M E (SESSIONAL)

1. Continuous Internal Assessment of 60 marks is to be carried out by the teachers throughout the Third Semester. **Distribution of marks:** Continuous evaluation:50 Marks; Class Attendance: 10 Marks

2. External Assessment (end Semester examination) of 40 marks shall be held at the end of the Third Semester on the entire syllabus. Assignment on the day of Viva-voce and practical report submission:20; Viva-voce:20

Course Code	:	
Course Title	:	DC Machines and Transformers
Semester		Third
Number of Credits	:	3 (L: 3, P: 0)
Prerequisites	:	NIL
Course Category	:	PC
Full Marks		100 [Internal :40+ External: 60]

Course Objective: The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences: Maintain dc motors and transformers.

	Contents (Theory):	Hrs/ unit
Unit:1	**GENERAL INTRODUCTION OF ROTATING MACHINE** 1.1. Mechanism of Electro-Mechanical energy conversion for generator & motor mode	01
Unit: 2	**D.C. Generator** 2.1 Working principles, Construction & Types of dc generator. 2.2 Armature winding types – Concept of Lap & Wave winding. 2.3 E.m.f equation 2.4 Generator characteristics: Open circuit characteristics; External Characteristics (Concept only); Methods of building up of e.m.f, Significance of Critical resistance and Critical speed (Numerical). 2.5 Concept of flux distribution in DC machine. 2.6 Armature reaction in DC machine and remedial measures. (Concept only). 2.7 Commutation method, Concept of reactance voltage. 2.8 Applications of different types of D.C. generator. 2.9 Parallel operation of dc generator.	08
Unit:3	**3. D.C. Motor** 3.1 Working principles, Back e.m.f., Speed and Torque equation. (Numerical) 3.2 Characteristics of Series, Shunt & Compound motors. 3.3 Methods of speed control of DC motors. (Numerical) 3.4 Starting methods of DC motor – 3-point & 4-point starter. 3.5 Losses and Efficiency (Numerical). 3.6 Braking methods of DC motor – Regenerative braking, Counter current braking, Dynamic braking. 3.7 Applications of different types of DC motor. 3.8 Brushless DC motors: Construction; working principle and applications.	09
Unit:4	**4.Single phase Transformer:** 4.1 Types of transformers: Shell type and Core type; construction; different parts of transformer and their function. Material used for different parts: CRGO, CRNGO, HRGO, amorphous core. 4.2 Principle of operation. 4.3 E.m.f. equation, Transformation ratio, rating of transformers. (Numerical) 4.4 Concept of ideal and practical transformer.	14

	4.5 Performance under no-load condition with phasor diagram. (Numerical) 4.6 Performance under load condition with phasor diagram. (Numerical) 4.7 Equivalent circuit. (Numerical) 4.8 Per unit representation of impedance (Numerical). 4.9 Voltage Regulation at upf, lagging pf & leading pf. (Numerical) 4.10 Polarity test of transformer. 4.11 Open Circuit and Short Circuit tests – Estimation of losses & Equivalent circuit parameters. (Numerical) 4.12. Losses, Efficiency, Maximum efficiency, All-day efficiency. (Numerical) 4.13 Parallel operation of single phase transformers (Numerical). 4.14 Tap-changing methods, Tap changers – Off load & On-load type. 4.15 Single-phase Auto transformer: Construction, Working principles and application.	
Unit:5	**5. Three phase Transformer**: 5.1 Types of three phase transformer: Bank of three single phase transformers, single unit three phase transformers. 5.2 Construction of 3-phase transformer –different types of Winding. 5.3 Connections of 3-phase transformer – As per IS:2026 (part-IV)-1977 (Classification & necessity). 5.4 Practical application of zig-zag connection in earthing transformer. 5.5 Concept of Tertiary winding and its utility. 5.6 Different cooling methods of transformer. 5.7 Three-phase Auto transformer – Construction; working principle and application. 5.8 Scott-connected transformer – working principle, connection Diagram, practical application. 5.9 Open delta connection – working principle, connection diagram and practical application. 5.10 Criteria for selection of Power transformer and distribution transformer. Amorphous Core type distribution transformer; Specification of three phase distribution transformer as per IS: 1180(Part I)-1989 5.11 Need of parallel operation of three phase transformer, Conditions for parallel operation. 5.12 Phasing out test on Three-phase transformer.	09
Unit:6	**Special Purpose Transformers** 6.1 Isolation transformer: Construction, working principle and applications. 6.2 Single phase welding transformer: Construction, working principle and applications. 6.3 Pulse transformer: Construction, working principle and applications. 6.4 'K' factor of transformers: overheating due to non-linear loads and harmonics.	04
		45 hrs

References:

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7. G.C. Garg & P.S. Bimbhra, Electrical Machines, Vol-I, II, Khanna Book Publishing House ISBN: 978-9386173-447, 978-93-86173-607, New Delhi

8. Mehta, V. K. and Mehta, Rohit, Principles of Electrical Machines, S. Chand and Co. Ltd., New Delhi, ISBN: 9788121930888

9. Bandyopadhyay, M. N., Electrical Machines Theory and Practice, PHI Learning Pvt. Ltd., New Delhi, ISBN: 9788120329973

10. S K Sen, Electrical Machinery, Khanna Book Publishing House, New Delhi; ISBN: 8174091521

Course Outcomes: After completion of this course, the students will be able to-

1. Describe the construction, classification and applications of DC Generator.
2. Identify features, classification and application of DC motors and staters.
3. Describe the working principle, construction, classification and application of single-phase transformer.
4. Explain the construction, classification, applications and different circuit connections of three phase transformer in real field.
5. Identify the construction, features and application of various Special Purpose Transformers.

Internal Assessment (40 Marks)		
Mid Semester Class Test:20 Marks	Quizzes, viva-voce, Assignment: 10 Marks	Attendance: 10

External Assessment (End Semester Examination:60 Marks)	
GROUP	UNIT
A	1,2,3
B	4
C	5,6

Course Code		
Course Title		DC Machines and Transformers Laboratory
Semester		Third
Number of credits		1 (L:0, T:0; P:2)
Prerequisites		NIL
Course Category		PC
Full Marks		100 [Internal :60; External:40]

Course Objective: The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Use dc machines and transformers.

List of Practical: **At least eight Experiments to be performed (four from dc machines and four from transformers)**
1. Dismantle a DC machine.
2. Plot the O.C.C. of a D.C. generator & find the critical resistance.
3. Starting and reversing of DC motor.
3. Control the speed of DC shunt motor above & below rated speed & draw the speed characteristics.
4. Perform the brake test on DC series motor.
5. Compute the efficiency of a D.C. motor by Swinburne's test.
6. Determine equivalent circuit parameters of single-phase transformer by performing O.C. test and S.C. test.
7. Determine the regulation & efficiency of single-phase transformer by direct loading method
8. Compute the efficiency of a single-phase transformer by Back-to-Back test.
9. Perform parallel operation of two single phase transformers to determine the sharing of load current, apparent and real power.
10. Check the functioning and testing of the isolation transformer.
11. Check the functioning and testing of pulse transformer
12. Study and check the connections (vector grouping) of three phase transformers

Course outcomes

The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

1. Maintain DC generators and determine the performance characteristics.
2. Maintain different types of DC motors and determine their performance characteristics.
3. Maintain single phase transformer and determine the performance characteristics.
4. Check different types of connections of three phase transformers.
5. Maintain different types of special purpose transformers used in different applications.

E X A M I N A T I O N S C H E M E (SESSIONAL)

1. Continuous Internal Assessment of 60 marks is to be carried out by the teachers throughout the Third Semester. **Distribution of marks:** Continuous evaluation:50 Marks; Class Attendance: 10 Marks

2. External Assessment (end Semester examination) of 40 marks shall be held at the end of the Third Semester on the entire syllabus. Assignment on the day of Viva-voce and practical report submission:20; Viva-voce:20

Course Code	:	
Course Title	:	ANALOG AND DIGITAL ELECTRONICS
Semester		Third
Number of Credits	:	3 (L: 3, T: 0, P: 0)
Prerequisites	:	NIL
Course Category	:	PC
Full Marks		100 [Internal:40; External:60]

Course Objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences –

- Maintain the proper functioning of analog and digital electronic devices.

Course Contents:

Contents (Theory):		Hrs./Unit
Unit : 1	**1. Diode** **1.1 Zener Diode** 1.1.1 Construction, Symbol, Circuit diagram for characteristics of Zener diode (Forward & Reverse), Zener & Avalanche Breakdown 1.1.2 Zener diode specifications – Zener voltage, power dissipation, break over current, dynamic resistance & maximum reverse current. **1.2 Rectifiers and Filters:** 1.2.1 Need of rectifier, Types of single-phase rectifiers - Half wave & full wave rectifier (Bridge & Centre tapped). 1.2.2 Circuit operation of the rectifiers, Input & output waveforms for voltage & current, Average and rms value of voltage & current (expression only), Ripple, Ripple factor, Ripple frequency, form factor, PIV of diode used, Rectifier efficiency. 1.2.3 Need of filters, Types of filters – a) Series inductor, b) Shunt capacitor, c) LC filter, d) π filter. 1.2.4 Circuit operation of the filters, limitations & advantages	**6**
Unit : 2	**2. Transistors:** **2.1 Bipolar Junction Transistor (BJT):** 2.1.1 Transistor configurations – CB, CE, CC, circuit diagram for input & output characteristics of each configuration, Input & output characteristics. Comparison between three configurations. 2.1.2 Transistor parameters – input & output resistance, α, β and relation between them, Related numerical problems. 2.1.3 Transistor specification – $V_{CE\ Sat}$, $I_{C\ Max}$, V_{CEO}, I_{CEO}, $V_{CE\ Breakdown}$, α, β, Power dissipation. 2.1.4 Basic Common Emitter Amplifier and D.C Load Line : Importance of DC load line and AC load line, Operating point, selection of Q point and stabilization, Related numerical problems.	**14**

	2.1.5 Need of biasing, Name of different biasing methods of transistor. 2.1.6 Voltage Divider biasing method 2.1.7 Power Amplifier: Classification of power amplifier – Class A, Class -B, Class AB and class C, Operation of Push – pull amplifier. **2.2 Field effect transistor (JFET):** 2.2.1 Symbol, Construction of JFET, working principle and V-I characteristics of JFET, pinch-off voltage, drain resistance, transconductance, amplification factor and their relationship. **2.2.2** Introduction to MOSFET-Types of MOSFET, construction, working principle and applications. **2.3 Unijunction transistor (UJT):** Symbol, Construction, Working principle and characteristics of UJT, Equivalent circuit, UJT as relaxation oscillator, Applications.	
Unit: 3	**3. Feedback Amplifiers and Oscillators** 3.1 Concept of Positive and Negative feedback, Voltage gain of amplifier for negative and positive feedback. **3.2** Introduction to oscillator, Block diagram of sine wave oscillator, requirement of oscillation, Barkhausen criterion. Wien bridge oscillator, Colpitt oscillator – operating principle, frequency of oscillation.	**05**
Unit: 4	**4. Combinational Logic Circuits:** **4.1** Half adder, Full adder, Half subtractor, Full subtractor, N bit parallel adder, Parity Generator and checker, Digital comparator **4.2** Code converter, Encoder, Decoder, Multiplexer, Demultiplexer.	**06**
Unit: 5	**5 Sequential Logic Circuits:** 5.1 Flip-flops – RS, D, T, JK, JK Master Slave Flip Flops using basic gates, preset and clear signals. 5.2 Counters - Asynchronous & Synchronous Counter, Mod-N counter, Up Down Counter, Ring counter 5.3 Registers - Shift register, Serial in Serial out, Serial in Parallel out, Parallel in serial out, Parallel in Parallel out.	**09**
Unit: 6	**6 Data Converters & Memory Devices:** 6.1 D/A Converter: Basic concepts, Weighted Resistor D/A converter, R-2R Ladder D/A converter. 6.2 A/D Converter: Successive approximation method, Dual slope method. 6.3 Concept of - Static Memory & Dynamic Memory, SDRAM, DDR RAM, PROM, EEROM, EPROM. 6.4 Comparison of Logic families – DTL, TTL and ECL Gates	**05**
	Total	**45**

References:

1. Electronic Principles, Albert Malvino & D.J.Bates , McGraw Hill Publisher; ISBN:13- 978-9354602399
2. David A. Bell; Electronic Device and Circuit; Oxford University Press, India; ISBN: 13-9780-0-19-569340-9
3. Electronics Engineering, J.B.Gupta, S.K.Kataria & Sons; ISBN:13-978-9350144374
4. V K Mehta & Rohit Mehta; Principles of Electronics, S. Chand Publishing; ISBN: 13- 978-9352838363
5. S Salivahanan, N Suresh Kumar, Electronic Devices and Circuits, McGraw Hill Education; ISBN:13- 978-9339219505
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7. Digital Electronics, G.K. Karate, Oxford University Press
8. Digital Circuits and Design: S. Salivahanan; Oxford University Press, India ISBN: 13-9780-0-19-948868-1

Course Outcomes

After completion of this course, the students will be able to-

1. Describe the features and applications of diode and rectifiers.
2. Explain construction, different types of circuit configurations and applications of Transistors.
3. Describe the concept, features and different application of Feedback Amplifiers and Oscillators through block diagram.
4. Describe and explain different combinational circuits and the practical applications in digital electronics.
5. Describe and explain different sequential circuits in digital electronics.
6. Describe and explain different data converter and memory devices.

Internal Assessment (40 Marks)		
Mid Semester Class Test:20 Marks	Quizzes, viva-voce, Assignment: 10 Marks	Attendance: 10

External Assessment (End Semester Examination:60 Marks)	
GROUP	UNIT
A	1,3,4
B	2
C	5,6

Course Code	:	
Course Title	:	ANALOG AND DIGITAL ELECTRONICS LABORATORY
Number of Credits	:	1 (L: 0, T: 0, P: 2)
Semester		Three
Prerequisites	:	NIL
Course Category	:	PC
Full Marks		100 [Internal:60; External:40]

Course Objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences –

- Maintain the efficient operation of various analog and digital electronic devices.

Practicals

1. Analog Electronics: (At least Four Experiments are to be performed) :

1.1 Construct full-wave rectifier circuit & draw input, output waveforms – with filters and without filters.

1.2 Plot the characteristics of Zener diode and find the breakdown voltage.

1.3 Plot the input & output characteristics of a BJT in CE or CB mode.

1.4 Plot the characteristics of JFET / MOSFET.

1.5 Construct a single stage CE amplifier circuit on a bread board to find out the gain and observe the input and output waveforms.

1.6 Construct Relaxation Oscillator using UJT and observe output waveform by CRO.

1.7 Construct a ±12V power supply on bread board and observe the output waveform by CRO with and without filter circuit. Also observe the output voltage using IC regulator 78XX & 79XX.

2. **Digital Electronics: (At least Four Experiments are to be performed) :**

2.1 Realization of Half Adder, Full Adder, Half Subtractor and Full Subtractor.

2.2 Verification of the function of SR, D, JK and T Flip-flops.

2.3 Realization of Encoder and Decoder circuit.

2.4 Realization of Multiplexer and Demultiplexer circuit.

2.5 Construction of binary Asynchronous or Synchronous counter.

2.6 Construction controlled shift register & verify SISO, SIPO, PISO, PIPO operation.

2.7 Implementation of D/A converter and A/D converter using trainer kit.

Course Outcomes

The theory, practical experiences and relevant soft skills associated with the course are to be taught and implementatecd so that the student demonstrates the following industry-oriented COs associated with the above-mentioned competency:

a) Analyze and describe the performance of Full wave rectifier, Zener Diode, BJT, JFET, MOSFET, UJT, Single stage Amplifier.
b) Construct and analyze the performance of dc power supply (± 12 V).
c) Explain and describe the performance of combinational circuits.
d) Explain and describe the performance of sequential circuits.

E X A M I N A T I O N S C H E M E (SESSIONAL)

1. Continuous Internal Assessment of 60 marks is to be carried out by the teachers throughout the Third Semester. **Distribution of marks:** Continuous evaluation:50 Marks; Class Attendance: 10 Marks

2. External Assessment (end Semester examination) of 40 marks shall be held at the end of the Third Semester on the entire syllabus. Assignment on the day of Viva-voce and practical report submission:20; Viva-voce:20

Internship-I

Course Objectives: Following are the objectives of this course:

- To understand industrial environment and Electrical Engineering activities.
- To get exposure to field level works.
- To get brief idea on drawings/ auto CAD for electrical etc.

After **2nd semester, for Internship I,** students are required to be involved in Inter/Intra Institutional activities viz; Training and simulation programmed with different Institute like workshop of ITI, Other Polytechnics and Technical Institutes Institutions; Soft skill training organized by Training and Placement Cell of the respective institutions; contribution at innovation/entrepreneurship cell of the institute; participation in workshops/ competitions etc; Learning at Departmental lab/ Institutional workshops.

After completion of Internship, the student should prepare a comprehensive report to indicate what he/she has observed and learnt in training period. The student may contact Industrial Supervisor/Faculty Mentor/ TPO for assigning topics and problems and should prepare the final report on the assigned topics. The training report should be signed by the Industrial Supervisor/ Internship Faculty Mentor, TPO and HOD.

Institute may follow

- Basic and applied Science in Inter/ Intra Institution.

OR

- Technical Drawings/ auto CAD for electrical in Inter/ Intra Institution.

OR

- Computer application like module course on "C Programming language "etc in Inter/ Intra Institution.

OR

- Mini project as beginner in Sci Lab/Tinkercad / any other simulation software in Inter/ Intra Institution.

OR

- Workshop practice in Inter/ Intra Institution.

OR

- Industrial Visit

OR

- Free online technical courses

OR

- Different combination of the above.
- **Activities may be conducted continuously for stipulated period of time or may be arranged in a staggered fashion.**

Course outcome

The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry-oriented COs associated with the above-mentioned competency:

- To learn new skills and supplement knowledge.
- To practice communication and teamwork skills.
- To learn strategies like time management, multi-tasking etc. in an industrial setup.
- To meet new people and learn networking skills

The Internship Report will be evaluated on the basis of following criteria (as applicable)

Sl. No.	Criteria for evaluation of Internship Report
1	Originality
2	Adequacy and purposeful write up
3	Organizations, format, drawing, sketches, style, language
4	Practical applications and relationships with basic theory
5	Concept taught in the course outcome
6	Practical applications, relationships with basic theory and concept taught in the course
7	Attendance record, daily diary, quality of Internship Report.

Seminars must be arranged for the student based on his/her training report, before an Internal Committee constituted by the concerned department of the Institute. The evaluation will be based on the following criteria:

Sl. No.	Criteria for evaluation of Internship Seminar
1	Quality of content presented
2	Proper Planning for presentation
3	Effectiveness of presentation
4	Depth of knowledge and skills
5	Viva voce

West Bengal State Council of Technical & Vocational Education and Skill Development
(Technical Education Division)



Syllabus
of

Diploma in Electrical Engineering [EE]

Part-II (4th Semester)

Revised 2022

4th Semester

Sl.No	Category of course	Code No	Course Title	Credits	Marks	Contact Hours per Week	
						L	P
1	Program Core Course	**EEPC202**	Power Electronics Converters and Application	3	100	3	0
2	Program Core Course	**EEPC204**	Power Electronics Converters and Application Laboratory	1	100	0	2
3	Program Core Course	**EEPC206**	Electric Power Transmission and Distribution	3	100	3	0
4	Program Core Course	**EEPC208**	Electric Power Transmission and Distribution Laboratory	1	100	0	2
5	Program Core Course	**EEPC210**	Induction, Synchronous and Special Electrical Machines	3	100	3	0
6	Program Core Course	**EEPC212**	Induction, Synchronous and Special Electrical Machines Laboratory	1	100	0	2
7	Program Core Course	**EEPC214**	Renewable Energy Power Plants	3	100	3	0
8	Program Core Course	**EEPC216**	Renewable Energy Power Plants Laboratory	1	100	0	2
9	Program Elective course I	**EEPE 202**	Switchgear and protection	3	100	3	0
10	Program Elective course I Lab	**EEPE204**	Switchgear and Protection Laboratory	1	100	0	2
11	Project	**PR202**	Minor Project	1	100	0	2
TOTAL				**21**	**1100**	**15**	**12**
Total contact hrs./ week =27							

- Student **contact hrs./ week =27**
- Theory and practical periods of 60 minutes each
- Abbreviation: L: Lecture class; P: Practical class
- **For Theoretical subjects**: Internal Assessment (40 Marks): Mid semester class test: 20 Marks; Quizzes, viva-voce, Assignment: 10 Marks; Attendance: 10; External Assessment: 60 Marks.
- **For Practical/ Sessional Subjects**: Internal Assessment-60 Marks [Continuous Evaluation:50; Class Attendance:10]; End Semester Assessment-40 Marks [Assignment on the day of Viva-voce and Practical Report submission:20; Viva-voce:20]
- To make the students more familiar with software, effort should be made to prepare laboratory report (like graph; data table etc.) in soft format in addition with traditional hard copy wherever possible.

Course Code	:	EEPC202
Course Title	:	Power Electronics Converters and Applications
Number of Credits	:	3 (L: 3, T: 0, P: 0)
Prerequisites	:	NIL
Course Category	:	PC

Course Objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Maintain the proper functioning of power electronic devices.

Course Contents:

Contents (Theory):		Hrs./Unit
Unit : 1	**1. Power semiconductor devices** **1.1 THYRISTOR (SCR)** 1.1.1 Construction, symbol, operation & two transistor analogy. 1.1.2 V-I characteristics of SCR (Holding current, Latching current, Break over voltage). 1.1.3 Thyristor specifications – voltage rating, current rating, power rating, dv/dt, di/dt, Gate current, temperature. 1.1.4. Utility of Snubber circuit, Freewheeling diode. 1.1.5. SCR mounting and cooling. 1.1.6 DIAC, TRIAC, SCS – Principle of operation, specification, characteristics & application. 1.1.7 IGBT - Principle of operation, specification, characteristics & application. 1.1.8 Review of Power BJT, Power MOSFET, IGBT and GTO. 1.1.9. Comparison among Thyristor, BJT, MOSFET, IGBT and GTO as switch. 1.1.10. Concept of single electron transistor (SET) - aspects of Nano- technology.	**10**
Unit : 2	**2. Turn-on and Turn-off Methods of Thyristors** 2.1 SCR Turn-On methods: High Voltage, Thermal triggering, Illumination triggering, dv/dt triggering, Gate triggering. 2.2 Gate trigger circuits – Resistance and Resistance-Capacitance circuits. 2.3 SCR triggering using UJT, PUT: Relaxation Oscillator and Synchronized UJT circuit. 2.4 Pulse transformer and opto-coupler based triggering. 2.5 SCR Turn-Off methods: Class A- Series resonant commutation circuit, Class B-Shunt-Resonant commutation circuit, Class C-Complimentary Symmetry commutation circuit, Class D –Auxiliary commutation, Class E- External pulse commutation, Class F- Line or natural commutation.	**10**
Unit: 3	**3.Converter and Inverter** **3.1. AC to DC Converter:** 3.1.1 Single phase fully controlled Half Wave Converter with a) Resistive load b) R-L load, 3.1.2 Single phase fully controlled Full Wave Converter a) Resistive load b) R-L load, 3.1.3 Three phase fully controlled Bridge Converter a) with Resistive load b) with RL load.	**12**

	3.2 Cycloconverter – Principle of operation of Single phase & Three phase cycloconverter, Basic circuit diagram, Input & Output waveforms, applications. **3.3 Inverter:** 3.3.1 Classification of Single phase & Three phase Inverter – Line commutated & Forced commutated Inverters, Series, Parallel, Bridge Inverter 3.3.2 Operation of basic Series Inverter. 3.3.3 Operation of basic Parallel Inverter. 3.3.4 Operation of Single-phase Bridge Inverter - a) Half Bridge Inverter b) Full Bridge Inverter 3.3.5 Pulse Width Modulated Inverter – a) Single pulse width Modulated Inverter. b) Multiple pulse width Modulated Inverter. **c)** Sinusoidal pulse width Modulated Inverter.	
Unit: 4	**4. DC Chopper:** 4.1Principles of chopper. 4.2 Classification – a) Step-up & Step-down chopper b) Second quadrant, Two quadrant & Four quadrant operation. 4.3 Type-A, B, C, D & E chopper – Operating Principle and applications. 4.4 Jones chopper. 4.5 Related numerical problems.	**04**
Unit: 5	**5.DC & AC Drives:** 5.1Speed control of separately excited DC motor by single phase fully controlled converter. 5.2 Speed control of separately excited DC motor with three phase fully controlled converter. 5.3Speed control of DC series motor with chopper control. 5.4 Speed control of Three phase Induction motor with variable frequency PWM VSI. 5.5 Speed control of Three phase Induction motor with variable voltage variable frequency control.	**06**
Unit 6	**6. Power Supply** 6.1 Uninterrupted power supply – Principle of operation & Block diagram of On load & Off load type UPS. 6.2 SMPS- Principle of operation and application of Forward converter and Flyback converter.	**03**
	Total	**45**

References:

1. Soumitra Kumar Mandal, Power Electronics McGraw Hill Education (India) Private Limited, ISBN-13:978-93-3290-114-8

2. Ramamoorty M., An Introduction to Thyristors and their applications, East-West Press Pvt. Ltd., New Delhi, ISBN: 8185336679.

3. Sugandhi, Rajendra Kumar and Sugandhi, Krishna Kumar, Thyristors: Theory and Applications, New Age International (P) ltd. Publishers, New Delhi, ISBN: 978-0-85226-852-0.

4.Bhattacharya, S.K., Fundamentals of Power Electronics, Vikas Publishing House Pvt. Ltd. Noida. ISBN: 978-8125918530.

5.Jain & Alok, Power Electronics and its Applications, Penram International Publishing (India) Pvt. Ltd, Mumbai, ISBN: 978-8187972228.

6. Rashid, Muhammad, Power Electronics Circuits Devices and Applications, Pearson Education India, Noida, ISBN: 978-0133125900.

7. Singh, M. D. and Khanchandani, K.B., Power Electronics, Tata McGraw Hill Publishing Co. Ltd, New Delhi, 2008 ISBN: 9780070583894.

8. Zbar, Paul B., Industrial Electronics: A Text –Lab Manual, McGraw Hill Publishing Co. Ltd., New Delhi, ISBN: 978-0070728226.

9. Grafham D.R., SCR Manual, General Electric Co., ISBN: 978-0137967711.

10.Agrawal; Power Electronic Systems: Theory & Desig; Pearson

11. Bimal K. Bose; Modern Power Electronics and AC Drives; Pearson

12. Asghar, Power Electronics; PHI Learning

Course Outcomes

The theory and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry-oriented COs associated with the above-mentioned competency:

a) Select power electronic devices for specific applications.

b) Know turn-on and turn-off circuits of Thyristors.

c) Understand the switching circuits and their applications.

d) Understand working and performance characteristics of phase-controlled rectifiers, cycloconverter and inverters.

e) Understand working and performance characteristics of chopper.

f) Apply different types of converters for industrial applications.

g) Understand the basic operation of different power supply.

Internal Assessment (40 Marks)		
Mid Semester Class Test:20 Marks	Quizzes, viva voce, Assignment: 10 Marks	Attendance : 10 marks
External Assessment (End Semester Examination:60 Marks)		
GROUP	**UNIT**	
A	1,2	
B	3,4	
C	5,6	

Course Code	:	EEPC204
Course Title	:	Power Electronics Converters and Applications LABORATORY
Number of Credits	:	1 (L: 0, T: 0, P: 2)
Prerequisites	:	NIL
Course Category	:	PC

Course Objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Maintain the proper functioning of power electronic devices.

Practicals (at least EIGHT are to be performed)

1. Test the proper functioning of power electronic switches – SCR, IGBT, SCS and TRIAC.
2. Test the proper functioning of DIAC to determine the break over voltage.
3. Determine the latching current and holding current using V-I characteristics of SCR.
4. Test the variation of R, C in R and RC triggering circuits on firing angle of SCR.
5. Test the effect of variation of R, C in UJT triggering technique.
6. Perform the operation of Class – A, B, C turn off circuits.
7. Perform the operation of Class –D, E, F turn off circuits.
8. Use CRO to observe the output waveform of half wave-controlled rectifier with resistive load and determine the load voltage.
9. Use CRO to observe the output waveform of Full wave-controlled rectifier with R load, RL load, free wheeling diode and determine the load voltage.
10. Determine the firing angle using DIAC and TRIAC phase-controlled circuit on output power under different loads such as lamp, motor or heater
11. Simulate above firing angle control on SCILAB software
12. Perform speed control of DC series / DC separately excited motor using SCR.
13. Perform speed control of 3-phase Induction motor using PWM inverter. Interpret speed-torque characteristics. Use variable voltage variable frequency drive.
14. Test the performance of given UPS.
15. Study on SMPS.

Course outcomes:

The practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry-oriented COs associated with the above-mentioned competency:

a) Perform the testing of power electronic devices for specific applications.

b) Apply different turn-on and turn-off circuits of Thyristors.

c) Maintain phase-controlled rectifiers.

d) Apply converters for industrial control circuits.

e) Understanding of performance of different power supply.

E X A M I N A T I O N S C H E M E (SESSIONAL)

1. **Continuous Internal Assessment of 60 marks** is to be carried out by the teachers throughout the fourth Semester. **Distribution of marks:** Continuous evaluation:50 Marks; Class Attendance: 10 Marks
2. **External Assessment (end Semester examination) of 40 marks** shall be held at the end of the fourth Semester on the entire syllabus. Assignment on the day of Viva-voce and practical report submission:20; Viva-voce:20

Course Code	:	EEPC206
Course Title	:	ELECTRIC POWER TRANSMISSION AND DISTRIBUTION
Number of Credits	:	3 (L: 3, T: 0, P: 0)
Prerequisites	:	NIL
Course Category	:	PC

Course Objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences –

- Maintain the proper functioning of the electrical transmission and distribution systems.

Contents (Theory):		Hrs./Unit
Unit : 1	Basics of Transmission and Distribution 1.1. Single line diagrams with components of the electric supply transmission and distribution systems. 1.2. Classification of transmission and distribution lines: Primary and secondary transmission; Primary and secondary distribution; standard voltage level used in India. 1.3. Classification of transmission lines: based on type of voltage, voltage level, length and others Characteristics of high voltage for power transmission. 1.4. Kelvin's laws for the economic choice of conductor size – related problem	**07**
Unit : 2	Transmission Line Parameters and Performance 2.1 Line Parameters: Concepts of R, L and C of line parameters and types of lines. Performance of short line: Efficiency, regulation and its derivation, effect of power factor Ferranti Effect (Related Numerical Problems). 2.2 Performance of medium line: representation, nominal 'T', nominal 'π' and end condenser methods (Related Numerical Problems). 2.3 Representation of long transmission line. 2.3.1. Surge impedance. 2.4 Transposition of conductors and its necessity. 2.5 Skin effect and proximity effect.	**08**
Unit: 3	Extra High Voltage Transmission 3.1 Extra High Voltage AC (EHVAC) transmission line: Necessity, advantages, limitations and applications and lines in India. Corona effect. 3.2 High Voltage DC (HVDC) Transmission Line: Necessity, components, advantages, limitations and applications. Layout of monopolar, bi-Polar and homo-polar transmission lines. HVDC lines in India. 3.3 Features of EHVAC and HVDC transmission line. 3.4 Flexible AC Transmission line: Features, Types of FACTS controller. 3.5 New trends in wireless transmission of electrical power.	**08**

Unit: 4	A.C Distribution System 4.1 AC distribution: Components, classification, requirements of an ideal distribution system. 4.2 Feeder and distributor, factors to be considered in design of feeder and distributor. 4.3 Types of different distribution schemes: radial, ring, and grid, layout, advantages, disadvantages and applications. 4.4 Voltage drop, sending end and receiving end voltage (Related Numerical Problems). 4.5 Methods of solving A.C.-1 phase & 3 phase -phase connection (balanced) distribution system. (Numerical based on 1-ph & 3-ph balanced distribution system) 4.6 Distribution Sub-Station: Classification, site selection, advantages, disadvantages and applications. 4.7 Single Line diagram (layout) of 33/11KV Sub-Station, 11KV/400V sub-station, Main connection schemes, Symbols and functions of their components.	**10**
Unit: 5	Components of Transmission and Distribution Line 5.1 Overhead Conductors: Properties of material, types of conductors with trade names, Sag in Overhead Lines, Calculation of Sag (Related Numerical Problems), Stringing chart and its uses, Spacing of conductors, length of span, Relevant I.E. Rules. 5.2 Line supports: Requirements, types of line structures and their specifications, methods of erection. 5.3 Line Insulators: Properties of insulating material, selection of material, types of insulators and their applications, failure of insulators, causes of insulator failure, creepage distance (definition & significance only), derivation of equation of string efficiency for string of three suspension insulator, methods of improving string efficiency. 5.4 Underground Cables: Requirements, classification, construction, Description of (i) PVC, (ii) PILC (iii) FRLS (Fire Retardant Low Smoke), (iv) XLPE cables & (v) Gas filled (SF6) cables, comparison with overhead lines, cable laying. 5.5 Power Factor Improvement Using Static condenser, and Synchronous condenser (Related Numerical Problems), static VAR compensator.	**12**
	Total	**45**

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7. Sivanagaraju; Electric Power Transmission and Distribution; Pearson.
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Course Outcomes

The theory and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry-oriented COs associated with the above-mentioned competency:

a) Draw and explain single line diagrams of transmission and distribution systems.
b) List and identify transmission line parameters and calculate its performance.
c) Compare and explain the EHVAC and HVDC transmission systems.
d) Explain distribution systems with distribution substation.
e) List and explain different components of transmission and distribution Line.

Internal Assessment (40 Marks)		
Mid Semester Class Test:20 Marks	Quizzes, viva voce, Assignment: 10 Marks	Attendance: 10 marks
External Assessment (End Semester Examination:60 Marks)		

GROUP	**UNIT**
A	1,2
B	3,4
C	5

Course Code	:	EEPC 208
Course Title	:	ELECTRIC POWER TRANSMISSION AND DISTRIBUTION LABORATORY
Number of Credits	:	1 (L: 0, T: 0, P: 2)
Prerequisites	:	NIL
Course Category	:	PC

Course Objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

• Maintain the proper functioning of the electrical transmission and distribution systems.

Practicals

1,2,3,4 compulsory and any three from rest	
1.	Students should conduct following activities (any two) Activity-I: Prepare a report based on transmission line network in West Bengal. Activity-II: Collect the information on components of transmission line. Activity-III: Evaluate transmission line performance parameters of a given line. Activity-IV: Library/ Internet survey of electrical high voltage line and HVDC lines. Activity-V: Visit to Transmission/ Distribution Substation and make a report on the visit.
2.	Prepare a model showing: (any two) Activity-I: Single line diagram of electric supply system.

	Activity-II: Single line diagram of a given distribution system. Activity-III: Short line and medium transmission line. Activity-IV: Write a report on the same by giving the details of lines in West Bengal.
3.	Prepare a power point presentation: (any two) Activity-I: Extra High Voltage AC Transmission line. Activity-II: High Voltage DC Transmission line. Activity-III: Flexible AC Transmission line. Activity-IV: New trends in wireless transmission of electrical power.
4.	Collect information on: Activity: A.C Distribution System adjacent to your institute.
5.	Study samples of Overhead Conductors, Underground Cables, Line supports and Line Insulators.
6.	Demonstrate various system faults by D.C. network analyzer.
7.	Demonstrate the improvement of p.f. using static condenser.
8.	Study of distribution simulator using power transmission trainer.

Course Outcomes

The theory, practical experiences and relevant soft skills associated with the course are to be taught and implementatecd so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

a) Interpret the normal operation of the electric transmission and distribution systems.

b) Demonstrate the functioning of the medium and high voltage transmission system.

c) Identify the components of the extra high voltage transmission system.

d) Illustrate the functioning of the low voltage AC distribution system.

e) Demonstrate the components of the transmission and distribution lines.

E X A M I N A T I O N S C H E M E (SESSIONAL)

1. Continuous Internal Assessment of 60 marks is to be carried out by the teachers throughout the fourth Semester. **Distribution of marks:** Continuous evaluation:50 Marks; Class Attendance: 10 Marks

2. External Assessment (end Semester examination) of 40 marks shall be held at the end of the fourth Semester on the entire syllabus. Assignment on the day of Viva-voce and practical report submission:20; Viva-voce:20

Course Code	:	EEPC210
Course Title	:	INDUCTION, SYNCHRONOUS AND SPECIAL ELECTRICAL MACHINES
Number of Credits	:	3 (L: 3, T: 0, P:0)
Prerequisites	:	NIL
Course Category	:	PC

Course objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Maintain Induction, Synchronous and FHP Machines used in different applications.

Contents (Theory):		Hrs./Unit
Unit : 1	**Three Phase Induction Motor** 1.1 Working principle: production of rotating magnetic field, Synchronous speed, rotor speed and slip. 1.2 Constructional details of 3 phase induction motors: Squirrel cage induction motor and Slip ring induction motor. 1.3 Rotor quantities: frequency, induced emf, power factor at starting and running condition. (Numerical) 1.4 Characteristics of torque vs. slip (or speed), Torques: starting, full load and maximum with relations among them. (Numerical) Effect of change in rotor circuit resistance and supply voltage on Torque-Slip characteristics. 1.5 Induction motor as a generalized transformer with phasor diagram and equivalent circuit. Four quadrant operation. 1.6 Power flow diagram, Losses, Efficiency. (Numerical) 1.7 Starters: need and types; DOL, stator resistance, auto transformer, star delta, rotor resistance and soft starters. 1.8 Speed control methods: stator voltage, pole changing, rotor resistance and Variable Voltage Variable Frequency (VVVF) speed control. 1.9 Braking ; Plugging, Rheostatic, Regenerative Method 1.10 Cogging & Crawling (simple idea) and remadial measures. 1.11 Concept of double cage rotor & deep-bar rotor. 1.12 Motor selection for different applications as per the load torque – speed requirements. 1.13 Maintenance of three phase induction motor.	**11**
Unit : 2	**Single phase induction motors** 2.1 Double field revolving theory, principle of making these motors self-starting. 2.2 Construction and working: Resistance start induction run, capacitor start induction run, capacitor start capacitor run, shaded pole, repulsion type, series motor, universal motor, and hysteresis motor. 2.3 Torque-speed characteristics for all of the above motors. 2.4 Motor selection for different applications as per the load torque-speed requirements. 2.5 Maintenance of single phase induction motor.	**06**

Unit: 3	**Three phase Alternators** 3.1 Principle of working, moving and stationary armatures. Advantages of Stationary armature and Rotating field system. 3.2 Methods of excitation systems of 3-phase alternator by – (a) Static excitation (b)Brushless Excitation (c) DC Generator 3.3 Constructional details: parts and their functions, rotor constructions; salient & nonsalient 3.4 Windings: Single and Double layer, Concentrated and Distributed 3.5 E.M.F. equation of an Alternator by considering short pitch factor and distribution factor. (Numerical) 3.6 Alternator loading: Factors affecting the terminal voltage of alternator; Armature resistance and leakage reactance drops. Armature reaction at various power factors and synchronous impedance. Phasor Diagram at lagging, leading and unity power factor. Terminal Voltage vs Load Current characteristics at different power factor. 3.7 Voltage regulation: direct loading and synchronous impedance method. Open and Short Circuit Characteristics (Numerical) 3.8 Active & Reactive power equations in terms of power angle at steady state for non-salient pole alternator.Power vs. Load angle characteristics. 3.9 Losses, efficiency and condition for maximum efficiency. 3.10 Parallel operation of alternators : reasons, advantages and conditions.(Numerical) 3.11 Synchronisation, Methods of synchronisation by (a) three lamp methd (b) using Synchroscope 3.12 Load-frequency characteristics of alternators running in parallel and sharing a common load. (numerical) 3.13 Active Power and Reactive Power control (concept only) 3.14 Maintenance of alternators.	**15**
Unit: 4	**Synchronous Motors** 4.1 Principle of working /operation, significance of load angle. 4.2 Methods of Starting of Synchronous Motor. 4.3 Torques: starting torque, running torque, pull in torque, pull out torque. 4.4 Synchronous motor on load with constant excitation (numerical) 4.5 Effect of excitation at constant load (numerical). 4.6 V-Curves and Inverted V-Curves. 4.7 Hunting and Phase swinging. 4.8 Losses in synchronous motors and efficiency (no numerical) 4.9 Application of synchronous motors.	**08**
Unit: 5	**Fractional horse power (FHP) Motors** 5.1 Construction and working: Synchronous Reluctance Motor, Switched Reluctance Motor, BLDC, Permanent Magnet Synchronous Motors, stepper motors, AC and DC servomotors. 5.2 Torque speed characteristics of above motors. 5.3 Applications of above motors.	**05**
	Total	**45**

References:

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Course Outcomes:

The theory and relevant soft skills associated with the course are to be taught and implementatecd so that the student demonstrates the following industry-oriented COs associated with the above-mentioned competency:

1. Understand the construction and working principle, explain the characteristics of three phase induction motor and select various types of three phase induction motor for different applications.
2. Understand the construction and working principle, explain the characteristics of different types of single-phase induction motor and select various types of single-phase induction motor for different applications.
3. Understand the construction and working principle, explain the characteristics of three phase alternator and their applications.
4. Understand the construction and working principle, explain the characteristics of three phase synchronous motor and their applications.
5. Understand the construction and working principle, explain the characteristics of different types of FHP motor and select various types of FHP motors in different applications.

Internal Assessment (40 Marks)		
Mid Semester Class Test:20 Marks	Quizzes, viva voce, Assignment: 10 Marks	Attendance : 10 marks

External Assessment (End Semester Examination:60 Marks)	
GROUP	**UNIT**
A	1,2
B	3
C	4,5

Course Code	:	EEPC212
Course Title	:	INDUCTION, SYNCHRONOUS AND SPECIAL ELECTRICAL MACHINES LABORATORY
Number of Credits	:	2 (L: 0, T: 0, P:2)
Prerequisites	:	NIL
Course Category	:	PC

Course Objective:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

Maintain Induction, Synchronous and FHP Machines used in different applications.

Practicals: (at least EIGHT are to be done)

1. Identify the different parts (along with function and materials) for the given single phase and three phase induction motor.
2. Perform the direct load test on the three phase squirrel cage induction motor and plot the i) efficiency versus output, ii) power factor versus output, iii) power factor versus motor current and iv) torque – slip/speed characteristics efficiency versus output, v) power factor versus output, vi) power factor versus motor current and vii) torque – slip/speed characteristics.
3. Conduct the No-load and Blocked-rotor tests on given 3-phase squirrel cage induction motor and determine the equivalent circuit parameters.
4. Control the speed of the given three phase squirrel cage induction motor using the applicable methods: i) auto-transformer, ii) VVVF.
5. Measure the open circuit voltage ratio of the three-phase slip ring induction motor and perform the speed control by insertion of resistance in rotor circuit for slip ring induction motor.
6. Conduct the direct load test to determine the efficiency and speed regulation for different loads on the given single phase induction motor; plot the efficiency and speed regulation curves with respect to the output power.
7. Perform the direct loading test on the given three phase alternator and determine the regulation and efficiency.
8. Determine the regulation and efficiency of the given three phase alternator from OC and SC tests (Synchronous impedance method)
9. Conduct the test on load or no load to plot the ‘V’ curves and inverted ‘V’ curves (at no-load) of 3-phase

synchronous motor.

10. Dismantling and reassembling of single phase motors used for ceiling fans, universal motor for mixer.
11. Control the speed and reverse the direction of stepper motor
12. Control the speed and reverse the direction of the AC servo motor
13. Control the speed and reverse the direction of the DC servo motor

Course outcomes:

The practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

a) Verify the performances of three phase induction motor experimentally and maintain it used in different applications.
b) Verify the performances of single-phase induction motor experimentally and maintain it used in different applications.
c) Verify the performances of three phase alternator experimentally and maintain it used in different applications.
d) Verify the performances of three phase synchronous motor experimentally and maintain it used in different applications.
e) Maintain FHP motors used in different applications

E X A M I N A T I O N S C H E M E (SESSIONAL)

1. **Continuous Internal Assessment of 60 marks** is to be carried out by the teachers throughout the fourth Semester.
   Distribution of marks: Continuous evaluation:50 Marks; Class Attendance: 10 Marks
2. **External Assessment (end Semester examination) of 40 marks** shall be held at the end of the fourth Semester on the entire syllabus. Assignment on the day of Viva-voce and practical report submission:20; Viva-voce:20

Course Code	:	EEPC214
Course Title	:	**Renewable Energy Power Plants**
Number of Credits	:	3 (L: 3, T: 0, P:0)
Prerequisites	:	NIL
Course Category	:	PC

Course objectives: The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:
Maintain the efficient operation of various types of renewable energy power plants.

Course contents:

Contents (Theory):		Hrs/ Unit
Unit1	1.1. Classification of energy resources. 1.2. Significance of Non-conventional energy resources. 1.3 Salient features of non-conventional sources and their availability in India.	**2**
Unit2	**Solar Photovoltaic system and Solar Power Plants** 2.1. Basic concept of Solar Radiation Geometry – Latitude, Longitude, Declination, Surface azimuth angle, Solar azimuth angle, Slope of surface, Hour angle, Angle of incidence. 2.2. Solar Map of India: Beam, Diffuse & Global solar power radiation. 2.3 Radiation on inclined surface (Expression only). 2.3 Measurement of solar radiation using Pyranometer. 2.4 Types of Solar Collectors: Liquid Flat plate collectors; Parabolic Trough, Parabolic Dish, Fresnel Reflectors. (Brief description) 2.5. Solar Photovoltaic (PV) system: 2.5.1 Principles for Electron-Hole Pair generation by Photon absorption, Photo-electric effect, Photo-conductive effect and Photovoltaic effect. 2.5.2 Solar cell and its classification. 2.5.3 Working principle of photovoltaic cell. 2.5.4 Current-voltage (I-V) and power voltage (P- V) characteristics of photovoltaic cell, Maximum power point (MPP),dark current, Fill factor, Short circuit current (Isc), Open circuit voltage (Voc), Cell efficiency. 2.5.5 Concept of PV module, PV panel, PV array and its formation. 2.5.6 Classification of PV system – Centralized Solar PV System. Distributed Solar PV System (Brief concept only) 2.6. Solar Photovoltaic (PV) power plant: Components, Layout, Construction, Working principle, Applications. 2.7 Concentrated Solar Power (CSP) plants – Components, Layout, Construction, Working principle, Applications. 2.8 Rooftop solar PV power system.	**15**
Unit 3	**Wind Energy and Wind Power Plants** 3.1 Wind Map of India: Wind power density in watts per square meter. 3.2. Lift and drag principle; long path theory.	**14**

	3.3. Energy estimation of wind, power coefficient (Cp); lift coefficient (C_L), drag coefficient (C_D); tip speed ratio. 3.2. Types of wind turbines 3.2.1. Horizontal axis small wind turbine: direct drive type, geared type, components and working. 3.2.2. Vertical axis small wind turbine: direct drive type, geared type, components and working. 3.3. Wind energy conversion system (WECS): 3.3.1. Constant Speed Electric Generators: Squirrel Cage Induction Generators (SCIG), Wound Rotor Induction Generator (WRIG). 3.3.2. Variable Speed Electric Generators: Doubly-fed induction generator (DFIG), wound rotor synchronous generator (WRSG), permanent magnet synchronous generator (PMSG). 3.4 Geared type wind power plants: Components, Layout and Working principle. 3.5 Direct drive type wind power plants: Components, Layout and Working principle.	
Unit 4	**Micro-hydro Power Plants** 4.1 Energy conversion process of hydro power plant. 4.2 Classification of hydro power plant: High, medium and low head. 4.3 Layouts of micro-hydro power plants. 4.4 Construction and working of hydro turbines used in different types of hydro power plant: High head – Pelton turbine Medium head – Francis turbine Low head – Kaplan turbine. 4.5 Safe Practices for micro hydro power plants.	**6**
Unit 5	**Biomass-based Power Plants** 5.1 Properties of solid fuel for biomass power plants: bagasse, wood chips, rice husk, municipal waste 5.2 Properties of liquid and gaseous fuel for biomass power plants: Jatropha, bio-diesel, gobar gas. 5.3 Layout & working of a Bio-chemical based (e.g. biogas) power plant. 5.4 Layout & working of a Thermo-chemical based (e.g. Municipal waste) power plant. 5.5 Layout & working of Agro-chemical based (e.g. bio-diesel) power plant.	**8**
	Total	**45**

References:

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Course outcomes:

The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

1. Classify non-conventional energy resources and their features.

2. Define the terms related to solar geometry and explain the working principle of solar PV and CS power plants.

3. Define the terms related to wind turbine and explain the working of wind power plants along with constant & variable speed generators used there.

4. Classify hydro power plants and hydro turbines used for the plant. Explain the energy conversion method in hydro power plant.

5. Explain the layout & working of different types of biomass-based power plants.

Internal Assessment (40 Marks)		
Mid Semester Class Test:20 Marks	Quizzes, viva-voce, Assignment: 10 Marks	Attendance: 10

External Assessment (End Semester Examination:60 Marks)	
GROUP	UNIT
A	1,2
B	3
C	4,5

Course Code	:	EEPC216
Course Title	:	**Renewable Energy Power Plants Laboratory**
Number of Credits	:	1 (L: 0, T: 0, P:2)
Prerequisites	:	NIL
Course Category	:	PC

Course objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

1. Identify the parts of various renewable energy power plants
2. Maintain the efficient operation of various renewable energy power plants.

Practicals: (at least EIGHT are to be done)

1. Perform experiment to measure solar radiation using Pyranometer on tilted surface at different angles of inclination and plot radiation vs. time characteristics for certain duration.
2. Perform experiment to plot I-V characteristics of photovoltaic cell module and find out the solar cell parameters (O.C. voltage, Short circuit current, Voltage-current-power at Maximum Power point, Fill factor, Efficiency).
3. Study different parts of a solar flat plate collector/ solar concentrating collector.
4. Perform experiment to measure thermal performance of a solar water heating system.
5. Perform experiment to measure thermal performance of a solar cooker with varying reflector.
6. Identify & study different components of solar street lighting system for AC supply.
7. Identify & study the specified components of a 1 KW Small Wind Turbine (SWT) system.
8. Identify & study the specified components of water turbine using cut section model.
9. Set up the solar PV plant to produce electricity.
10. Set up the wind power plant of with a Small Wind Turbine to produce electricity.
11. Set up the Biogas power plant to produce electricity.
12. Integrate electrical power from solar PV plant, wind power plant and biogas power plant and apply to Microgrid system.
13. Study on assembling and dismantling of CSP plant using parabolic trough / parabolic dish.
14. Study on assembling of vertical axis / horizontal axis small wind turbine to produce electric power.
15. Study on assembling of a micro hydro power plant.
16. Study on assembling of a small biogas plant to generate electric power.

Course outcomes:
The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

1. Identify different components of solar PV and CS power plant and construct solar PV and CS power plant.
2. Identify different components of wind power plant and construct a small wind power plant.
3. Identify different components of micro hydro power plant and construct micro hydro power plant.
4. Identify different components of biomass based power plant and construct a biogas plant.
5. Integrate the power from different sources of renewable energy power plant and supply through Microgrid system.

E X A M I N A T I O N S C H E M E (SESSIONAL)

1. Continuous Internal Assessment of 60 marks is to be carried out by the teachers throughout the fourth Semester. **Distribution of marks:** Continuous evaluation:50 Marks; Class Attendance: 10 Marks

2. External Assessment (end Semester examination) of 40 marks shall be held at the end of the fourth Semester on the entire syllabus. Assignment on the day of Viva-voce and practical report submission:20; Viva-voce:20

Course Code	:	EEPE202
Course Title	:	SWITCHGEAR AND PROTECTION
Semester		Four
Number of Credits	:	3 (L: 3, P: 0)
Prerequisites	:	NIL
Course Category	:	PE
Full Marks		100 [Internal :40 Marks+ External: 60 Marks]

Course Objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Maintain switchgear and protection schemes used in electrical power systems.

Contents (Theory):		Hrs./Unit
Unit : 1	**Fundamental:** 1.1Necessity & functions of protective system. 1.2 Normal & abnormal conditions. 1.3 Types of faults & their causes. 1.4 Symmetrical and Asymmetrical fault- concept of positive, negative and zero sequence. 1.5.1 Use of current limiting reactors & their arrangements. 1.5.2 Short circuit fault calculations in lines fed by generators through transformers 1.5.3 Short- circuit KVA calculations for symmetrical faults – Numerical problems.	**5**
Unit : 2	**Circuit interrupting devices:** 2.1 Basic fuse terminology: fuse element, rated current, fusing current, fusing factor, prospective current, cut-off current, arcing time, rupturing capacity, total operating time. Fuse Characteristics 2.1.1 HRC fuses – construction, types, working, characteristics, selection and applications 2.2 Isolators- vertical break, horizontal break & pantograph type 2.3 Arc formation process, methods of arc extinction (High resistance and Low resistance), Arc voltage, Recovery voltage, Re-striking voltage, RRRV. 2.4 Circuit breakers- Concept, Classification, Working principle, Construction, Specification & Applications of 2.4.1 E.H.V/H.V – Minimum oil circuit breakers (M.O.C.B.), Air Blast Circuit Breaker (A.B.C.B),Sulphur Hexafluoride circuit breaker (SF6). Vacuum circuit breaker. 2.4.2 L.V.- Air circuit breakers (ACB), miniature circuit breakers (M C B), Moulded case circuit breakers (M C C B) , Earth leakage circuit breaker (E L C B or R C CB), Comparison of fuse & MCCB 2.5 Selection of MCCB for motor. 2.6 Selection and rating of HT and LT circuit breakers – breaking capacity, making capacity, rated operating duty, rated voltage. 2.7 Elementary idea of Auto-reclosing. 2.8 Concept of Gas insulated switchgear.	**12**
Unit: 3	**Protective Relaying:** 3.1 Fundamental quality requirements: Selectivity, Speed, Sensitivity, Reliability, Simplicity, Economy. 3.2 CT & PT used in protection: rating of CT and PT; Accuracy Class of CT and their significance, PT; Requirements, Basic circuit diagram, working principle & application of	**12**

	CVT and CCVT. 3.3 Zones of protection, primary & back-up protection. 3.4 Operating principles and construction (in brief) of: Electromagnetic relays (Attracted armature type, Solenoid type, Watt-hour meter type), thermal relays, 3.5 Block diagram and working of static relays and Microprocessor based relays 3.6 Over current relay--- Time-current characteristics of definite time, instantaneous, inverse time and IDMT Relays. Use of very inverse-type O/C relay and extremely inverse type O/C relay. Time-setting, current-setting, PSM – Numerical problems. 3.7 Directional Relay - Construction, Characteristics: Constant product characteristics, Polar characteristics, Concept of dead zone, 30^0, 60^0 and 90^0 connection. 3.8 Distance Protection Relay: Area of applications, Impedance relays, Reactance relay, MHO relay: operating characteristics, effect of arc resistance on their characteristics. 3.9 Differential Relay: Introduction, Current differential protection for an internal fault – fed from single & both ends. Voltage balance differential protection – Schematic diagram & operation (in brief). Mention the position of operating coil and the restraining coil for both the cases.	
Unit: 4	**Equipment Protection:** 4.1 P**rotection of Alternators** – Types of faults; Percentage differential stator protection, brief idea of: - rotor protection due to loss of excitation, protection against rotor overheating because of unbalance in load, overspeed protection, protection against motoring and field suppression. 4.2 **Transformer protection** – Types of faults; Percentage biased differential protection – numerical problems, Buchholz Relay: Construction, Operation, merit and demerits, rate of rise of pressure relay, WTI and OTI; over-fluxing protection, O/C protection, Earth fault protection and restricted earth fault protection scheme. **4.3 Protection of Motor**: Abnormalities & faults. Short circuit protection, Overload protection, Single phase preventer. **4.4 Protection of Busbar & transmission line** 4.4.1 Bus bar protection: Differential and Fault bus protection. 4.4.2 Transmission line protection: Over current protection (Time graded, current graded), Distance and Pilot wire protection.	**10**
Unit: 5	**Over voltage Protection:** 5.1 Causes of over voltages. 5.2 Lighting phenomena & over voltage due to lightning. 5.3 Protection of transmission line & substation from direct stroke. 5.4 Types of lightning arresters & surge absorbers & their Construction & principle of operation. 5.5 Protection against traveling waves. 5.6 Insulation co-ordination.	**6**
	Total	**45**

References:

1. Mehta V K; Rohit Mehta, Principles of Power System – S. Chand & Company Ltd., New Delhi.ISBN-13- 978-8121924962; ISBN:10- 9788121924962

2. Rao. Sunil S., Switchgear and Protection, Khanna Publishers, New Delhi, ISBN: 978-81-7409- 232-3.

3. Singh, R. P., Switchgear and Power System Protection, PHI Learning, New Delhi, ISBN: 978-81-203-3660-5.

4. Gupta. J. B. Switchgear and Protection, S. K. Kataria and Sons, New Delhi, ISBN: 978-93-5014- 372-8.

5. Bhavesh R. Bhalja, R. P. Maheshwari,Nilesh G Chothani ; Protection and Switchgear (Second edition) Oxford University Press; New Delhi; ISBN: 13: 978-0-19-947067-9

6. C L Wadhwa; Electrical Power System (Seventh Edition); New Age International Publishers; New Delhi; ISBN: 978-93-86070-19-7

7. Ram, Badri; Vishwakarma D. N., Power System Protection and Switchgear, McGraw-Hill, New Delhi. ISBN : 9780071077743

8. Ingole Arun; Switchgear and Protection; Pearson

Course Outcomes

The theory and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

1. Identify various types of faults in power system and apply the knowledge to calculate the fault current.

2. Understand the construction and working of various types of current interrupting devices and apply suitably for different applications.

3. Understand construction and working of various types of protective relays and their applications.

4. Know about the protection systems of alternators, transformers. Busbars, Motors.

5. Understand different protection schemes for power system against over voltages.

Internal Assessment (40 Marks)		
Mid Semester Class Test:20 Marks	Quizzes, viva-voce, Assignment: 10 Marks	Attendance: 10

External Assessment (End Semester Examination:60 Marks)	
GROUP	UNIT
A	1,2
B	3
C	4,5

Course Code	:	EEPE204
Course Title	:	SWITCHGEAR AND PROTECTION LABORATORY
Number of Credits	:	1 (L: 0, T: 0, P: 2)
Prerequisites	:	NIL
Course Category	:	PE

Course objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Maintain switchgear and protection schemes used in electrical power systems.

Course contents

List of Practical: (**at least EIGHT are to be done)**
1. Identify various components of different switchgears (through visit or video or model) viz A. Circuit Breaker [: i) LT air circuit breaker; ii) Minimum Oil Circuit Breaker; iii) Air Blast Circuit Breaker; iv) SF6 Circuit Breaker; v) Vacuum Circuit Breaker] B. Isolator [vertical break, horizontal break] C. CTs and PTs D. Power and distribution Transformers **and write their specifications**
2. Test HRC fuse by performing the load test
3. Test MCB by performing the load test
4. Dismantle MCCB/ELCB/ RCCB and identify various parts
5. Testing of Induction type/ Microprocessor Based Over Current relay using Relay Testing Kit to plot the inverse characteristics.
6. Testing of static distance protection relay using Relay Testing Kit.
7. Testing of static Overcurrent protection relay using Relay Testing Kit.
9. Testing of Directional Over Current Relay (DOCR) by Relay Testing Kit.
10. Testing of percentage Differential Protection of Transformer Using Transformer Differential Relay (Electromagnetic/Microprocessor based/ static).
11. Simulate differential protection scheme for transformer with power system simulation kit.
12. Demonstrate the operation of single phasing preventer by creating single phasing fault for a given 3-ph induction motor with D.O.L. starter.
13. Simulate transmission line protection by using the impedance relay/over current relay for various faults. (On transmission line protection simulation Kit).
14. Study of different lightning arrester and identify different parts.
15. Study of ACB/VCB and identify different parts.

Course outcomes:

The practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry-oriented COs associated with the above-mentioned competency:

a) Select suitable switchgears for different applications.

b) Test the performance of different protective relays.

c) Maintain protection systems of alternators and transformers.

d) Apply the knowledge of different protection schemes through simulation.

e) Select suitable arrester against lightning.

E X A M I N A T I O N S C H E M E (SESSIONAL)

1. **Continuous Internal Assessment of 60 marks** is to be carried out by the teachers throughout the fourth Semester. **Distribution of marks:** Continuous evaluation:50 Marks; Class Attendance: 10 Marks
2. **External Assessment (end Semester examination) of 40 marks** shall be held at the end of the fourth Semester on the entire syllabus. Assignment on the day of Viva-voce and practical report submission:20; Viva-voce:20

Course Code	:	PR 202
Course Title	:	Minor Project
Number of Credits	:	1 (L: 0, T: 0, P: 2)
Prerequisites	:	Knowledge of subjects up to 3rd Semester of Electrical Engineering.
Course Category	:	PR

Course Objective:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Design and development of small electrical and electronics device/equipment.

Project group:

1. Formation of project group: Maximum **6 students per batch**.
2. Each project group should select work by consulting the guide.

Activity:

1. a) Study of IC 555 and develop different multivibrator circuits using IC 555 and IC741.
b) Design of single-phase transformer of different voltage ratio and different rating upto 1KVA

AND

2. a) Develop different analog and digital electronics-based circuit as assigned by teacher.

OR

b) Develop model/ circuit/ software simulation for verification of different electrical laws / network theorem or as assigned by teacher.

OR

c) Construct single phase transformer of different voltage ratio and different rating upto 1KVA.

References:

1. A K Sawhney; A course in Electrical Machine Design; Dhanpat Rai & Co.
2. Raina Bhattacharya; Electrical Design, Estimating and Costing; New Age International Publishers
3. V. Rajini and V.S. Nagarajan; Electrical Machine Design; Pearson
4. Bhattacharya Chatterji; Projects in Electrical, Electronics, Instrumentation and computer Engineering.

Course outcome:

i. Develop proper planning to achieve the project goal.
ii. Collect relevant information and resources.
iii. Identify and apply proper techniques.
iv. Analyse the performance of project output.
v. Organize the Written documentation of the project work

E X A M I N A T I O N S C H E M E (SESSIONAL)

1. Continuous Internal Assessment of 60 marks is to be carried out by the teachers throughout the fourth Semester. **Distribution of marks:** Continuous evaluation:50 Marks; Class Attendance: 10 Marks

2. External Assessment (end Semester examination) of 40 marks shall be held at the end of the fourth Semester on the entire syllabus. Assignment on the day of Viva-voce and practical report submission:20; Viva-voce:20

West Bengal State Council of Technical & Vocational Education and Skill Development
(Technical Education Division)



Syllabus
of

Diploma in Electrical Engineering [EE]

Part-III (5th Semester)

Revised 2022

Sl.No	Category of course	Code No	Course Title	Credits	Marks	Total Contact Hours per Week	
						L	P
1	Program Core Course	**EEPC301**	Microcontroller and its Applications	3	100	3	0
2	Program Core Course	**EEPC303**	Microcontroller and its Applications Laboratory	1	100	0	2
3	Program Core Course	**EEPC305**	Building Electrification	3	100	3	0
4	Program Core Course	**EEPC307**	Building Electrification Laboratory	1	100	0	2
5	Program Elective course II		**Any one of the following subjects to be chosen**	3	100	3	0
		EEPE301/1	1.IndustrialAutomation & Control				
		EEPE301/2	2. Industrial Drives				
6	Program Elective course II lab		**Any one of the following laboratories to be chosen**	1	100	0	2
		EEPE303/1	1. Industrial Automation &Control Laboratory				
		EEPE303/2	**2.** Industrial Drives Laboratory				
7	Program Elective course III		**Any one of the following subjects to be chosen**.	3	100	3	0
		EEPE305/1	1. Illumination Practices				
		EEPE305/2	2. Electric Traction				
		EEPE305/3	3. Solar Power Technologies				
8	Program Elective course III Lab		**Any one of the following laboratories to be chosen**	1	100	0	2
		EEPE307/1	1. Illumination Practices Laboratory				
		EEPE307/2	2. Electric Traction Laboratory				
		EEPE307/3	3. Solar Power Technologies Laboratory				
9	Internship		Internship II	1	100	0	
10	Major Project	**PR301**		2	100	0	4
TOTAL				**19**	**1000**	**12**	**12**

- Student **contact hrs./ week =24**
- Theory and practical periods of 60 minutes each
- Abbreviation: L: Lecture class; P: Practical class
- **For Theoretical subjects**: Internal Assessment (40 Marks): Mid semester class test: 20 Marks; Quizzes, viva-voce, Assignment: 10 Marks; Attendance: 10; External Assessment: 60 Marks.
- **For Practical/ Sessional Subjects**: Internal Assessment-60 Marks [Continuous Evaluation:50; Class Attendance:10]; End Semester Assessment-40 Marks [Assignment on the day of Viva-voce and Practical Report submission:20; Viva-voce:20]
- To make the students more familiar with software, effort should be made to prepare laboratory report (like graph; data table etc.) in soft format in addition with traditional hard copy wherever possible.

Course Code	:	EEPC301
Course Title	:	MICROCONTROLLER AND IT'S APPLICATIONS
Number of Credits	:	3 (L: 3, T: 0, P: 0)
Prerequisites	:	NIL
Course Category	:	PC

Course Objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences –

• Maintain different types of microcontroller-based systems.

Contents (Theory):		Hrs./Unit
Unit : 1	Introduction to Microcontrollers 1.1. Basic concept of 8085 microprocessor architecture with functional block diagram. 1.2. Evolution of Microcontrollers. 1.3. Block diagram of Microcomputer, elements of Microcomputer, types of buses. 1.4. Von Neuman and Harvard Architecture. 1.5. Compare Microprocessor and Microcontrollers. 1.6. Need of Microcontroller. 1.7. Family of Microcontrollers and their specifications. 1.8. Versions of Microcontroller 8951, 89C1051, 89C2051, 89C4051 with their specifications and comparison.	**07**
Unit : 2	Architecture of Microcontroller 8051 2.1 Block diagram of 8051, function of each block. 2.2 Pin diagram, function of each pin. 2.3 Concept of Internal memory and External memory (RAM and ROM). 2.4 Internal RAM structure. 2.5 Reset and clock circuit. 2.6 Various registers and SFRs of 8051	**08**
Unit: 3	8051 Instruction Set and Programs 3.1 Overview of 8051 instruction set. 3.2 Various addressing modes. 3.3 Classification of instructions. 3.4 Data transfer instructions. 3.5 Arithmetic and Logical instructions. 3.6 Branching instructions. 3.7 Bit manipulation instructions. 3.8 Stack, subroutine and interrupt related instructions. 3.9 Programs based on above instructions such as Addition, Subtraction, Multi-byte addition, Multiplication of two numbers, BCD to Hex conversion, Hex to BCD conversion, HEX to ASCII conversion etc.	**08**

Unit: 4	Assembly Language Programming 4.1 Software development steps. 4.2 Software development tools like Editor, Assembler, Linker, Loader and Hex converters. 4.3 Role of various files created at various levels in running an Assembly program using simulators like RIDE or KEIL. 4.4 Various directives of Assembly language programming. 4.5 Programs using directives.	**11**
Unit: 5	8051 Internal Peripherals and Related Programs 5.1 I/O ports- List, diagram, read write operation, instructions and related SFRs. 5.2 Timers/counters – list, related SFRs, programming modes, operations with diagram. 5.3 Serial communication- Basics of serial communication, baud rate, related SFRs, programming modes, operations with diagram. 5.4 Interrupts- related SFRs, types, operations with diagram. Power saving operation- modes, related SFR. 5.5 Some applications of microcontroller a. Measurement of voltage, current, frequency. b. Generation of square, triangular and staircase waveform. c. Over current Relay operation. d. Speed control of D.C. motor.	**11**
	Total	**45**

References:

1. Mazidi, Mohmad Ali; Mazidi, Janice Gelispe; MckinlayRoline D., The 8051 Microcontroller and Embedded system, Pearson Education, Delhi, ISBN 978-8177589030
2. Pal, Ajit, Microcontroller Principle and Application, PHI Learning, New Delhi,ISBN13:978-81-203-4392-4
3. Kenneth, Ayala, 8051 Microcontroller Architecture Programming and Application, PHI Learning, New Delhi, ISBN: 978-1401861582
4. Deshmukh, Ajay, Microcontroller Theory and Application, McGraw Hill., New Delhi, ISBN- 9780070585959
5. Kamal, Raj, Microcontroller Architecture Programming, Interfacing and System Design, Pearson Education India, Delhi, ISBN: 9788131759905
6. Mathur; Panda, Microprocessors and Microcontrollers, PHI Learning, New Delhi, ISBN:978-81- 203-5231-5
7. Krishna Kant, Microprocessors and Microcontrollers: Architecture programming and System Design, PHI Learning, New Delhi, ISBN:978-81-203-4853-0

Course Outcomes

The theory and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry-oriented COs associated with the above-mentioned competency:

a) Interpret the salient features of various types of microcontrollers.
b) Interpret the salient features of architype of types microcontrollers IC 8051.
c) Maintain the program features of the Microcontroller based application.
d) Develop assembly language program.
e) Develop programs to interface 8051 microcontrollers with LED/SWITCH/MOTOR.

Internal Assessment (40 Marks)		
Mid Semester Class Test:20 Marks	Quizzes, viva-voce, Assignment: 10 Marks	Attendance: 10

External Assessment (End Semester Examination:60 Marks)	
GROUP	UNIT
A	1,2
B	3,4
C	5

Course Code	:	EEPC 303
Course Title	:	MICROCONTROLLER AND IT'S APPLICATIONS LABORATORY
Number of Credits	:	1 (L: 0, T: 0, P: 2)
Prerequisites	:	NIL
Course Category	:	PC

Course Objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

• Maintain microcontroller-based systems.

Practicals

List of Practical:(at least EIGHT are to be done)	
1.	Interpret details of Hardware kit for Microcontroller and practice to write and execute programs.
2.	Identify different menus available in a simulator software RIDE/KEIL and demonstrate their use.
3.	Develop and execute Assembly language programs using Arithmetic Instructions and demonstrate outcome for a given input data.
4.	Develop and execute Assembly language programs using Logical Instructions and demonstrate outcome for a given input.
5.	Develop and execute an Assembly language program for Addition of series of 8 bit nos, 16 bit result and demonstrate outcome for a given input data.
6.	Develop and execute Assembly language program for addition/subtraction of 16 bit no/multibyte nos. and demonstrate outcome for a given input data.
7.	Develop and execute Assembly language program for Block transfer from and to Internal/External memory using directives and demonstrate outcome for a given input data.
8.	Develop and execute Assembly language program Largest/smallest of given series of no. from Internal/External memory and demonstrate outcome for a given input data.
9.	Develop and execute Assembly language program arrange no in ascending/descending order from Internal/External memory and demonstrate outcome for a given input data.
10.	Develop and execute Assembly language program for LED blinking/LED sequences using delay/timer mode.
11.	Develop and execute Assembly language program to interface LED with microcontroller
12.	Using 8051, develop, run and test Over voltage/under voltage or over current/under current relay circuit with suitable hardware circuit.
13.	Using 8051, Develop, run and test Speed Control of a D.C. motor and note speed vs. Load characteristics at open loop condition.
14.	Using 8051, develop, run and test the operation of a stepper motor with a fixed number of steps and determine the angular displacement per step by measuring the total angular rotation.
15.	Using 8051, develop, run and test Traffic light Control using 8051.
16.	Measurement of ac voltage, current, frequency and phase angle difference (either between two voltages or between voltage and current) using suitable PT, CT, Zero crossing detectors, A/D converters etc. using 8051.

Course Outcomes

The theory, practical experiences and relevant soft skills associated with the course are to be taught and implementatecd so that the student demonstrates the following industry-oriented COs associated with the above-mentioned competency:

a) Interpret the salient features of various types of microcontrollers.

b) Interpret the salient features of architype of types microcontrollers IC 8051.

c) Maintain the program features of the Microcontroller based application

d) Develop assembly language program.

e) Develop program to interface 8051 microcontrollers with LED/SWITCH/MOTOR.

E X A M I N A T I O N S C H E M E (SESSIONAL)

1. **Continuous Internal Assessment of 60 marks** is to be carried out by the teachers throughout the fourth Semester. **Distribution of marks:** Continuous evaluation:50 Marks; Class Attendance: 10 Marks
2. **External Assessment (end Semester examination) of 40 marks** shall be held at the end of the fourth Semester on the entire syllabus. Assignment on the day of Viva-voce and practical report submission:20; Viva-voce:20

Course Code	:	EEPC 305
Course Title		BUILDING ELECTRIFICATION
Number of Credits	:	3 (L: 3, T: 0, P: 0)
Prerequisites	:	NIL
Course Category	:	PC

Course objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Design electrical installation systems in building complexes.

Contents (Theory):		Hrs/Unit
Unit : 1	1.1 Indian Electricity Rules (1956): Rule 28: Voltage level definitions. Rule 30: Service lines & apparatus on consumer premises. Rule 31: Cut-out on consumer's premises. Rule46: Periodical inspection & testing of consumer's installation. Rule 47: Testing of consumer's installation. Rule 54: Declared voltage of supply to consumer. Rule 55: Declared frequency of supply to consumer. Rule 56: Sealing of meters & cut-outs. Rule 77: Clearances above ground of the lowest conductor. Rule 79: Clearances between conductors & trolley wires. Rule 87: Lines crossing or approaching each other. Rule 88: Guarding 1.2 Classification of electrical accessories- controlling, holding, safety, outlet BIS symbols of following electrical accessories. 1.2.1 **Switch** – Their types according to construction such as surface switch, flush switch, and pull switch, rotary switch, knife switch, pendent switch, Main-switch (ICDP, ICTP). Their types according to working such as single pole, double pole, two-way, two-way centre off, intermediate, series parallel switch 1.2.2 **Holders-** Their types such as bayonet cap lamp holder, pendent holder, batten lamp holder, angle holder, bracket holder, tube light holder, screw type Edison and goliath Edison lamp holder, swivel lamp holder. 1.2.3 **Socket outlets and plugs-** two pin, three-pin, multi pin sockets, two-pin and three-pin plug. 1.2.4 **Others-** Iron connector, adaptor, and ceiling rose, distribution box, neutral link, bus-bar chamber. Wooden/ mica boards, Moulded/ MS Concealed boxes of different sizes. Modular accessories.	**06**

Unit : 2	**Electrical Wires and Underground Cables** 2.1Conductors: - wire, cable, bus bar, stranded conductor, cable, armored cable, flexible cable, solid conductor, PVC wires, CTS wire, LC wire, FR (Fire retardant) wire. 2.2 Size of wire according to BIS. Tools used for measurement of wire size, Wire jointing methods. 2.3 Classification of cables: low tension, high tension, and extra high tension cables, solid, oil filled and gas filled type. 2.4 Cable insulation materials –Class of Insulation; vulcanized rubber (VIR), polyvinyl chloride (PVC), cross linked polythene (XLPE), impregnated paper. 2.5 Selection of suitable cable size and type from standard data. 2.6 Cable jointing methods and Cable laying methods.	
Unit: 3	**Illumination in Residential Installation** 3.1. Concept of Luminous flux, Luminous intensity, Lumen, Illumination or illuminance, Lux, Space-height ratio, utilization factor, depreciation factor, luminous efficacy. 3.2 Laws of Illumination-Inverse Square Law, Cosine Law (related numerical). 3.3 Factors affecting the illumination. Different types of lighting arrangements, 3.4 Luminous flux of different types of light sources. 3.5 Lux level required for different places as per SP 72: 2010. 3.6 Lighting design of a room for estimation of lamps.	06
Unit: 4	**Wiring Methods and wiring layout** **4.1 Classification of wiring methods:** 4.1.1 PVC casing-capping wiring- wiring rules according to IS: 732-1983 4.1.2 Conduit wiring- Types of conduits, comparison between Metal and PVC conduit, types of conduit wiring (Surface/Concealed). Conduit wiring accessories, BIS rules for Metal and PVC conduit wiring. 4.1.3 Factors determining the selection of wiring methods. 4.1.4 Comparison of various wiring systems. 4.1.6 General BIS rules for domestic installations. **4.2 Design, working and drawing of following electrical circuits:** 4.2.1 Simple light and fan circuits 4.2.2 Stair case wiring, Go-down wiring circuit 4.2.3 Bedroom lighting circuit 4.2.4 Corridor lighting circuit 4.2.5 Series parallel circuit, Master switch control circuit 4.2.6 Different lighting circuit using - Intermediate switch, Call bell circuit using bell indicator, Design of wiring circuits according to user's requirement.	**07**
Unit: 5	**Residential Building Electrification** 5.1 Domestic Dwellings/Residential Buildings: reading of Civil Engineering building drawing, Interpretation of electrical installation plan and electrical diagrams, electrical symbols as per IS: 732. 5.2 Electrical installation for residential building as per part I section 9 of NEC-2011 5.3 Difference between residential and industrial load, rules/requirements related to lighting load followed in electrical installations, Positioning of equipment. 5.4 Lighting and power circuits: Light and fan circuit, Power circuit 5.5 Wiring and circuit Schematic diagram according to IS: 2042(Part-I)-1962: multiline and single line representation 5.6 Load assessment: Selection of size of conductor, Selection of rating of main switch and protective switch gear.	**12**

	5.7 Design and drawing, estimation and costing of a residential installation having maximum 5 KW load; Sequence to be followed for preparing estimate; Calculation of length of wire and other materials, labour cost. 5.8 Testing of wiring installation as per IS: 732-1982: Insulation resistance - between earth and conductors, between conductors, polarity test of single pole switches. Testing of earth continuity path. 5.9 Residential building Service Connection- types Underground and overhead. Calculation of Material required for service connection.	
Unit 6	**Protection of Electrical Installation** 6.1 Fuse in electric circuit: Re-wirable, cartridge fuses (HRC and LRC), Fuse material, Selection of fuse and rating. 6.2 Miniature circuit Breaker (MCB)-Construction, Principal rating and uses, Earth Leakage Circuit Breaker (ELCB)-Construction, Principal rating and uses. 6.3 System and equipment earthing and its requirements, Earth, earth electrode, earth current, earth terminal, earthing wire, earthing lead, fault current, leakage current, Measurement of earth resistance using earth tester, Methods of reducing earth resistance, Methods of earthing as per IS 3043: 1987 and their procedure- Driven pipe, pipe and plate earthing, modern methods of earthing	**06**
	Total	**45**

References:

1. Raina, K.B. and S.K.Bhattacharya, Electrical Design Estimating and Costing, New Age International Ltd., New Delhi, ISBN 978-81-224-0363-3
2. Allagappan,N. S.Ekambarram, Electrical Estimating and Costing, TMH New Delhi, ISBN-13: 9780074624784
3. Singh, Surjit, Electrical Estimating and Costing, Dhanpat Rai and Co. New Delhi, ISBN: 1234567150995
4. Gupta, J B: A Course in Electrical Installation Estimating and Costing, S K Kataria and Sons,New Delhi, ISBN:978-93-5014-279-0
5. Bureau of Indian Standard, IS: 732-1989, Code of practice for electrical wiring installation
6. Bureau of Indian Standard, SP 30 National Electrical Code 2010
7. Bureau of Indian Standard, SP 72 National Lighting Codes 2010
8. E-REFERENCES:-
   - http://nptel.ac.in/courses/108108076/1 , assessed on 18th January 2016
   - http://www.electrical4u.com, assessed on 18th January 2016
   - https://www.youtube.com/watch?v=A9KSGAnjo2U, assessed on 18th January 2016
   - http://www.electricaltechnology.org/2015/09, assesed on 30 Jan 2016
   - www.slideshare.net/bawaparam/made-by-paramassesed on 30 Jan2016
   - www.electricaltechnology.org/2013/09/electrical-wiring.html assessed on 16 March 2016.

Course outcomes:

The theory and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

a) Understand the functioning of accessories for electrification and significance of IE rules.

b) Selection of wires, cables, wiring systems and cable laying.

c) Design proper illumination scheme for residential unit.

d) Design wiring layouts on wiring board.

e) Design electrical wiring installation system for residential unit.

f) Understand proper earthing and other protection devices for building electrification.

Internal Assessment (40 Marks)		
Mid Semester Class Test:20 Marks	Quizzes, viva-voce, Assignment: 10 Marks	Attendance: 10
External Assessment (End Semester Examination:60 Marks)		
GROUP	UNIT	
A	1,2	
B	3,4	
C	5,6	

Course Code	:	EEPC307
Course Title	:	BUILDING ELECTRIFICATION LABORATORY
Number of Credits	:	1 (L: 0, T: 0, P: 2)
Prerequisites	:	NIL
Course Category	:	PC

Course objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Design electrical installation systems in building complexes.

Practicals: (At least eight Experiments to be performed)

1. Prepare series testing board for testing fan or other electrical gadgets.
2. Select the electric wire using measuring and testing instruments for particular applications.
3. Study and Identify cables (PVC, XLPE, and VIR) and catenary cable of different current ratings.
4. Prepare wiring installation on a board from the commencement of supply showing energy meter, MCB, control of one lamp, one fan and one socket from one switch board in PVC surface conduit wiring system.
5. Prepare wiring installation on a board showing control of one lamp, one fan and one socket from one switch board using Batton wiring system.
6. Control one lamp from two different places using PVC surface conduit wiring system.
7. Make a report on design and estimation of lighting load of a class room

8. Design 2 BHK residential installation scheme and estimate the material required. Draw the detail wiring layout for installation and make a chart for the materials required.
9. Test wiring insulation using megger.
10. Draw different types of earthing system and make a chart of materials required.
11. Measure earth resistance using earth megger.
12. Measurement of three phase energy using static energy meter which can show maximum demand, reactive power, TOD in addition to active power.
13. Measurement of energy using CT or CT and PT.
14. Selection of transformer rating and associated equipments for indoor substation of commercial building/ complex.
15. Study of smart meter and associated system for commercial building/ complex.
16. Design electrical installation system of single building complexes.

Course outcomes:

The practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

a) Select accessories, wires, cables and wiring systems for electrification.
b) Design electrical wiring installation system for residential unit.
c) Design proper illumination scheme for residential unit.
d) Prepare wiring layouts on wiring board.
e) Locate and diagnose faults in electrical wiring installation.
f) Test for proper earthing for building electrification.

. E X A M I N A T I O N S C H E M E (SESSIONAL)

3. **Continuous Internal Assessment of 60 marks** is to be carried out by the teachers throughout the fourth Semester. **Distribution of marks:** Continuous evaluation:50 Marks; Class Attendance: 10 Marks
4. **External Assessment (end Semester examination) of 40 marks** shall be held at the end of the fourth Semester on the entire syllabus. Assignment on the day of Viva-voce and practical report submission:20; Viva-voce:20

Course Code	:	EEPE 301/1
Course Title	:	INDUSTRIAL AUTOMATION AND CONTROL
Number of Credits	:	3 (L: 3, T: 0, P:0)
Prerequisites	:	NIL
Course Category	:	PE

Course objectives:

Theaimofthiscourseistohelpthestudenttoattainthefollowingindustryidentifiedcompetency through various teaching learning experiences:

Contents (Theory):		Hrs/Unit
Unit : 1	**Introduction to Industrial Automation** 1.1 Automation: Need and benefits. 1.2 Types of automation system: Fixed, Programmable, Flexible 1.3 Different systems used for Industrial automation: Magnetic control PLC, SCADA,	**04**
Unit : 2	**2.1 Magnetic Control Systems:** Operation & Applications of Contactor control circuit components – (i) Switches – Push button type, Selector type, Limit switch, Pressure, Float type, Proximity, (ii) Electromagnetic Contactor (iii) Time delay relays (OFF delay, ON delay). **2.2 Magnetic control of ac motor:** Operation of Control circuit & Power circuits of – (i) Reversing the direction of rotation of induction motor with Interlocking systems (ii) Simple ON-OFF motor control circuit, (iii) Automatic Sequential control of motor. (iv) DOL starter, (v) Automatic Star-Delta starter	**10**
Unit: 3	**PLC Fundamentals** Evolution of PLC. Building blocks of PLC**:** CPU, Memory organization, Input-output modules (discrete and analog), Specially I/O Modules, Power supply ,Fixed and Modular PLC and their types, Redundancy in PLC module, I/O module selection criteria ,	**06**
Unit: 4	**PLC Programming and Applications** 4.1 PLC I/O addressing 4.2 PLC programming Instructions**:** Relay type instructions, Timer instructions: On delay, off delay, Counter instructions: Up, Down, High speed, Logical instructions, Data handling Instructions, Arithmetic instructions. 4.3 PLC programming language**:** Functional Block Diagram (FBD), Instruction List. Structured text, Sequential Function Chart (SFC), Ladder Programming. 4.4 Simple Programming examples using ladder logic based on relay, timer counter, logical, comparison, arithmetic and data handling instructions. 4.5 PLC Based Applications: Motor sequence control, Traffic light control, Elevator control, Tank Level control, Conveyor system, Stepper motor control.	**10**
Unit: 5	**Electric Drives and special machines** 5.1 Electric drives: Types, functions, characteristics, four quadrant operation. 5.2 DC and AC drive controls: V/F control, Parameters, direct torque control. Drives: Specifications, Applications-Speed control of AC motor /DC Motor.	**05**

Unit 6	**Supervisory Control and Data Acquisition System (SCADA)** 6.1.1 Basic components of Data Acquisition System. 6.1.2 Components of a PC-based Data Acquisition System. 6.2 Introduction to SCADA: 6.2.1 Typical SCADA architecture/ block diagram, 6.2.2 Benefits of SCADA. 6.2.3 Brief introduction to hardware and software used in SCADA. 6.2.4. Applications of SCADA.	**10**
	Total	**45**

References:

1. Dunning, G., Introduction to Programmable Logic Controllers, Thomson /Delmar learning, New Delhi, 2005,ISBN 13 :9781401884260
2. Jadhav, V. R., Programmable Logic Controller, Khanna publishers, New Delhi, 2017, ISBN : 9788174092281
3. Petruzella,F.D., Programmable Logic Controllers, McGraw HillIndia, NewDelhi, 2010,ISBN: 9780071067386
4. Hackworth ,John; Hackworth,Federic, Programmable Logic Controllers, PHI Learning, NewDel- hi, 2003, ISBN :9780130607188
5. StenersonJon, Industrial automation and Process control, PHI Learning, NewDelhi,2003, ISBN : 9780130618900
6. Mitra, Madhuchandra; Sengupta,Samarjit, Programmable Logic Controllers and Industrial Auto- mation An introduction, Penram International Publication,2015, ISBN:9788187972174
7. Boyar, S. A., Supervisory Control and Data Acquisition, ISA Publication, USA, ISBN: 978- 1936007097
1. BaileyDavid; WrightEdwin, Practical SCADA for industry, Newnes (animprintofElsevier), UK 2003,ISBN: 0750658053

Course outcomes:

The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented Cos associated with the above mentioned competency:

a) Identify different types of automation systems.

b) Interface I/O devices with the PLC modules.

c) Develop PLC ladder programs for various applications.

d) Select the suitable motor drives for different applications

e) Prepare simple SCADA applications.

Internal Assessment (40 Marks)		
Mid Semester Class Test:20 Marks	Quizzes, viva-voce, Assignment: 10 Marks	Attendance: 10

External Assessment (End Semester Examination:60 Marks)	
GROUP	UNIT
A	1,2
B	3,4
C	5,6

Course Code	:	EEPE 303/1
Course Title	:	INDUSTRIAL AUTOMATION AND CONTROL LABORATORY
Number of Credits	:	1 (L: 0, T: 0, P:2)
Prerequisites(Course code)	:	NIL
Course Category	:	PE

Course objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Maintain Industrial Automation Systems.

Practicals:

1. Testing push button switches, timer and electromagnetic contactor.
2. Make & test the control and power circuit for forward and reverse rotation of sq. cage induction motor using contactor circuit.
3. Make & test the control and power circuit for automatic star-delta starter operation of sq. cage induction motor using contactor control
4. Identify various parts of the given PLC and front panel status indicators.
5. Use PLC to test the START STOP logic using two inputs and one output.
6. Develop/Execute a ladder program for the given application using following:-timer, counter,
   comparison, logical, arithmetic instructions.
7. Use PLC to control the following devices like lamp, motor, pushbutton switches, proximity sensor
8. Measure the temperature of the given liquid using RTD or Thermocouple and PLC.
9. Develop/test ladder program to blink the LED/lamp.
10. Develop/test the Ladder program for sequential control application of lamps/DC motors.
11. Develop ladder program for Traffic light control system.
12. Develop and test ladder program for pulse counting using limit switch/Proximity sensor.
13. Develop/test ladder program for Automated elevator control.
14. Develop/test ladder program for rotating stepper motor in forward and reverse direction at constant speed.
15. Develop/test ladder program for tank water level control.
16. Develop/test ladder program for control of speed of stepper motor with suitable drivers.

17. To study DC chopper circuit using SCR.
18. Identify various front panel controls of VFD (smart drive).
19. Control speed of AC motor using VFD.(VFD-Variable Frequency Drive)
20. Use various functions of SCADA simulation editors to develop simple project.
21. Develop a SCADA mimic diagram for Tank level control.
22. Develop SCADA mimic diagram for Flow control in a given system.
23. Develop a SCADA mimic diagram for traffic light control.

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Course outcomes:

The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

a) Identify different types of automation systems.
b) Interface I/O devices with the PLC modules.
c) Develop PLC ladder programs for various applications.
d) Select the suitable motor drives for different applications.
e) Prepare simple SCADA applications.

.E X A M I N A T I O N S C H E M E (SESSIONAL)

5. **Continuous Internal Assessment of 60 marks** is to be carried out by the teachers throughout the fourthSemester. **Distribution of marks:** Continuous evaluation:50 Marks; Class Attendance: 10 Marks
6. **External Assessment (end Semester examination) of 40 marks** shall be held at the end of the fourth Semester on the entire syllabus. Assignment on the day of Viva-voce and practical report submission:20; Viva-voce:20

Course Code	:	EEPE 301/2
Course Title	:	INDUSTRIAL DRIVES
Number of Credits	:	3 (L: 3, T: 0, P: 0)
Prerequisites	:	NIL
Course Category	:	PE

Course Objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences –

• Maintain electric AC and DC Drives.

Contents (Theory):		Hrs./Unit
Unit : 1	AC and DC Motors 1.1 Need of Electric Drives, Functional Block diagrams of an electric drives. 1.2 Load characteristics, Starting method, speed control and braking of a. Series, Shunt and compound DC motors. b. Three phase Induction Motors i) Squirrel cage Induction motor ii) Slip ring Induction Motor c. Single phase AC Motors i) Capacitor run motors ii) Capacitor start motors d. Three phase Synchronous motor. 1.3 Determination of Motor Rating. 1.4 Duty class of motor. 1.5 Specification of motors.	**12**
Unit : 2	Special Motors 2.1 Characteristics, Control strategies and application of a. DC servo motor b. AC servomotor c. Stepper motor e. Brushless DC Motors. 2.2 Determination of Motor Rating. 2.3 Specification of motor.	**07**
Unit: 3	DC Drives 3.1 Speed control and torque control of DC drive (separately excited dc motor) using A. Single phase SCR Drives a) Semi control and Full controlled converter B. Three Phase SCR Drives a) Semi control and Full controlled converter b) Dual converter. 3.2 Chopper Controlled DC Drives for dc series motor. 3.3 Specification of drives.	**09**
Unit: 4	AC Drives 4.1 Speed control of three phase induction motor: A. Variable Frequency Control of B. Voltage Source Inverter Control C. Current Source Inverter Control D. Rotor Resistance Control E. Slip Power Recovery using static converter. 4.2 Synchronos motor drive. 4.3 Specification of drives. 4.4 Selection of AC and DC drive.	**10**
Unit: 5	Techniques of Motor Control 5.1 **Magnetic control of ac motor:** Operation of Control circuit & Power circuits of – (i) Reversing the direction of rotation of induction motor with Interlocking systems (ii) Simple ON-OFF motor control circuit, (iii) Automatic Sequential control of motor. (iv) DOL starter, (v) Automatic Star-Delta starter.	**07**

	5.2 AC/DC motor drive using Microcomputer block diagram. 5.3 AC/DC motor drive using Microcontroller block diagram. 5.4 Concept of Phase locked loop in dc motor speed control using block diagram.	
	Total	**45**

References:

1. P.S. Bimbhra, Electric Machines, Khanna Book Publishing Co., New Delhi (ISBN: 978-93-86173-294)

2. Saxena, S.B Lal ;Dasgupta, K., Fundamentals of Electrical Engineering, Cambridge university press pvt. Ltd., New Delhi, ISBN: 9781107464353

3. Theraja, B. L. Theraja, A. K., A Text Book of Electrical Technology Vol-II, S. Chand and Co. Ramnagar, New Delhi, ISBN :9788121924405

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5. Sen P.C., Power Electronics, McGraw-Hill Publishing Company Limited, New Delhi. ISBN:9780074624005

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8. Agrawal , Jai P., Power Electronic Systems Theory and Design, Pearson Education, Inc. ISBN 9788177588859.

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10. Pillai, S.K., A first course on Electrical Drives, Wiley Eastern Ltd. New Delhi, ISBN :13: 978- 0470213995

Course Outcomes

The theory and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry-oriented COs associated with the above-mentioned competency:

a) Understand the performance of DC and AC motor for various electric drive applications.
b) Understand the performance of special purpose motor for various electric drive applications.
c) Know the operations of DC Drives and its application.
d) Know the operations of AC Drives and its application.
e) Know the operations of microcomputer/micro controller based electric motors its application.

Internal Assessment (40 Marks)		
Mid Semester Class Test:20 Marks	Quizzes, viva-voce, Assignment: 10 Marks	Attendance: 10

External Assessment (End Semester Examination:60 Marks)	
GROUP	UNIT
A	1
B	2,3
C	4, 5

Course Code	:	EEPE 303/2
Course Title	:	INDUSTRIAL DRIVES LABORATORY
Number of Credits	:	1 (L: 0, T: 0, P: 2)
Prerequisites	:	NIL
Course Category	:	PE

Course Objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

• Maintain microcontroller-based systems.

Practicals

List of Practical:(at least EIGHT are to be done)	
1.	Control the speed of DC Motor using armature voltage control method.
2.	Control the speed of DC Motor using field current control method.
3.	Measure the output voltage of chopper for resistive load by varying the frequency and /or duty cycle of chopper.
4.	Control the speed of three phase squirrel cage induction motor using stator voltage control method.
5.	Effect on speed of given D.C. series motor by varying armature voltage using step down chopper.
6.	Observe the effect on speed of the given D.C. separately excited motor by varying voltage using step down chopper.
7.	Control the speed of the given separately exited motor by changing the firing angle of SCR using single phase full converter and measure the speed
8.	Control the speed of the given three phase induction motor by using constant V/f method and plot the graph between speed and frequency.
9.	Control the speed of the given three phase induction motor by varying frequency and plot the graph between speed and frequency.
10.	Control the speed of the given synchronous motor drives using microcontroller
11.	Demonstrate High power SCR/power device and Heat sink and write their specifications and rating.
12.	Control the speed of single-phase capacitor split phase induction motor using DIAC –TRIAC circuit.
13.	Control the speed of DC motor drives using microcontroller.
14.	Make & test the control and power circuit for forward and reverse rotation of sq. cage induction motor using contactor circuit.
15.	Make & test the control and power circuit for automatic star-delta starter operation of sq. cage induction motor using contactor control.
16.	Dynamic braking of three phase squirrel cage induction motor using contactor control.

Course Outcomes

The theory, practical experiences and relevant soft skills associated with the course are to be taught and implementatecd so that the student demonstrates the following industry-oriented COs associated with the above-mentioned competency:

a) Select relevant DC motor for various electric drive applications.

b) Select relevant AC motor for various electric drive applications.

c) Implement DC Drives for various applications..

d) Implement AC Drives for various applications

e) Perform speed control of motor by micro controller based drive.

E X A M I N A T I O N S C H E M E (SESSIONAL)

1. **Continuous Internal Assessment of 60 marks** is to be carried out by the teachers throughout the fourth Semester. **Distribution of marks:** Continuous evaluation:50 Marks; Class Attendance: 10 Marks
2. **External Assessment (end Semester examination) of 40 marks** shall be held at the end of the fourth Semester on the entire syllabus. Assignment on the day of Viva-voce and practical report submission:20; Viva-voce:20

Course Code	:	EEPE 305/1
Course Title	:	ILLUMINATION PRACTICES
Number of Credits	:	3 (L: 3, T: 0, P: 0)
Prerequisites	:	NIL
Course Category	:	PE

Course objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Design illumination schemes and associated electrification of buildings.

Course contents:

Contents (Theory):		Hrs./Unit
Unit : 1	**Fundamentals of illumination** 1.1 Electromagnetic radiation & Light; Electromagnetic spectrum – Ultraviolet, Visible, Infrared spectrum. 1.2 Definition of (As per CIE): Light, Luminous Flux, Luminous Intensity, Lumen, Candela, Illuminance, Lux, luminance, brightness, contrast, luminous efficacy. 1.3 Glare: Discomfort and disability glare. 1.4 Blackbody radiator, Selective radiator, CRI, CCT. 1.5 Human eye as an optical system – basic concept. Spectral sensitivity of human eye – Photopic, Scotopic, Mesopic vision. 1.6 Laws of illumination: Inverse Square Law; Lambert's Cosine Law; Related numerical.	**09**
Unit : 2	**Measurement** 2.1 Polar curves and its applications. 2.2 Construction and working principle of Luxmeter. 2.3. Measurement of illuminance by luxmeter; measurement of luminous flux by integrating sphere (concept only). 2.4 Application of Polar Photometer & Goniophotometer. 2.5. CIE standard source of illuminant.	**05**

Unit: 3	**Lamps and Luminaires** 3.1.1. Principle of incandescence and gaseous discharge lamps. 3.1.2. Electromagnetic & Electronic ballast – Operation & comparison; Ignitor – its function in lamps. 3.2 Construction, working principle, connection diagram, technical features and application of a) Incandescent lamp and Halogen lamps. b) Low pressure discharge lamps and related : Fluorescent lamp; CFL, sodium vapour lamp. c) High pressure discharge lamps: sodium vapour lamp, mercury vapour lamp, metal halide lamp. d) LED, LASER (concept only), Optical fibre. 3.3. Luminaire – Types of luminaires, Design consideration, Ingress protection (IP).	**10**
Unit: 4	**Illumination Control and Control Circuit** 4.1 Lighting control and energy conservation. 4.2. Dimmer: resistance type; auto-transformer type. 4.3 Electronic Dimmer: working principle and operation of: a. Thyristor operated dimmer b. Triac operated dimmer. 4.4.1 Photo cell : Construction and working principle. 4.4.2 Occupancy sensor: PIR; Ultrasonic sensors : Range, rating and applications.	**05**
Unit: 5	**Interior lighting design** 5.1. Standard illumination level for various interior application as per IS 3646 (Part-I). 5.2 Space height ratio, utilization factor, light loss factor overall uniformity, glare index, CRI, Light power density (LPD); ECBC. 5.3 Lighting calculation methods: Watt /m^2 method, Lumens or light flux method, Point to point method. (Numerical) 5.4. Design considerations for interior lighting of - (a) Residential complex. (b) Commercial complex. (c) Industrial premises.	**08**
Unit 6	**Exterior lighting design** 6.1 Road Lighting: 6.1.1 Related terms: Throw, spread and tilt angle, outreach, overhang. 6.1.2 Classification of roads according to luminance level. 6.1.3. Types of lighting arrangement: Single side, Staggered, opposite and central arrangement. 6.1.4. Classification of Luminaires: Symmetrical, Asymmetrical and double asymmetrical. 6.2 Flood lighting 6.2.1 Related terms: Beam factor, beam angle, waste light factor, NIMA classification. 6.2.2 Design of flood lighting; related numerical problems. 6.2.3 Selection of lamps and luminaires for application of flood lighting in railway yard, sports ground.	**08**
	Total	**45**

References:

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2. Simons, R. H., Bean, Robert; Lighting Engineering: Applied Calculations, Architectural Press. ISBN: 0750650516.
3. Casimer M Decusatis, Handbook of Applied Photometry, Springer, ISBN 1563964163.
4. Butterworths, Lyons Stanley, Handbook of Industrial Lighting, Butterworths
5. Simpson Robert S, Lighting Control Technology and Applications, Focal Press
6. Kao Chen, Energy Management in Illuminating Systems, CRC Press

Course outcomes:

The theory and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

a. Understand the fundamental concept of illumination.
b. Understand the measurement process of different quantities related to illumination.
c. Identify various characteristics of different lamps and select relevant lamps for various applications .
d. Understand different techniques for illumination control.
e. Design relevant illumination schemes for interior applications.
f. Design relevant illumination schemes for exterior applications.

Internal Assessment (40 Marks)		
Mid Semester Class Test:20 Marks	Quizzes, viva-voce, Assignment: 10 Marks	Attendance: 10
External Assessment (End Semester Examination:60 Marks)		
GROUP	UNIT	
A	1,2	
B	3,4	
C	5,6	

Course Code	:	EEPE 307/1
Course Title	:	ILLUMINATION PRACTICES LABORATORY
Number of Credits	:	1 (L: 0, T: 0, P: 2)
Prerequisites	:	NIL
Course Category	:	PE

Course objectives:

The aim of this course is to help the student to attain the following industry identified

competency through various teaching learning experiences:

- Design illumination schemes and associated electrification of buildings.

Practicals: (At least eight Experiments to be performed)

1. Verify the laws of illumination.
2. Prepare control circuit of twin fluorescent lamp.
3. Prepare control circuit of HPSV lamp.
4. Prepare control circuit of LED lamp.
5. Prepare light dimmer arrangement using the relevant dimmer type of transformer
6. Prepare light dimmer arrangement using solid state dimmer circuit.
7. Study of PIR and ultrasonic sensors.
8. Conduct illumination level assessment in workplace using lux meter.
9. Fit the given lamps in the selected mounting and calculate space height ratio.
10. Interpret the polar curves of the given type of lamp.
11. Measure the illumination output of different lamps (Incandescent, fluorescent, CFL, LED, HPSV, MH) and calculate their luminous efficacy.
12. Plot the iso-lux contour of indoor lamps.
13. Design an illumination scheme of a conference hall of medium size.
14. Design an illumination scheme of a workshop of your Institute.
15. Prepare an indoor room design using lighting design software.

Course outcomes:

The practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

a) Verify laws of illumination.
b) Select relevant lamps for various applications
c) Select the lighting accessories required for specific lamps.
d) Select relevant dimmer circuit for various applications.
e) Design relevant illumination schemes for interior applications.
f) Design Illumination schemes for various exterior applications.

E X A M I N A T I O N S C H E M E (SESSIONAL)

3. **Continuous Internal Assessment of 60 marks** is to be carried out by the teachers throughout the fourth Semester. **Distribution of marks:** Continuous evaluation:50 Marks; Class Attendance: 10 Marks
4. **External Assessment (end Semester examination) of 40 marks** shall be held at the end of the fourth Semester on the entire syllabus. Assignment on the day of Viva-voce and practical report submission:20; Viva-voce:20

Course Code	:	EEPE 305/2
Course Title	:	ELECTRIC TRACTION
Number of Credits	:	3 (L: 3, T: 0, P: 0)
Prerequisites	:	NIL
Course Category	:	PE

Course objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Maintain electric traction systems.

Course contents:

Contents (Theory):		Hrs/Unit
Unit : 1	**Basics of Traction** 1.1General description of Electrical Traction system in India : Surface railway and metro railway. 1.2 Advantages and Disadvantages of Electric Drive, Diesel Electric Drive, Battery Drive. 1.3Problems associated with AC traction System and remedies for it. 1.4Voltage balance, current balance, production of harmonics. Metro rail system, features.	**05**
Unit : 2	**Traction Mechanics:** 2.1 Types of Services. 2.2 Simplified Speed Time Curve (No Derivation) Average Speed and Schedule Speed; Factors Affecting the Schedule Speed. Related numerical. 2.3 Tractive Effort. Related numerical. 2.4 Specific Energy Consumption; Factors Affecting Specific Energy Consumption. Related numerical 2.5. Mechanics of train movement, Adhesion & coefficient of Adhesion.	**10**

Unit: 3	**Power Supply Arrangements** 3.1 **Constituents of supply system: -** 31.1 Substation: layout, list of equipment and their functions 3.1.2 Feeding post: list of equipment and their functions 3.1.3 Feeding and sectioning Arrangements 3.1.4 Sectioning and paralleling post 3.1.5 Sub sectioning and Paralleling post 3.1.6 Sub sectioning post 3.1.7 Elementary section 3.1.8 Protection system for traction transformer 25 kV catenary construction **3.2 Overhead Equipment** 3.2.1 Different types of overhead equipments: Pentagonal OHE Catenary Construction. 3.2.2 Overhead system- Trolley collector, bow collector; Pantograph Collector Types and construction of pantograph	**10**
Unit: 4	**Electric Locomotive** 4.1 Classification and Nomenclature of Electric Locomotive. 4.2 Block diagram of AC locomotive; Power Circuit of AC Locomotive Equipment (List and Function only) used in auxiliary circuit of AC Locomotive 4.3 Loco bogie classification according to wheel arrangements. 4.4 Maintenance of AC systems.	**06**
Unit: 5	**Traction Motors and Train Lighting** 5.1 Desirable characteristics of traction motor. 5.2 Types of AC and DC motors used for traction with their characteristics and features. 5.3Control of Traction motors: Series-parallel starting; drum controller; contactor type controller; Thyristor control. 5.4 Requirements of braking, types of Electrical braking: Regenerative braking; Plugging; Dynamic breaking. 5.5 Systems of train lighting, Single battery, double battery parallel block system Self Generation (SG), Head On Generation (HOG), End on generation (EOG).	**08**
Unit 6	**Signaling and Supervisory Control** 6.1 Requirements of signaling systems. 6.2 Types of signals, track circuits. 6.3 Advantages of remote control Systems, 6.4 Equipment and network Metro rail-supply systems, advantages, schemes in India.	**06**
	Total	**45**

References:

1. G.C. Garg, Utlization of Electric Power & Electric Traction, Khanna Book Publishing Co., New Delhi (ISBN: 978-93-86173-355) Revised Ed. 2018

2. Gupta J.B., S.K.Kataria and Son, Utilization of Electric power and traction
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6. Open Shaw Taylor, Orient Longman ltd., Utilisation of electrical energy.

Course outcomes:

The theory and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

a) Interpret the traction layout and its systems.
b) Understand train movement features.
c) Describe the power supply arrangements and overhead equipment for electric traction.
d) Maintain the different components of the electric locomotive.
e) Understand function of traction motors and train lighting system.
f) Classify signaling and supervisory control systems.

Internal Assessment (40 Marks)		
Mid Semester Class Test:20 Marks	Quizzes, viva-voce, Assignment: 10 Marks	Attendance: 10
External Assessment (End Semester Examination:60 Marks)		
GROUP	UNIT	
A	1,2	
B	3,4	
C	5,6	

Course Code	:	EEPE 307/2 LAB
Course Title	:	ELECTRIC TRACTION LABORATORY
Number of Credits	:	1 (L: 0, T: 0, P: 2)
Prerequisites (Course code)	:	NIL
Course Category	:	PE

Course objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Maintain electric traction systems

Practicals: (At least eight Experiments to be performed)

1. Dismantle and Assemble traction motor through watching video.
2. Perform the testing of dc series motor to obtain i) torque vs current; ii) speed vs current; iii) speed vs torque characteristics.
3. Troubleshoot a traction motor.
4. Visit / watching video on electric-traction train lighting system installation, identify components of system and prepare report
5. Visit/ watching video on electric-traction loco shed, investigate working of each section & prepare report
6. Visit to / watching video on Traction Substation or feeding post (for layout and OHE) and write a report
7. Visit to / watching video on Railway Station (for signaling and train lighting) and writing a report on visit
8. Draw traction substation Layout on drawing sheet and prepare report
9. Draw Pentagonal OHE Catenary, different Catenaries according to speed limit, OHE supporting structure on drawing sheet and prepare report
10. Draw Power Circuit of AC Locomotive on drawing sheet and prepare report.

Course outcomes:

The practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

a) Interpret the traction layout and its systems
b) Maintain the power supply arrangements.
c) Maintain the function of the overhead equipment for electric traction
d) Maintain the different components of the electric locomotive.
e) Maintain the traction motor and train lighting system
f) Maintain the signaling and supervisory control systems.

. E X A M I N A T I O N S C H E M E (SESSIONAL)

7. **Continuous Internal Assessment of 60 marks** is to be carried out by the teachers throughout the fourth Semester. **Distribution of marks:** Continuous evaluation:50 Marks; Class Attendance: 10 Marks
8. **External Assessment (end Semester examination) of 40 marks** shall be held at the end of the fourth Semester on the entire syllabus. Assignment on the day of Viva-voce and practical report submission:20; Viva-voce:20

Course Code	:	EEPE 305/3
Course Title	:	**SOLAR POWER TECHNOLOGIES**
Number of Credits	:	3 (L: 3, T: 0, P:0)
Prerequisites	:	NIL
Course Category	:	PC

Course objectives: The aim of this course is to help the student to attain the following industry

identified competency through various teaching learning experiences:
Maintain the efficient operation of various types of renewable energy power plants.

Course contents:

Contents (Theory):		Hrs/ Unit
Unit1	**Solar Energy:** 1.1 Solar energy availability in India 1.2 Daily Global solar power radiation, beam radiation, diffuse radiation. 1.3 Factors affecting the amount of solar energy reaching the earth. 1.4 Different types of Solar water heaters: Construction, working, 1.5 specifications and installation Solar Heating systems used to provide hot water in a residential building 1.6 Different types of Solar cookers and their use. 1.7 Solar drying system and its utility 1.8 Solar lighting system in residential building 1.9 Troubleshooting and preventive maintenance of the above solar powered apparatus.	**8**
Unit2	**Concentrated Solar Power (CSP):** 2.1 Concentrated Solar Power (CSP) plants or solar thermal electric systems 2.2 Parabolic Trough: Construction, working and specifications 2.3 Parabolic trough systems that generate electricity 2.4 Parabolic Dish: Construction, working and specifications 2.5 Power Tower. 2.6 Fresnel Reflectors: Construction, working and specifications 2.7 Solar Stirling engine: construction, operation and its use. 2.8 Troubleshooting and preventive maintenance of all of the above equipments.	**8**
Unit 3	**Solar PV Systems:** 3.8 Site selection and planning for solar PV power plant. 3.2 Solar PV cell: Types, construction, working, Typical specifications of solar cells 3.3 Solar PV working principle: Series and parallel connections of solar modules 3.4 Solar Photovoltaic (PV) system: components layout and working. 3.5 Designing of solar PV installation, Installing solar panel mounting hardware, Formation of solar array, Mounting solar panels at proper place & angle. 3.6 Solar modules, arrays and their standard specifications 3.7 Electrical wiring between solar panel array, Junction box, Meter, Battery, Charge controller and Inverter 3.8 Connection of electrical safety devices, Earthing provision. 3.9 Roof top solar PV systems: its components, installation method and working 3.10 Streetlight solar PV systems: its components, installation and working	**11**

Unit 4	**Solar PV Electronics** 4.1 Solar Charge controllers: working and specifications, switchgear and cables. 4.2 Battery energy storage system – Energy stored, Time remaining to full charge/discharge, Number of charge/discharge cycles, Storage efficiency, Temperature of battery. 4.3 Connection of batteries in series and parallel for solar banks. 4.4 Solar Inverters: types of inverter, their working and specifications 4.5 Signal conditioning systems: working and specifications 4.6 Solar Power tracking: construction, working, tilt angle, solar radiation, I-V, P-V characteristics, maximum power point tracking (MPPT) 4.7 Troubleshooting and Maintenance of these systems.	**10**
Unit 5	**Solar PV Off-grid and Grid Tied Systems** 5.1 Concept of grid and smart grid systems 5.2 Solar off grid systems: layout and specifications 5.3 Solar Grid tied (on grid) systems: Working principle of grid-tied dc-ac inverter, grid synchronization and active power export. 5.4 Net metering: main features and working 5.5 Solar-wind Hybrid systems: Layout and specifications.	**8**
	Total	**45**

References:

1. Deambi, Suneel: From Sunlight to Electricity: a practical handbook on solar photovoltaic application; TERI, New Delhi ISBN:9788179935736.
2. David M. Buchla, Thomas E. Kissell, Thomas L. Floyd - Renewable Energy Systems, Pearson Education New Delhi , ISBN: 9789332586826
3. Kothari, D.P. et al: Renewable Energy Sources and Emerging Technologies, PHI Learning, New Delhi, ISBN: -978-81-203-4470-9
4. O.P. Gupta, Energy Technology, Khanna Publishing House, New Delhi (ISBN: 978-9386173-683)
5. B H Khan, Non-Conventional Energy Resources; McGraw Hill Education PVT Ltd.
6. Shobh Nath Singh; Non-Conventional Energy Resources; Pearson
7. Solanki, Chetan Singh, - Solar Photovoltaics: Fundamentals, Technologies andApplications, PHI Learning, New Delhi, ISBN: 9788120351110
8. Solanki, Chetan Singh, - Solar Photovoltaic Technology and Systems - A Manual For Technicians, Trainers and Engineers, PHI Learning, New Delhi, ISBN: 9788120347113

Course outcomes:

The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

1. Know about the solar energy and its use in the operation of solar powered equipments.
2. Know the components and operations of CSP plants.
3. Know solar PV systems and its components to generate electricity from solar energy.
4. Know the electronic components required for solar PV systems and MPPT systems.
5. Know about smart grid power systems and differentiate between off-grid and on-grid solar power plants and power transmission.

Internal Assessment (40 Marks)		
Mid Semester Class Test:20 Marks	Quizzes, viva-voce, Assignment: 10 Marks	Attendance: 10

External Assessment (End Semester Examination:60 Marks)	
GROUP	UNIT
A	1,2
B	3
C	4,5

Course Code	:	EEPE 307/3
Course Title	:	SOLAR POWER TECHNOLOGIES LABORATORY
Number of Credits	:	1 (L: 0, T: 0, P: 2)
Prerequisites	:	NIL
Course Category	:	PE

Course objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

1. Extract solar power using PV system and apply solar power technology for various applications.
2. Maintain the efficient operation of CSP plant and its accessories.
3. Measure different parameters of solar PV system and analyse them.

Practical: (At least Eight experiments are to be performed)

Sl. No.	Item
1.	Experiment to plot V-I Characteristics of solar panel/cell and determine the fill factor.
2.	Perform experiment to study the effect of shading on solar cell current when PV cells are connected in Series-Parallel combination.
3.	Perform experiment to study the effect of tilt angle on solar cell parameters (Voc, Vmp, Isc, Imp, Mpp, Fill factor, Efficiency).
4.	Determination of the rating of accessories (preparation of solar panel from solar module, charge controller, battery, inverter) of Solar PV system to generate electricity.
5.	Experiment to determine efficiency measurement of standalone solar PV system.
6.	Experiment to measure beam and diffuse solar radiation using Pyranometer.
7.	Experiment to generate electricity from parabolic dish CSP plant.
8.	Study of off-grid solar inverter and its troubleshooting.
9.	Study of solar smart metering system and its troubleshooting.
10.	Troubleshoot solar PV MPPT system and identify its remedy.
11.	Study of solar-wind hybrid systems.
12.	Study and troubleshoot solar off-grid systems.
13.	Assemble solar power heaters and implement it for heating.
14.	Identify and Troubleshoot solar signal conditioners.
15.	Assemble different components of solar street lighting system for AC supply and implement it for lighting.
16.	Troubleshoot solar PV panel and arrays and identify its remedy.
17.	Assemble the components of solar home lighting system & implement it for lighting.
18.	Experiment to run water pumping system using solar power.
19.	Application of Solar Simulator for designing solar PV power system.

Course outcomes:

The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

1. Identify different components of solar PV system and their applications.
2. Identify and maintain different components of CSP plant.
3. Identify different components of solar powered equipments and their applications.
4. Measure solar radiation and analyse its components.
5. Maintain off-grid and on-grid solar power plants.
6. Maintain solar PV electronics and MPPT systems

Internship-II

Course Objectives: Following are the objectives of this course:

- To learn industrial activities and apply accordingly.
- To get exposure to field level works.

During the summer vacation after 4th semester, students are ready for industrial experience. Therefore, they may choose to undergo Internship / Innovation / Entrepreneurship related activities. Students may choose either to work on innovation or entrepreneurial activities resulting in start-up or undergo internship with industry/ NGO's/ Government organizations/ Micro/ Small/ Medium enterprises to make themselves ready for the industry. In case student wants to pursue their family business a declaration by a parent may be submitted directly to the TPO.

In addition to above students may execute training activities with different Institute like workshop of ITI, Other Polytechnics and Technical Institutions; Soft skill training / free online technical courses/ industrial visit organized by Training and Placement Cell/ respective department of the respective institutions.

After completion of Internship, the student should prepare a comprehensive report to indicate what he/she has observed and learnt in training period. The student may contact Industrial Supervisor/Faculty Mentor/ TPO for assigning topics and problems and should prepare the final report on the assigned topics. The training report should be signed by the Industrial Supervisor/ Internship Faculty Mentor, TPO and HOD.

- **Activities may be conducted continuously for stipulated period of time or may be arranged in a staggered fashion.**

Course outcome

The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry-oriented COs associated with the above-mentioned competency:

- To learn new skills from industrial visit/ industrial training and supplement knowledge.
- To improve communication skill and teamwork.
- To learn strategies like time management, multi-tasking etc. in an industrial setup.
- To meet new people and learn networking skills with industry personnel.

The Internship Report will be evaluated on the basis of following criteria (as applicable)

Sl. No.	Criteria for evaluation of Internship Report
1	Originality
2	Adequacy and purposeful write up
3	Organizations, format, drawing, sketches, style, language
4	Practical applications and relationships with basic theory
5	Concept taught in the course outcome
6	Practical applications, relationships with basic theory and concept taught in the course
7	Attendance record, daily diary, quality of Internship Report.

Seminars must be arranged for the student based on his/her training report, before an Internal Committee constituted by the concerned department of the Institute. The evaluation will be based on the following criteria:

Sl. No.	Criteria for evaluation of Internship Seminar
1	Quality of content presented
2	Proper Planning for presentation
3	Effectiveness of presentation
4	Depth of knowledge and skills
5	Viva voce

E X A M I N A T I O N S C H E M E (for Internship- II): Evaluation will be based on internal assessment.

Course Code	:	PR301
Course Title	:	Major Project
Number of Credits	:	2 (L: 0, T: 0, P: 4)
Prerequisites	:	Knowledge of subjects up to 4th Semester of Electrical Engineering.
Course Category	:	PR

Course Objective:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Design and development of small electrical and electronics device/equipment.

Project group:

1. Formation of project group: Maximum **6 students per batch**.
2. Each project group should select work by consulting the guide.

Activity (atleast one of the following):

1. Load survey of your institute.
2. Fabrication of inverter.
3. Fabrication of battery charger.
5. Fabrication of Electrical circuit using Solar panel.
6. Fabrication of water level controller.
7. Simulation project using SCILAB/ MATLAB or any other simulation software.
8. Drawing based on following topics using AutoCAD
    i. Control and power circuit for electrical machines.
    ii. Single line diagram of power transmission/ distribution network.
    iii. Single line diagram of Substation.
    iv. Single line diagram of Generating station.
9. Any other topics suggested by concern teacher.

References:

1. A K Sawhney; A course in Electrical Machine Design; Dhanpat Rai & Co.
2. Raina Bhattacharya; Electrical Design, Estimating and Costing; New Age International Publishers
3. V. Rajini and V.S. Nagarajan; Electrical Machine Design; Pearson
4. Bhattacharya Chatterji; Projects in Electrical, Electronics, Instrumentation and computer Engineering.

Course outcome:

i. Develop proper planning to achieve the project goal.
ii. Collect relevant information and resources.
iii. Identify and apply proper techniques.
iv. Analyse the performance of project output.
v. Organize the Written documentation of the project work

E X A M I N A T I O N S C H E M E (SESSIONAL)

1. Continuous Internal Assessment of 60 marks is to be carried out by the teachers throughout the fourth Semester. **Distribution of marks:** Continuous evaluation: 30 Marks; Project report:20 Class Attendance: 10 Marks

2. External Assessment (end Semester examination) of 40 marks shall be held at the end of the fourth Semester on the entire syllabus. Project and project report:20; Viva-voce:20

West Bengal State Council of Technical & Vocational Education and Skill Development
(Technical Education Division)

Syllabus
of

Diploma in Electrical Engineering [EE]

Part-III (6th Semester)

2023

6th Semester

Sl.No	Category of course	Code No	Course Title	Credits	Marks	Total Contact Hours per Week	
						L	P
1	Program Core Course	**EEPC302**	Energy conservation and Audit	3	100	3	0
2	Program Core Course	**EEPC304**	Energy conservation and Audit Laboratory	1	100	0	2
3	Program Elective course IV	**EEPE 302/1**	**Any one of the following subjects to be chosen** 1. Industrial Instrumentation and Condition Monitoring	3	100	3	0
		EEPE 302/2	2. Electrical Testing and Commissioning				
		EEPE 302/3	**3.** Electric vehicles				
4	Program Elective course IV Lab	**EEPE 304/1**	**Any one of the following subjects to be chosen** 1.Industrial Instrumentation and Condition Monitoring lab	1	100	0	2
		EEPE 304/2	2. Electrical Testing and Commissioning Laboratory				
		EEPE 304/3	3. Electric vehicles Laboratory				
5	Humanities and Social Science	**HS302**	Entrepreneurship and Start-ups	3	100	3	0
6	Open Elective course-I	**OE302**	Engineering Economics and Project Management	3	100	3	0
7	Open Elective course- II	**OE 304**	**Any one of the following subjects to be chosen.** i. Mechatronics ii. Disaster management iii. Internet of Things iv. Environmental Engineering and Science v. Industrial Management vi. Sustainable development vii. Industrial Safety Engineering viii. Medical Electronics	3	100	3	0
8	Major Project	PR302		2	100	0	4
9	Seminar	SE302		2	100	0	4
	Total			**21**	**900**	**15**	**12**
Total contact hrs= 27 hrs/week							

- Student **contact hrs./ week =27**
- Theory and practical periods of 60 minutes each
- Abbreviation: L: Lecture class; P: Practical class
- **For Theoretical subjects**: Internal Assessment (40 Marks): Mid semester class test: 20 Marks; Quizzes, viva-voce, Assignment: 10 Marks; Attendance: 10; External Assessment: 60 Marks.
- **For Practical/ Sessional Subjects**: Internal Assessment-60 Marks [Continuous Evaluation:50; Class Attendance:10]; End Semester Assessment-40 Marks [Assignment on the day of Viva-voce and Practical Report submission:20; Viva-voce:20]
- To make the students more familiar with software, effort should be made to prepare laboratory report (like graph; data table etc.) in soft format in addition with traditional hard copy wherever possible.

Course Code	:	EEPC302
Course Title	:	**ENERGY CONSERVATION AND AUDIT**
Semester		6
Number of Credits	:	3 (L:3,T:0,P:0)
Prerequisites	:	NIL
Course Category	:	PC

Course Objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

1. Know energy conservation in various electrical machines and electrical installation systems.
2. Know Energy conservation through Cogeneration and Tariff.
3. Know energy audit of electrical systems.

Contents (Theory):		Hrs./Unit
Unit : 1	**Energy Conservation Basics:** 1.1 Energy Scenario: Primary and Secondary Energy, Energy demand and supply, National scenario. 1.2 Energy conservation and Energy audit – concept, need and difference. 1.3 Indian Electricity Act 2003; relevant clauses of energy conservation 1.4 BEE and its Roles in energy conservation 1.5 Star Labelling: Concept, Need and its benefits.	**04**
Unit : 2	**Energy Conservation in Electrical Machines:** 2.1 Need for energy conservation in induction motor and transformer. 2.2 Energy conservation techniques in induction motor by – i) Improving Power quality. ii) Motor survey iii) Matching motor with loading. iv) Minimizing the idle and redundant running of motor. v) Operating in star mode lower output power. vi) Rewinding of motor. vii) Replacement by energy efficient motor viii) Periodic maintenance 2.3 Energy conservation techniques in Transformer by – i) Load sharing ii) Parallel operation iii) Isolating techniques. iv) Replacement by energy efficient transformers. v) Periodic maintenance. 2.4 Energy Conservation Equipment: i) Soft starter ii) Automatic star delta converter	**11**

	iii) Variable Frequency Drives iv) Automatic p. f. controller (APFC) v) Intelligent p. f. controller (IPFC) 2.5 Energy efficient motor – features, advantages, applications and limitations. 2.6 Energy efficient transformers, amorphous transformers, epoxy Resin cast transformer / Dry type of transformer.	
Unit: 3	**Energy conservation in Electrical Installation systems:** 3.1 Aggregated Technical and commercial losses (AT&C) – Power system at state, regional, national and global level. 3.2 Causes of Technical losses and measures to reduce it – i) Controlling I^2R losses ii) Optimizing distribution voltage iii) Balancing phase currents iv) Compensating reactive power flow v) Demand side management 3.3 Causes of Commercial losses and measures to reduce it – i) In meter reading ii) In metering iii) Theft of electricity by any means 3.4 Energy conservation equipment: Maximum Demand Controller, KVAR Controller, Automatic Power Factor controller (APFC); Active harmonic filter. 3.5 Energy Conservation in Lighting System – i) Replacing Lamp sources. ii) Using energy efficient luminaries. iii) Using light controlled gears. iv) Installation of separate transformer / servo stabilizer for lighting. v) Periodic survey and adequate maintenance programs. 3.6 Energy Conservation techniques in fans, Electronic regulators. 3.7 Techniques of Energy Saving in Ventilating systems and Air Conditioners 3.8 Techniques of Energy Saving in Furnace, Ovens and Boiler.	**12**
Unit: 4	**Energy conservation through Cogeneration and Tariff:** 4.1 Co-generation and Tariff – concept, significance for energy conservation 4.2 Co-generation – Types of cogeneration on basis of sequence of energy use (Topping cycle,Bottoming cycle) 4.2.1 Types of cogeneration basis of technology (Steam turbine cogeneration, Gas turbine cogeneration, Reciprocating engine cogeneration). 4.2.2 Factors governing the selection of cogeneration system. 4.2.3 Advantages of cogeneration. 4.3 Tariff: Types of tariff structure: Special tariffs; Time-off-day tariff, Peak-off-day tariff, Power factor tariff, Maximum Demand tariff, Load factor tariff. Application of tariff system to reduce energy bill. 4.4 Energy conservation by improving load factor and power factor.	**09**
Unit: 5	**Energy Audit of Electrical System:** 5.1 Energy audit (definition as per Energy Conservation Act). 5.2 ABC analysis – its need and application.	**09**

	5.3 Energy audit instruments and their use. 5.4 Questionnaire for energy audit projects. 5.5 Energy flow diagram (Sankey diagram) and its importance. 5.6 Calculation of simple payback period for energy conservation equipment. Energy Audit procedure (walk through audit and detailed audit). 5.7 Energy Audit report format. 5.8 Numericals on energy audit.	
	Total	**45**

References:

1. Guide Books No. 1 and 3 for National Certification Examination for Energy Managers and Energy Auditors, Bureau of Energy Efficiency (BEE), Bureau of Energy Efficiency (A Statutory body under Ministry of Power, Government of India) (Fourth Edition 2015).

2. O.P. Gupta, Energy Technology, Khanna Publishing House, New Delhi

3. Henderson, P. D., India - The Energy Sector, University Press, Delhi, 2016. ISBN: 978-0195606539

4. Turner, W. C., Energy Management Handbook, Fairmount Press, 2012, ISBN 9781304520708

5. Sharma, K. V., Venkataseshaiah; P., Energy Management and Conservation, I K International Publishing House Pvt. Ltd; 2011 ISBN 9789381141298

6. Mehta ,V. K., Principles of Power System, S. Chand &Co.New Delhi, 2016, ISBN 9788121905947

7. Singh, Sanjeev; Rathore, Umesh, Energy Management, S K Kataria&Sons,New Delhi ISBN-13: 9789350141014.

8. Desai, B. G.; Rana, J. S.; A. Dinesh, V.; Paraman, R., Efficient Use and Management of Electricity in Industry, Devki Energy Consultancy Pvt. Ltd.

9. Chakrabarti, Aman, Energy Engineering And Management, e-books Kindle Edition

Course Outcomes:

The theory and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

a) Interpret energy conservation policies in India.
b) Know energy conservation techniques in electrical machines & implement it.
c) Know different types of measuring instruments for measuring electric power.
d) Apply energy conservation techniques in electrical installations.
e) Use Co-generation and relevant tariff for reducing losses in facilities.
f) Know energy audit for electrical system and apply it for real cases.

Internal Assessment (40 Marks)		
Mid Semester Class Test:20 Marks	Quizzes, viva-voce, Assignment: 10 Marks	Attendance: 10
External Assessment (End Semester Examination:60 Marks)		
GROUP	UNIT	
A	1,2	
B	3	
C	4,5	

Course Code	:	EEPC304
Course Title	:	**ENERGY CONSERVATION AND AUDIT LABORATORY**
Number of Credits	:	1 (L:0,T:0,P:2)
Prerequisites	:	NIL
Course Category	:	PC

Course Objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences –

1. Apply energy conservation in various electrical machines and electrical installation systems.
2. Apply Energy conservation through Cogeneration and Tariff.
3. Apply energy audit in electrical systems.

List of Practicals: (At least EIGHT are to be performed)

1. Experiment to compare power consumption of different types of TL with electromagnetic ballast, electronic ballast and LED lamps by direct measurements and estimate energy saving.
2. Experiment to determine the reduction in power consumption in star mode operation of Induction motor compared to delta mode at different load conditions.
3. Experiment to estimate energy saving by improving power factor using PFC/APFC for an electrical load.
4. Experiment to estimate energy saving by improving load factor for an establishment.
5. Perform experiment to determine the reduction in power consumption by replacement of Fans and regulators in a class room / laboratory.
6. Collect electricity bill of a residential consumer and suggest suitable means for energy conservation and reduce consumption.
7. Prepare an energy audit report (Phase-I, Phase-II, Phase-III) for a Workshop/ Institute.
8. Identify star labeled electrical apparatus and compare the data for various star ratings.
9. Collect electricity bill of a commercial consumer and suggest suitable tariff for energy conservation and reduction of its energy bill.
10. Collect electricity bill of an industrial consumer and suggest suitable tariff for energy conservation and its impact on energy bill.
11. Prepare a sample energy audit questionnaire for the given industrial/Commercial facility.
12. Energy flow diagram (Sankey diagram) through EMS software Web Link : demo.ecostruxure-power-monitoring-expert.se.app/Web ;User ID : demo; Password : demo

Course Outcomes

The theory, practical and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

a) Interpret energy conservation policies in India.
b) Implement energy conservation techniques in electrical machines.
c) Apply energy conservation techniques in electrical installations.

d) Use Co-generation and relevant tariff for reducing losses in facilities.
e) Apply energy audit for electrical system.

E X A M I N A T I O N S C H E M E (SESSIONAL)

1. **Continuous Internal Assessment of 60 marks** is to be carried out by the teachers throughout the fourth Semester. **Distribution of marks:** Continuous evaluation:50 Marks; Class Attendance: 10 Marks
2. **External Assessment (end Semester examination) of 40 marks** shall be held at the end of the sixth Semester on the entire syllabus. Assignment on the day of Viva-voce and practical report submission:20; Viva-voce:20

Course Code	:	EEPE 302/1
Course Title	:	INDUSTRIAL INSTRUMENTATION AND CONDITION MONITORING
Number of Credits	:	3 (L: 3, T: 0, P: 0)
Prerequisites	:	NIL
Course Category	:	PE

Course Objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences –

- Knowledge about different transduces.
- Know about of signal conditioning.
- Know about condition monitoring.

Contents (Theory):		Hrs./Unit
Unit : 1	**Fundamentals of instrumentation** 1.1 Basic purpose of instrumentation. 1.2 Basic block diagram (transduction, signal conditioning, signal presentation) and their functions. 1.3 Construction, working and application of switching devices- Push button, limit switch, float switch, pressure switch, thermostat, electromagnetic relay.	**4**
Unit : 2	**Transducers** 2.1 Concept of Transducers. 2.2 Distinguish between Transducers: 2.2.1 Primary and Secondary Transducers. 2.2.2 Electrical and Mechanical Transducers. 2.2.3 Analog and Digital Transducers. 2.2.4 Active and passive Transducers. 2.3 Advantages of electric transducers. 2.4 Required characteristics of transducers. 2.5 Factors affecting the choice of transducers. 2.6 Construction, working principle and application (with diagram & explanation) of following transducers: 2.6.1 RTD, Thermistor, Thermocouple. 2.6.2 Potentiometer (various types) 2.6.3 Strain gauge (No derivation only formula) Types of strain gauges like unbonded, bonded and semiconductor 2.6.4 LVDT and RVDT, measurement for displacement, Piezoelectric transducer. 2.6.5 Contact and non-contact type tachometer for speed measurement.	**10**

	2.6.6 Construction and Working of Flow measurement by electromagnetic and Turbine Flow meter.	
Unit: 3	**Operational Amplifier** 3.1 Different Parameters of op-amp: Input offset voltage, Input offset current, Input bias current, Differential input resistance, CMRR, SVRR, voltage gain, output voltage, slew rate, gain band- width. Output, short circuit current. 3.2 Use of op-amp (IC-741) as adder, subtractor, integrator, differentiator, comparator. 3.3 Instrumentation Amplifier.	**6**
Unit: 4	**Signal conditioning:** 2.1 Basic Concept of signal conditioning System. 2.2 Block diagram of AC and DC signal conditioning and working. 2.3. V to I converter, I to V converter, V to F converter. 2.4 Filters - Types and frequency response (No derivation) and circuits. 2.5 Multiplexing – Fundamentals, different types. 2.6 Sample and hold circuits - operation and its application	**8**
Unit: 5	**Data Acquisition System** 5.1 Generalized DAS- Block diagram and description of Transducer, signal conditioner, multiplexer, converter and recorder. 5.2 Draw Single Channel and Multi-channel DAS- Block diagram only. Difference between Signal Channel and Multi-Channel DAS. 5.3 Concept of electrical and electronic data transmission- serial, parallel, synchronous, asynchronous. 5.6 Digital display device- operation and its application of seven segment display, dot matrix display and concept of 3½, 4½ digits, LED and LCD applications.	**7**
Unit: 6	**Condition Monitoring and Diagnostic Analysis** 6.1 Definition of condition monitoring 6.2 Condition monitoring of: power transformer, electrical motors, alternators and circuit breakers - purpose and key benefits. 6.3 Test for condition monitoring: Insulation resistance, tan delta test and Polarization index, partial discharge test, transformer oil breakdown voltage test. 6.4 Concept of thermal imaging for condition monitoring.	**10**
	Total	**45**

References:

1. Sawhney, A.K. Electric and Electronic Measurement and instrumentation, Dhanpat Rai and Co.Author, Nineteenth revised edition 2011 reprint, 2014**,** ISBN:10: 8177001000
2. Rangan, C.S. G.R.Sharma. andV.S.V.Mani, Instrumentation devices and system, Pen ram International *Publishing* India Pvt. Ltd. Fifth edition, ISBN:10: 0074633503
3. Mehta, V.K. Electronics and instrumentation, Third edition-S.Chand and company Pvt Ltd Re-print, 2010, ISBN:81-219-2729-3
4. Singh, S.K. Industrial instrumentation and control, Tata McGraw-Hill, 1987. ISBN: 007451914X, 9780074519141.
5. J.G. Joshi, Electronic Measurement and Instrumentation, Khanna Publishing House, New Delhi(ISBN: 978-93-86173-621).

6. Rao, B V S Asia Club House, First Reprint, 2011, Operation and Maintenance of Electrical Equipment Vol-I, ISBN No 8185099022
7. Tarlok Singh; S. K. Kataria and Sons; Installation, Commissioning and Maintenance of Electrical Equipments.

Course outcomes:

The theory and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associatedwith the above mentioned competency:

a) Understand electrical transducers for measurement of electrical quantities and to select relevant transducers according to the applications.
b) Understand non-electrical transducers for measurement of non-electrical quantities and to select relevant transducers according to the applications.
c) Know about operational amplifier and its use in the field of instrumentation.
d) Know about signal conditioning system for their proper functioning.
e) Interpret data acquisition systems in various applications.
f) Undertake condition monitoring for diagnostic analysis of electrical equipment.

Internal Assessment (40 Marks)		
Mid Semester Class Test:20 Marks	Quizzes, viva-voce, Assignment: 10 Marks	Attendance: 10
External Assessment (End Semester Examination:60 Marks)		
GROUP	UNIT	
A	1,2	
B	3,4	
C	5,6	

Course Code	:	EEPE 304/1
Course Title	:	INDUSTRIAL INSTRUMENTATION AND CONDITION MONITORINGLABORATORY
Number of Credits	:	1 (L: 0, T: 0, P: 2)
Prerequisites	:	NIL
Course Category	:	PE

Course objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Use instrumentation equipment for condition monitoring and control.

Practicals

List of Practical:(at least EIGHT are to be done)	
1.	Identify float, proximity, limit switches, push button, pressure switch, thermostat, electromagnetic relay used in instrumentation system.
2.	Measure linear displacement by L.V.D.T.
3.	Measure the strain with the help of strain gauge.
4.	Measure temperature by PT-100, thermistor, thermocouple along with simple resistance bridge.
5.	Use Thermocouple to control the temperature of a furnace/machine
6.	Measure angular speed of rotating machine using stroboscope and tachometer.
7.	Measure the flow using flow meter (Electromagnetic/ turbine).
8.	Use op-amp(IC -741) as differentiator.
9.	Use op-amp(IC -741) as integrator.
10.	Use op-amp(IC -741) as instrumentation amplifier.
11.	Break Down Voltage test of transformer oil.
12.	Visit to testing center of electrical testing lab for tan delta and diagnostic tests and determine polarization index.
13.	Prepare a Report on various tools and equipment used for condition monitoring of electrical machines.
14.	Use of infrared camera for thermal imaging of electrical equipment.

Course outcomes:

The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associatedwith the above mentioned competency:

a) Select relevant instruments used for measuring electrical and non-electrical quantities.

b) Select relevant transducers/sensors for various applications.

c) Use relevant instruments for measuring non-electrical quantities.

d) Check the signal conditioning system for their proper functioning.

e) Use data acquisition systems in various applications.

f) Undertake condition monitoring for diagnostic analysis of electrical equipment.

Course Code	:	EEPE 302/2
Course Title	:	ELECTRICAL TESTING AND COMMISIONING
Number of Credits	:	3 (L: 3, T: 0, P: 0)
Prerequisites	:	NIL
Course Category	:	PC

Course objective

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Follow standard safety procedures in testing and commissioning of electrical equipments.

Contents (Theory):		Hrs./Unit
Unit : 1	1.1. Do's and don'ts regarding safety in domestic electrical appliances. 1.2. Electrical safety in industry/power stations/ substations at the time of operation/ control/ maintenance. 1.3. Procedure for rescuing the person who has received an electric shock, methods of providing artificial respiration (CPR). 1.4. Class of fire; Fire detection alarm, fire-fighting equipments. Precautions to be taken to avoid fire due to electrical reasons.	**04**
Unit : 2	**Installation and Erection** 2.1 Concept of foundation for installation of machinery. Requirements of foundation for static and rotating electrical machinery. 2.2 Concept of leveling and aligning Procedure for leveling and alignment of direct coupled drive, effects of misalignment. 2.3 **Installation of transformer as per** I.S. 10028(part II): 1981 reaffirmed 2021. 2.3.1 INSTALLATION: Precautions, Site Preparation, Cabling, Bushings and Cable Boxes, Connections, Precautions against Risk of Fire, Safety Precautions. 2.3.2. DRYING OF TRANSFORMERS: Precautions When Drying and Methods of Drying. 2.3 **Requirements of installation of induction motor** as per I.S. 900 – 1992: INSTALLATION WORK: Location of Motors and Control Apparatus; Drying Out; Commissioning of motor.	**10**
Unit: 3	**Testing and Commissioning** 3.1 Objectives of testing, Types of tests and concepts: Routine test, type test, supplementary test, special tests. Methods of testing - Direct/Indirect/Regenerative testing. 3.2 Factors affecting life of insulating materials. Classifications of insulating materials as per IS:1271-1985 Reaffirmed 2001. Ageing Factors and Thermal Classes 3.3 Insulating oil - properties of insulating oil: viscosity, purity, acidity, flash point and fire point; causes of deterioration of oil. **3.4 Tests before and after Commissioning**: 3.4.1 Testing of transformer oil: a) dielectric strength test; b) acidity test; c) sludge test; d) moisture test, e) flash point test. 3.4.2. Testing of transformer: Impedance voltage, load losses, Insulation resistance, induced over voltage withstand test, Impulse voltage withstand test, Temperature	**15**

	rise test of oil & winding, Different methods of determining temperature rise- back-to-back test, open delta (delta – delta) test. 3.4.3 Testing of three-phase induction motor as per I.S.4029 -2010: High voltage test; Temperature-Rise Test; No load and locked rotor test. 3.4.4 List of type, routine and acceptance tests of single-phase induction motor as per I.S.996-2009. 3.4.5 Testing methods of synchronous machines as per IS 7132-1973: High voltage tests.	
Unit: 4	**Troubleshooting Plans** 4.1 Internal and external causes for failure / abnormal operation of equipment. 4.2 List of mechanical faults, electrical faults and magnetic faults in the electrical equipment and their remedies. 4.3 Preparation of trouble shooting charts for D.C. Machines, AC Machines and transformers, batteries.	**08**
Unit: 5	**Maintenance** 5.1 Concept of maintenance, types of maintenance, time based and condition based preventive maintenance, breakdown maintenance. 5.2 Preventive maintenance schedules for electrical machines, Factors affecting preventive maintenance schedules 5.3 Concept of Total productive maintenance (TPM), Pillars of TPM 5.4 Maintenance schedules of the following: i. Power and Distribution transformer. ii. Three phase Induction motors. iii. LV and HV switchgear. iv. Station Batteries.	**08**
	Total	**45**

References:

1. Deshpande.M. V. PHI Learning Pvt. Ltd., 2010, Design and Testing of Electrical Machines ISBN No 8120336453, 9788120336452.
2. Rao, B V S Asia Club House, First Reprint, 2011, Operation and Maintenance of Electrical Equipment Vol-I, ISBN No 8185099022
3. Rosenberg. Mc GRAW-HILL, 1st Edition, May 2003, Maintenance and Repairs, ISBN No 9780071396035
4. Sharotri, S.K. Glencoe/ Mcgraw- Hill; 2ndEdition , June 1969; Preventive Maintenance of Electrical Apparatus, ISBN No 10: 007030839X 13: 978-0070308398
5. Tarlok Singh; S. K. Kataria and Sons; Installation, Commissioning and Maintenance of Electrical Equipments.
6. P.P Gupta; Dhanpat Rai Publication Ltd; Installation, Commissioning and Maintenance of Electrical Equipments.

Course outcomes:

The theory and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

a) Follow safety procedures with respect to earthing and insulation of electrical equipment
b) Select proper tools, equipment, for installation, testing, maintenance of electrical machines and transformers
c) Test and commission electrical equipment in accordance with IS codes
d) Make plans for troubleshooting electrical machines.
e) Undertake regular preventive and breakdown maintenance.

Internal Assessment (40 Marks)		
Mid Semester Class Test:20 Marks	Quizzes, viva-voce, Assignment: 10 Marks	Attendance: 10
External Assessment (End Semester Examination:60 Marks)		
GROUP	UNIT	
A	1,2	
B	3	
C	4,5	

Course Code	:	EEPE304/2
Course Title	:	ELECTRICAL TESTING AND COMMISIONING LABORATORY
Number of Credits	:	1 (L: 0, T: 0, P: 2)
Prerequisites	:	NIL
Course Category	:	PE

Course objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Follow standard safety procedures in testing and commissioning of electrical equipment.

Practicals:

List of Practical:(at least EIGHT are to be done)	
1.	Determine breakdown strength of transformer oil.
2.	Perform insulation resistance test on any one motor/transformer.
3.	Prepare trouble shooting charts for electrical machines such as Transformer, D.C. machines, Induction motor and Synchronous machines.
4.	Measure impedance voltage and load losses of three-phase transformer.
5.	Find regulation and efficiency of single-phase transformer using back-to-back connection method.
6.	Determine efficiency of D.C. motor by direct loading or by electrical loading.
7.	Determine efficiency of D.C. machine by Hopkinson's test.
8.	Perform reduced voltage running up test on three-phase Induction motor as per I.S.325.
9.	Measure no load power, losses, current of a single-phase transformer upto 110% of rated voltage.
10.	Perform no load test on single phase Induction motor for the measurements of no load current, power input, and speed at rated voltage.
11.	Methods of providing artificial respiration (CPR) and prepare a report.
12.	Study of different types of fire extinguisher.

Course outcomes:

The practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

a) Select proper tools, equipment, for installation, testing, maintenance of electrical machines and transformers

b) Test the performance of insulating oil and transformers

c) Test the performance of induction machine.

d) Test the performance of dc machines.

e) Make plans for troubleshooting electrical machines.

E X A M I N A T I O N S C H E M E (SESSIONAL)

3. **Continuous Internal Assessment of 60 marks** is to be carried out by the teachers throughout the fourth Semester. **Distribution of marks:** Continuous evaluation:50 Marks; Class Attendance: 10 Marks
4. **External Assessment (end Semester examination) of 40 marks** shall be held at the end of the sixth Semester on the entire syllabus. Assignment on the day of Viva-voce and practical report submission:20; Viva-voce:20.

Course Code	:	EEPE 302/3
Course Title	:	ELECTRIC VEHICLES
Number of Credits	:	3 (L: 3, T: 0, P: 0)
Prerequisites	:	NIL
Course Category	:	PE

Course objectives:
The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

i. To know the salient features of Hybrid electric vehicles and its auxiliaries.
ii. To explain the working of dc to dc, and dc to ac converter applied in Electric vehicles.
Iii. To know the parameters of battery used in EV and its charging system.

Contents (Theory):		Hrs./Unit
Unit : 1	**Unit – I Introduction to Hybrid Electric Vehicles** 1.1 Evolution of Electric vehicles. 1.2 Advanced Electric drive vehicle technology: 1.2.1 Vehicles-Electric vehicles (EV), 1.2.2 Hybrid Electric drive (HEV), 1.2.3 Plug in Electric vehicle (PIEV). 1.3 Components used Hybrid Electric Vehicle. 1.4 Economic and environmental impacts of Electric hybrid vehicle: i. Parameters affecting Environmental; ii. Comparative study of vehicles for economic, environmental aspects.	**10**
Unit : 2	**Dynamics of hybrid and Electric vehicles** 2.1 General description of vehicle movement. 2.2 Factors affecting vehicle motion- Vehicle resistance, tyre ground adhesion, rolling resistance, aerodynamic drag, equation of grading resistance, dynamic equation. Drive train configuration, Automobile power train, classification of vehicle power plant , Performance characteristics of IC engine, , need of gear box, electric motor. 2.3 Classification of motors used in Electric vehicles 2.3.1 Basic architecture of hybrid drive trains, types of HEVs ; Energy saving potential of hybrid drive trains. HEV Configurations-Series, parallel, Series-parallel, complex.	**10**
Unit: 3	**DC-DC Converters for EV and HEV Applications** 3.1 EV and HEV configuration based on power converters 3.2 Classification of converters –unidirectional and bidirectional 3.3 Application of Buck, Boost and Buck- Boost converters in EV using block diagram.	**05**

Unit: 4	**DC-AC Inverter & Motors for EV and HEVs** 4.1 Principle, operation and Characteristics of permanent magnet synchronous motors, BLDC and switched reluctance motor. 4.2 Applications of DC-AC Converters in EV. 4.3 Application & control of induction motor, permanent magnet synchronous motors, BLDC and switched reluctance motors used in EVs and HEVs. 4.4 Application of regenerative braking in EV.	**10**
Unit: 5	**Batteries** 5.1 Overview of batteries: Battery Parameters, types of batteries 5.2 Battery Charging and Battery Management System. 5.3 Alternative novel energy sources-solar photovoltaic cells, fuel cells, super capacitors, flywheels. 5.4. Control system for EVs and HEVs, overview, Electronic control unit ECU Schematics of hybrid drive train, control architecture.	**10**
	Total	**45**

References:

1. A.K. Babu, Electric & Hybrid Vehicles, Khanna Publishing House, New Delhi (Ed. 2018)
2. Fuhs, A. E. Hybrid Vehicles and the Future of Personal Transportation, CRC Press,
3. Gianfranco, *Electric and Hybrid Vehicles:* Power Sources, Models, Sustainability, Infrastructure And The Market, Pistoia Consultant, Rome, Italy,
4. Ehsani, M. *Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design*, CRC Press
5. Husain, I. *Electric and Hybrid Electric Vehicles*, CRC Press
6. Chan C. C. and K. T. Chau, *Modern Electric Vehicle Technology*, Oxford Science Publication,
7. Lechner G. and H. Naunheimer, *Automotive Transmissions: Fundamentals, Selection,Design and Application*, Springer
8. Rashid, M. H. *Power Electronics: Circuits, Devices and Applications*, 3rd edition, Pearson,
9. Moorthi, V. R. *Power Electronics: Devices, Circuits and Industrial Applications*, Oxford University Press
10. Krishnan, R. *Electric motor drives: modelling, analysis, and control*, Prentice Hall
11. Krause, O. P. ; C. Wasynczuk, S. D. Sudhoff, *Analysis of electric machinery*, IEEE Press

Course outcomes:

The theory and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

a) Interpret the salient features of Hybrid electric vehicles.
b) Interpret the Dynamics of hybrid and Electric vehicles
c) Analyse the DC-DC converters used in EV applications.
d) Analyse and apply the DC-AC converters in EV applications.
e) Select the batteries for EV applications.

Course Code	:	EEPE304/3
Course Title	:	ELECTRIC VEHICLES LABORATORY
Number of Credits	:	1 (L: 0, T: 0, P: 2)
Prerequisites (Course code)	:	NIL
Course Category	:	PE

Course objectives:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

Maintain electric vehicles

Practicals:

List of Practicals: (At least EIGHT are to be performed)	
1.	Develop block diagram of Electric vehicle and identify parts.
2	State of Charge (SOC) estimation by open circuit voltage.
3	Develop schematic diagram of hybrid electric vehicle and identify the components.
4.	Prepare report on Plug in Electric vehicle by visiting a charging station.
5.	Experiment on inverter connected with lead acid/ lithium-ion battery.
6.	Experiment on Buck Boost converters.
7.	Experiment on bidirectional converter.
8.	Lithium-ion battery modelling and fault detection design.
9.	Prepare test procedure for equipment used in Electric vehicle.
10.	List safety procedures and schedule for handling HEVs and EVs.
11.	Assembly of EV charging point and study of charging modes.

Course outcomes:

The theory, practical experiences and relevant soft skills associated with this course are to be taught and implemented, so that the student demonstrates the following industry oriented COs associated with the above mentioned competency:

a) Interpret the salient features of Hybrid electric vehicles.
b) Interpret the Dynamics of hybrid and Electric vehicles
c) Maintain the DC-DC converters in EV applications.
d) Maintain the DC-AC converters in EV applications
e) Select the batteries for EV applications.

Course Code	:	PR302
Course Title	:	Major Project
Number of Credits	:	2 (L: 0, T: 0, P: 4)
Prerequisites	:	Knowledge of subjects up to 5th Semester of Electrical Engineering.
Course Category	:	PR

Course Objective:

The aim of this course is to help the student to attain the following industry identified competency through various teaching learning experiences:

- Design and development of small electrical and electronics device/equipment.

Project group:
1. Formation of project group: Maximum **6 students per batch**.
2. Each project group should select work by consulting the guide.

Activity (Atleast one):

i. Speed control of stepper motor/ dc motor using microcontroller.

ii. Robotic car using microcontroller.

iii. Over voltage/ over current protection using microcontroller/ static system.

iv. Battery management system using microcontroller.

v. Home automation using microcontroller.

vi. Any topic relevant to Diploma in Electrical Engineering course as suggested by the Department/supervisor(s).

References:

1. A K Sawhney; A course in Electrical Machine Design; Dhanpat Rai & Co.
2. Raina Bhattacharya; Electrical Design, Estimating and Costing; New Age International Publishers
3. V. Rajini and V.S. Nagarajan; Electrical Machine Design; Pearson
4. Bhattacharya Chatterji; Projects in Electrical, Electronics, Instrumentation and computer Engineering.

Course outcome:

i. Develop proper planning to achieve the project goal.
ii. Collect relevant information and resources.
iii. Identify and apply proper techniques.
iv. Analyse the performance of project output.
v. Organize the Written documentation of the project work

E X A M I N A T I O N S C H E M E (SESSIONAL)

1. Continuous Internal Assessment of 60 marks is to be carried out by the teachers throughout the fourth Semester. **Distribution of marks:** Continuous evaluation: 30 Marks; Project report:20 Class Attendance: 10 Marks

2. External Assessment (end Semester examination) of 40 marks shall be held at the end of the sixth Semester on the entire syllabus. Project and project report:20; Viva-voce:20

Course Code	:	SE302
Course Title	:	Seminar
Number of Credits	:	2 (L: 0, T: 0, P: 4)
Prerequisites	:	Knowledge of subjects up to 5th Semester of Electrical Engineering.
Course Category	:	SE

The course 'Seminar' is intended to enable a student to read, understand, prepare and present report about an academic document. The learner shall search in the literature including various journals, books, project reports, online resources etc., and identify an appropriate paper /report in her/his area of interest, in consultation with her/his seminar guide. This course can help the learner to experience how a presentation can be made about a selected academic document and also empower her/him to prepare a technical report.

Course Objectives:

- To do literature survey in a selected area of study.
- To understand an academic document from the literature and to give a presentation about it.
- To prepare a technical report.

General Guidelines: It's advisable to choose topics for the Seminar to be closely linked with following topics.
Seminar1: Based on any theoretical paper/ laboratory/ previous semester project or any other topics as instructed by concerned teacher.
Seminar2 : Based on final semester project . (group/ individual student)

(Every student has to submit report and presentation on Seminar1 and Seminar2.)

The Electrical Engineering Department shall form an Internal Evaluation Committee for the seminar. During the seminar presentation of a student, all members of committee shall be present. Formation of group of students and corresponding guide allotment shall be completed as earlier after completion of 5th semester examination.

Course Outcomes: After successful completion of the course, the students will be able to:

- Identify academic documents from the literature which are related to her/his areas of interest.
- Read and apprehend an academic document from the literature which is related to her/ his areas of interest.
- Prepare a presentation about an academic document.
- Give a presentation about an academic document.
- Prepare a technical report.

E X A M I N A T I O N S C H E M E (SESSIONAL)

1. Continuous Internal Assessment of 60 marks is to be carried out by the teachers throughout the fourth Semester. **Distribution of marks:** Continuous evaluation: 30 Marks; Presentation report:20 Class Attendance: 10 Marks

2. External Assessment (end Semester examination) of 40 marks shall be held at the end of the sixth Semester: Presentation:20; Viva-voce:20

Course Title	Entrepreneurship and Start-ups
Course Code	HS 302
Number of Credits	3
Pre-Requisites	None
Total Contact Hours	3(L: 2; T: 1)/week = 45 hrs
Course Category	HS

Course Learning Objectives

1. To raise awareness, knowledge and understanding of enterprise/ entrepreneurship.
2. To motivate and inspire students toward an entrepreneurial career.
3. To understand venture creation process and to develop generic entrepreneurial competences.
4. To introduce students to the basic steps required for planning, starting and running a business.
5. To familiarise students with the different exit strategies available to entrepreneurs.

Course Outcomes:

After completing the course students will able to:

CO 1	Identify qualities of entrepreneurs, develop awareness about entrepreneurial skill and mindset and express knowledge about the suitable forms of ownership for small business
CO 2	Comprehend the basics of Business idea, Business plan, Feasibility Study report, Project Report and Project Proposal
CO 3	Understand the concept of start-up business and recognise its challenges within legal framework and compliance issues related to business.
CO 4	Make a Growth Plan and pitch it to all stakeholders and compare the various sources of funds available for start-up businesses

Detailed Course Content

Unit	Name of the Topic	Hours
1.	**ENTREPRENEURSHIP – INTRODUCTION AND PROCESS** • Concept, Competencies, Functions and Risks of entrepreneurship • Entrepreneurial Values& Attitudes and Skills • Mindset of an employee/manager and an entrepreneur • Types of Ownership for Small Businesses o Sole proprietorship o Partnerships o Joint Stock company- public limited and private limited companies • Difference between entrepreneur and Intrapreneur	10
2.	**PREPARATION FOR ENTREPRENEURIAL VENTURES** • Business Idea- Concept, Characteristics of a Promising Business Idea, Uniqueness of the product or service and its competitive advantage over peers.	20

	• Feasibility Study – Concept – Locational, Economic, Technical and Environmental Feasibility. Structure and Contents of a standard Feasibility Study Report • Business Plan – Concept, rationale for developing a Business Plan, Structure and Contents of a typical Business Plan • Project Report- Concept, its features and components • Basic components of Financial Statements- Revenue, Expenses (Revenue & capital exp), Gross Profit, Net Profit, Asset, Liability, Cash Flow, working capital, Inventory. Funding Methods-Equity or Debt. Students are just expected to know about the features and key inclusions under, Business Plan and Project Report. <u>They **may not** be asked to prepare a Business Plan/ Project Report/ Project Feasibility Report in the End of Semester Examination.</u>	
3.	**ESTABLISHING SMALL ENTERPRISES** • Legal Requirements and Compliances needed for establishing a New Unit- o NOC from Local body o Registration of business in DIC o Statutory license or clearance o Tax compliances	03
4.	**START-UP VENTURES** • Concept & Features • Mobilisation of resources by start-ups: Financial, Human, Intellectual and Physical • Problems and challenges faced by start-ups. • Start-up Ventures in India – Contemporary Success Stories and Case Studies to be discussed in the class. Case studies have been included in the syllabus to motivate and inspire students toward an entrepreneurial career from the success stories. <u>No questions are to be set from the case studies.</u>	04
5.	**FINANCING START-UP VENTURES IN INDIA** • Communication of Ideas to potential investors – Investor Pitch • Equity Funding, Debt funding – by Angel Investors, Venture Capital Funds, Bank loans to start-ups • Govt Initiatives including incubation centre to boost start-up ventures • MSME Registration for Start-ups –its benefits	06
6.	**EXIT STRATEGIES FOR ENTREPRENEURS** • Merger and acquisition exit, Initial Public Offering (IPO), Liquidation, Bankruptcy – **<u>Basic Concept only</u>**	02

Examination Scheme

- **End Semester Examination: 60 marks**

Suggested Question Paper Scheme for End Semester Examination

Group A: 20marks

Question Type	umber of questions to be set	umber of questions to be answered
MCQ, Fill in the blanks, True or False (Carrying 1 mark each)	25	20

Group B: 40marks

Question Type	umber of questions to be set	umber of questions to be answered
Subjective Type questions (Carrying 8 marks each)	10	5

- **Internal Assessment: 40 marks**
  - Class test : 20 marks
  - Assignment: 10 marks
  - Class attendance: 10 marks

Suggested Learning Resources

Sl. No.	Title of Book	Author	Publication
1.	Entrepreneurship Development	Sangeeta Sharma	Prentice Hall of IndiaLearning Private Ltd
2.	Entrepreneurship Development	S. Anil Kumar	New Age International
3.	Fundamentals of Entrepreneurship	Sangram Keshari Mohanty	Prentice Hall of India Learning Private Ltd
4.	Fundamentals of Entrepreneurship	Dr. G.K. Varshney	Sahitya Bhawan Publication
5.	Managing New Ventures: Concepts and Caseson Entrepreneurship	Anjan Raichaudhuri	Prentice Hall of India Learning Private Ltd
6.	How to Start a Business in India	Simon Daniel	Buuks, Chennai
7.	Entrepreneurship and Small Business Management	S.S. Khanka	S. Chand & Sons, New Delhi
8.	Entrepreneurship Development	Abhik Kumar	Oxford University

	and Business Ethics	Mukherjee & Shaunak Roy	Press
9.	Entrepreneurship Development and Business Ethics	Dr B Chandra & Dr B Biswas	Tee Dee Publications
10.	Entrepreneurship Development Small Business Entrepreneurship	Poornima Charantimath	Pearson Education India

Engineering Economics & Project Management

Course Code:	OE302
Course Title:	Engineering Economics & Project Management
No. of Credits:	3 (L:3, T:0,P: 0)
Prerequisites:	NIL
Course Category:	Open Elective (Compulsory for all branches)

Course Objectives:

- To acquire knowledge of basic economics to facilitate the process of economic decision making.
- To acquire knowledge on basic financial management aspects.
- To develop the idea of project plan, from defining and confirming the project goals and objectives, identifying tasks and how goals will be achieved.
- To develop an understanding of key project management skills and strategies.

Group-A

Unit-I (INTRODUCTION, THEORY OF DEMAND & SUPPLY) [9 hours]

1.1 Introduction to Engineering Economics, the relationship between Engineering and Economics

1.2 Resources, scarcity of resources, and efficient utilization of resources.

1.3 Opportunity cost, rationality costs, and benefits

1.4 Theory of Demand: the law of demand, different types of demand, determinants of demand, demand function, price elasticity of demand.

1.5 Theory of Supply: determinants of supply, supply function.

1.6 Market mechanism: Equilibrium, basic comparative static analysis (Numerical problems)

Unit-II (THEORY OF PRODUCTION & COSTS) [10 hours]

2.1 Concept of production (goods & services), Different factors of production (fixed and variable factors), Short-run Production function (Graphical illustration), and Long run production function (returns to scale).

2.2 Theory of Cost: Short-run and long-run cost curves with graphical illustration, basic concept on total cost, fixed cost, variable cost, marginal cost, average cost etc.

2.3 Economic concept of profit, profit maximization (numerical problems)

UNIT-III (DIFFERENT TYPES OF MARKET AND ROLE OF GOVERNMENT) [4 hours]

3.1 Perfect Competition: Features of Perfectly Competitive Market.

3.2 Imperfect Competition: Monopoly, Monopolistic Competition, and Oligopoly.

3.3 Role of government in Socialist, Capitalist and Mixed Economy structure with example.

Group-B

Unit-I (CONCEPT OF PROJECT) [4 hours]

1.1 Definition and classification of projects)

1.2 Importance of Project Management.

1.3 Project life Cycle [Conceptualization→Planning→Execution→Termination]

Unit-II (FEASIBILITY ANALYSIS OF A PROJECT) [10 hours]

2.1 Economic and Market analysis.

2.2 Financial analysis: Basic techniques in capital budgeting– Payback period method, Net Present Value method, Internal Rate of Return method.

2.3 Environmental Impact study–adverse impact of the project on the environment.

2.4 Project risk and uncertainty: Technical, economical, socio-political, and environmental risks.

2.5 Evaluation of the financial health of a project–Understanding the basic concept of Fixed & Working Capital, Debt & Equity, Shares, Debentures etc., and different financial ratios like Liquidity Ratios, Activity Ratios, Debt-equity ratio & Profitability Ratio (Basic concept only).

N.B: Knowledge of financial statements is not required; for the estimation of ratios the values of the relevant variables will be provided.

Unit-III (PROJECT ADMINISTRATION) [8 hours]

3.1 Gantt Chart– a system of bar charts for scheduling and reporting the progress of a project (basic concept).

3.2 Concept of Project Evaluation and Review Technique (PERT) and Critical Path method (CPM): basic concept and application with real-life examples.

Examination Scheme:

A. Semester Examination pattern of 60 marks:

1. Objective type Question (MCQ, Fill in the blanks, and Very Short question-1 mark each): At least five questions from each unit. [Total marks: 20]

2. Subjective questions: Eight questions to be answered taking at least three from each group. (Two questions should be given from each unit). [Total marks: 40]

B. Assignment (10Marks)

Guide line for Assignment (10 Marks)

Students should be instructed to prepare a report on a project (preferably the Major Project they prepare in 6th Semester), using a popular project management software in IT/ Computer Laboratory, under the guidance of the Lecturer in Computer Science & Technology and Lecturer in Humanities.

C. Class Test: Two examinations 20 marks each. Take best of two.

D. Attendance: 10 Marks

Suggested reference books:

1 *Principles of Economics – Case and Fair, Pearson Education Publication*
2 *Principles of Economics – Mankiw, Cengage Learning*
3 *Project planning, analysis, selection, implementation and review – Prasannachandra– Tata McGraw Hill.*
4 *Project Management – Gopala Krishnan – Mcmillan India Ltd*

West Bengal State Council of Technical & Vocational Education and Skill Development
(Technical Education Division)



Syllabus
of

Compulsory Subjects & Open Electives as per New Syllabus [OE]

Part-III (6th Semester)

2023

List of Compulsory Subjects & Open Electives as per New Syllabus 2023

Compulsory Subject		
Subject Code	**Name of the Subject**	**Page Number**
HS 302	Entrepreneurship & Start-ups	3-6
Open Elective - I		
Subject Code	**Name of the Subject**	**Page Number**
OE 302	Engineering Economics & Project Management	7-9
Open Elective - II		
Subject Code	**Name of the Subject**	**Page Number**
OE 304/1	Electric Vehicle Technology	10-12
OE 304/2	Industrial Management	13-16
OE 304/3	Industrial Safety	17-22
OE 304/4	Disaster Management	23-25
OE 304/5	Environmental Science & Engineering	26-29
OE 304/6	Renewable Energy	30-32
OE 304/7	Mechatronics	33-35
OE 304/8	Internet of Things	36-37
OE 304/9	Sustainable Development	38-40
OE 304/10	Medical Electronics	41-42
OE 304/11	Occupational Health & Safety Engineering	43-49
OE 304/12	Industrial Hazards & Modern Waste Management	50-54
OE 304/13	Export & Import Management	55-60
OE 304/14	Industrial Management and Safety	61-66
OE 304/15	Electrical Machines & Control	67-73
OE 304/16	Artificial Intelligence	74-75
OE 304/17	Operations Research	76-77
OE 304/18	Soft Computing Techniques	78-79
OE 304/19	Construction Management	80-82
OE 304/20	Solid Waste Management	83-85
OE 304/21	Sustainable Architecture	86-87
OE 304/22	Machine Learning	88-90
OE 304/23	Web Designing	91-94
OE 304/24	Energy & Environment Control in Metallurgical Industries	95-97
OE 304/25	Network Security Management and Administration	98-101
OE 304/26	Internet of Things (for CFS)	102-103
OE 304/27	Network Security Management and Administration Lab (for CFS)	104-105
OE 304/28	Internet of Things Lab (for CFS)	106
OE 304/29	Economic Policies in India	107-108

Course Title	Entrepreneurship and Start-ups
Course Code	HS 302
Number of Credits	3
Pre Requisites	None
Total Contact Hours	3(L: 2; T: 1)/week = 45 hrs
Course Category	HS

Course Learning Objectives

1. To raise awareness, knowledge and understanding of enterprise/ entrepreneurship.
2. To motivate and inspire students toward an entrepreneurial career.
3. To understand venture creation process and to develop generic entrepreneurial competences.
4. To introduce students to the basic steps required for planning, starting and running a business.
5. To familiarise students with the different exit strategies available to entrepreneurs.

Course Outcomes:

After completing the course students will able to:

CO 1	Identify qualities of entrepreneurs, develop awareness about entrepreneurial skill and mindset and express knowledge about the suitable forms of ownership for small business
CO 2	Comprehend the basics of Business idea, Business plan, Feasibility Study report, Project Report and Project Proposal
CO 3	Understand the concept of start-up business and recognise its challenges within legal framework and compliance issues related to business.
CO 4	Make a Growth Plan and pitch it to all stakeholders and compare the various sources of funds available for start-up businesses

Detailed Course Content

Unit	Name of the Topic	Hours
1.	**ENTREPRENEURSHIP – INTRODUCTION AND PROCESS** • Concept, Competencies, Functions and Risks of entrepreneurship • Entrepreneurial Values& Attitudes and Skills • Mindset of an employee/manager and an entrepreneur • Types of Ownership for Small Businesses o Sole proprietorship o Partnerships o Joint Stock company- public limited and private limited	10

	companies • Difference between entrepreneur and Intrapreneur	
2.	**PREPARATION FOR ENTREPRENEURIAL VENTURES** • Business Idea- Concept, Characteristics of a Promising Business Idea, Uniqueness of the product or service and its competitive advantage over peers. • Feasibility Study – Concept – Locational, Economic, Technical and Environmental Feasibility. Structure and Contents of a standard Feasibility Study Report • Business Plan – Concept, rationale for developing a Business Plan, Structure and Contents of a typical Business Plan • Project Report- Concept, its features and components • Basic components of Financial Statements- Revenue, Expenses (Revenue & capital exp), Gross Profit, Net Profit, Asset, Liability, Cash Flow, working capital, Inventory. Funding Methods-Equity or Debt. Students are just expected to know about the features and key inclusions under, Business Plan and Project Report. <u>They **may not** be asked to prepare a Business Plan/ Project Report/ Project Feasibility Report in the End of Semester Examination.</u>	20
3.	**ESTABLISHING SMALL ENTERPRISES** • Legal Requirements and Compliances needed for establishing a New Unit- o NOC from Local body o Registration of business in DIC o Statutory license or clearance o Tax compliances	03
4.	**START-UP VENTURES** • Concept & Features • Mobilisation of resources by start-ups: Financial, Human, Intellectual and Physical • Problems and challenges faced by start-ups. • Start-up Ventures in India – Contemporary Success Stories and Case Studies to be discussed in the class. Case studies have been included in the syllabus to motivate and inspire students toward an entrepreneurial career from the success stories. <u>No questions are to be set from the case studies.</u>	04

5.	**FINANCING START-UP VENTURES IN INDIA** • Communication of Ideas to potential investors – Investor Pitch • Equity Funding, Debt funding – by Angel Investors, Venture Capital Funds, Bank loans to start-ups • Govt Initiatives including incubation centre to boost start-up ventures • MSME Registration for Start-ups –its benefits	06
6.	**EXIT STRATEGIES FOR ENTREPRENEURS** • Merger and acquisition exit, Initial Public Offering (IPO), Liquidation, Bankruptcy – **Basic Concept only**	02

Examination Scheme

- **End Semester Examination: 60 marks**

  Suggested Question Paper Scheme for End Semester Examination

  Group A: 20marks

Question Type	Number of questions to be set	Number of questions to be answered
MCQ, Fill in the blanks, True or False (Carrying 1 mark each)	25	20

  Group B: 40marks

Question Type	Number of questions to be set	Number of questions to be answered
Subjective Type questions (Carrying 8 marks each)	10	5

- **Internal Assessment: 40 marks**
  - Class test : 20 marks
  - Assignment: 10 marks
  - Class attendance: 10 marks

Suggested Learning Resources

Sl. No.	Title of Book	Author	Publication
1.	Entrepreneurship Development	Sangeeta Sharma	Prentice Hall of IndiaLearning Private Ltd
2.	Entrepreneurship Development	S. Anil Kumar	New Age International
3.	Fundamentals of Entrepreneurship	Sangram Keshari Mohanty	Prentice Hall of India Learning Private Ltd
4.	Fundamentals of Entrepreneurship	Dr. G.K. Varshney	Sahitya Bhawan Publication
5.	Managing New Ventures: Concepts and Caseson Entrepreneurship	Anjan Raichaudhuri	Prentice Hall of India Learning Private Ltd
6.	How to Start a Business in India	Simon Daniel	Buuks, Chennai
7.	Entrepreneurship and Small Business Management	S.S. Khanka	S. Chand & Sons, New Delhi
8.	Entrepreneurship Development and Business Ethics	Abhik Kumar Mukherjee & Shaunak Roy	Oxford University Press
9.	Entrepreneurship Development and Business Ethics	Dr B Chandra & Dr B Biswas	Tee Dee Publications
10.	Entrepreneurship Development Small Business Entrepreneurship	Poornima Charantimath	Pearson Education India

Draft Syllabus of Engineering Economics & Project Management

Course Code:	OE301
Course Title:	Engineering Economics & Project Management
No. of Credits:	3 (L:3, T:0,P: 0)
Prerequisites:	NIL
Course Category:	Open Elective (Compulsory for all branches)

Course Objectives:

- To acquire knowledge of basic economics tofacilitate theprocess ofeconomic decision making.
- To acquire knowledge on basic financial management aspects.
- To develop the idea of project plan, fromdefining and confirming theproject goals and objectives, identifying tasks andhow goalswill be achieved.
- To develop an understanding of key project management skills and strategies.

Group-A

Unit-I (INTRODUCTION, THEORY OF DEMAND & SUPPLY) [9 hours]

1.1 Introduction to Engineering Economics, the relationship between Engineering and Economics

1.2 Resources, scarcity of resources, and efficient utilization of resources.

1.3 Opportunity cost, rationality costs, and benefits

1.4 Theory of Demand: the law of demand, different types of demand, determinants of demand, demand function, price elasticity of demand.

1.5 Theory of Supply: determinants of supply, supply function.

1.6 Market mechanism: Equilibrium, basic comparative static analysis (Numerical problems)

Unit-II (THEORY OF PRODUCTION & COSTS) [10 hours]

2.1 Concept of production (goods & services), Different factors of production (fixed and variable factors), Short-run Production function (Graphical illustration), and Long run production function (returns to scale).

2.2 Theory of Cost: Short-run and long-run cost curves with graphical illustration, basic concept on total cost, fixed cost, variable cost, marginal cost, average cost etc.

2.3 Economic concept of profit, profit maximization (numerical problems)

UNIT-III (DIFFERENT TYPES OF MARKET AND ROLE OF GOVERNMENT) [4 hours]

3.1 Perfect Competition: Features of Perfectly Competitive Market.

3.2 Imperfect Competition: Monopoly, Monopolistic Competition, and Oligopoly.

3.3 Role of government in Socialist, Capitalist and Mixed Economy structure with example.

Group-B

Unit-I (CONCEPT OF PROJECT) [4 hours]

1.1 Definition and classification of projects)

1.2 Importance of Project Management.

1.3 Project life Cycle [Conceptualization→Planning→Execution→Termination]

Unit-II (FEASIBILITY ANALYSIS OF A PROJECT) [10 hours]

2.1 Economic and Market analysis.

2.2 Financial analysis: Basic techniques in capital budgeting– Payback period method, NetPresentValue method, InternalRate of Returnmethod.

2.3 Environmental Impact study–adverse impact of the project on the environment.

2.4 Project risk and uncertainty: Technical, economical, socio-political, and environmental risks.

2.5 Evaluation of the financial health of a project–Understanding the basic concept of Fixed & Working Capital, Debt & Equity, Shares, Debentures etc., and different financial ratios like Liquidity Ratios, Activity Ratios, Debt-equity ratio & Profitability Ratio (Basic concept only).

N.B: Knowledge of financial statements is not required; for the estimation of ratios the values of the relevant variables will be provided.

Unit-III (PROJECT ADMINISTRATION) [8 hours]

3.1 Gantt Chart– a system of bar charts for scheduling and reporting the progress of a project (basic concept).

3.2 Concept of Project Evaluation and Review Technique (PERT) and Critical Path method (CPM): basic concept and application with real-life examples.

ExaminationScheme:

A. SemesterExaminationpatternof60 marks:

1. Objective type Question (MCQ, Fill in the blanks, and Very Short question-1 mark each): At least five questions from each unit. [Total marks: 20]

2. Subjective questions: Eight questions to be answered taking at least three from each group. (Two questions should be given from each unit). [Total marks: 40]

B. Assignment (10Marks)

Guidelinefor Assignment (10 Marks)

Students should be instructed to prepare a report on a project (preferably the Major Project they prepare in 6th Semester), using a popular project management software in IT/ Computer Laboratory, under the guidance of the Lecturer in Computer Science & Technology and Lecturer in Humanities.

C. Class Test: Two examinations 20 marks each. Take best of two.

D. Attendance: 10 Marks

Suggested reference books:

1 *Principles of Economics – Case and Fair, Pearson Education Publication*
2 *Principles of Economics – Mankiw, Cengage Learning*
3 *Project planning, analysis, selection, implementation and review – Prasannachandra–Tata McGraw Hill.*
4 *Project Management – Gopala Krishnan – Mcmillan India Ltd*

PROPOSED SYLLABUS FOR ENVIRONMENTAL SCIENCE & ENGINEERING

Course Code	OE
Course Name	ENVIRONMENTAL SCIENCE & ENGINEERING
Number of Credits and L-T-P	3 [L – 3, T – 0, P - 0]
Course Category	OE
Prerequisites	NA

Course Objectives:

After completing this course, the students will be able

1. To increase the awareness towards Environmental Science and Engineering.
2. To recognize and apply the role of technology towards Environmental Science and Engineering.
3. To know the method and tools used for Environmental Science and Engineering.
4. To know about the environmental pollution management act.

Course Contents:

Unit No.	Description of Topic	Contact Hrs.
01	**Environment and Ecology** 1.1 Classification of Environment 1.2 Environmental descriptors 1.3 Environmental quality and descriptive parameters 1.4 Ecology: Definition and classification 1.5 Environmental impact on ecology	08
02	**Water pollution and pollutants (Natural and Anthropogenic)** 2.1 Ground water: Sources and quality analysis 2.2 Surface water: Sources and quality analysis 2.3 Quality parameters in water treatment along with flow-sheets 2.4 Basic processes for potable water supply (Detailed technology not necessary) 2.5 Water pollution: Surface and ground water pollution, types of pollutants 2.6 Mode of water pollution 2.7 Parameters to be assessed for water pollution (Turbidity, pH, total suspended solids, total solids, BOD and COD:	10

	Definition, calculation) 2.8 Chemistry aspect for water pollution 2.9 Control of water pollution (Description only) 2.10 Fundamental of water treatment techniques.	
03	**Air quality, Air Pollution and Control, Noise Pollution** 3.1 Definition of pollution and pollutant, Natural and manmade sources of air pollution (Refrigerants, I.C., Boiler) 3.2 Air Pollutants: Types, Units of air pollutants 3.3 Atmospheric physics for air pollution 3.4 Particulate Pollutants: Effects and control strategies (Bag filter, Cyclone separator, Electrostatic Precipitator) 3.5 Advanced air pollution control methods 3.6 Noise pollution: sources of pollution, measurement of noise pollution 3.7 Noise measuring devices and their demonstration	10
04	**Solid waste and Soil pollution** 4.1 Definition of solid waste 4.2 Classification of solid waste 4.3 Overview on municipal, industrial, hazardous, hospital, plastic, E-waste.etc. 4.4 Solid waste management and disposal process. 4.5 Soil pollution ,Poor Fertility, Septicity, Concentration of Infecting Agents in Soil 4.6 Leaching and its impact on soil pollution.	06
05	**Renewable sources of Energy** 5.1 Energy Resources: Energy scenario, national and international status. 5.2 Solar Photovoltaics: Solar radiation and types, basic working principle of solar PV, solar cells and types, water pumping and applications of solar PV. 5.3 Solar Thermal system: basic working principle and applications of solar thermal energy, solar water	06

	heater and types, solar cooking, solar pond, Solar still etc. 5.4 Wind energy systems: basic principle, types of wind turbines, application of wind energy, 5.5 Bio-energy systems: bio thermal and chemical basic principle, gasifier and digesters. 5.6 Hydro energy systems: small and micro hydro systems and its basic working. 5.7 Geothermal energy: Basic working principle, types and application of geothermal energy. 5.8 Ocean & Tidal Energy: Basic working principle, applications and types of different types of energy generation through ocean and tidal systems	
06	**Environment Legislation system and Rules** 6.1 Environmental protection rules 6.2 Sustainable environmental management	02
Total Hours		**42**

Weightage distribution in both objective, short and broad answer type questions:

Group	Unit Number	Weightage (%)
A	1 & 2	50
B	3 & 4	30
C	5 & 6	20

Course Outcomes:

At the end of the course, the student will be able to:

CO1	Recognize the relevance and the concept of Environmental Science and Engineering and different world-wide activities on this area.
CO2	Illuminate the different types of environmental pollutant, their effects and their sustainable solutions.
CO3	Discuss the environmental regulations act. and standards
CO4	Gather basic idea about conventional and non-conventional energy resources
CO5	Demonstrate the broad perspective of Environmental Science practices by utilizing engineering knowledge and principles

Text Books:

1. **Environmental Studies- By N.N.Basak**
2. **Environmental Studies-By D .Srivastava**
3. **Introduction to Environmental Engineering— By Dr.Manindra Nath Patra.**
4. **Environmental Engineering- By A.K.Jain**

Reference Books:

1. **Environmental Engineering---By G.Killy**
2. **Environmental Engineering--- By Peavy, Rowe**
3. **Water and Waste Water Engineering— By S.Garg**
4. **Waste Water Engineering--By -Panmia**
5. **Non-conventional Energy Sources-4th Edition, By Prasad Rajesh K and Ojha**
6. **Non-conventional Energy Resources—By Chauhan and Srevastava**
7. **Non-conventional Energy Sources---By G.D.Rai (Khanna Publisher)**
8. **Ecology --By -Odum**
9. **Ecology---By -Das & Das**
10. **Environmental Law ---By -Gurdip Sing**
11. **Environmental Law----By Jaiswal Jaiswal Jaiswal**
12. **Environmental Law in India ---By -P.Leela Krishnan**
13. **Environment Impact Assessment Guidelines, Notification of Government of India, 2006**
14. **Mackenthun, K.M., Basic Concepts in Environmental Management, Lewis Publication, London, 1998**
15. **ECBC Code 2007, Bureau of Energy Efficiency, New Delhi Bureau of Energy Efficiency Publications-Rating System, TERI Publications - GRIHA Rating System**